

Schematic / PCB #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-1160	1	SCH_MLB,J94	SCH	CRITICAL	
820-4274	1	PCBF_MLB,J94	PCB	CRITICAL	
685-00030	1	PCBA_MLB,COMMON PARTS,J94	CMNPPTS		MLB_CMNPPTS

Main BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
985-0889	PCBA_MLB,DEV,J94	DEVELOPMENT,J94_DEVEL
939-00653	PCBA_MLB,DEV,J94,CPU_INT	MLB_CMNPPTS,ALTERNATE,X403:Y,CPU:SOCKET,DDR3:HYNIX_8GB_1866,SSD:Y
639-00915	PCBA_MLB,HY,8GB,29NM,1866,SSD,J94	MLB_CMNPPTS,ALTERNATE,X403:Y,CPU:BGA,DDR3:HYNIX_8GB_1866,SSD:Y
639-00916	PCBA_MLB,EL,8GB,25NM,1866,SSD,J94	MLB_CMNPPTS,ALTERNATE,X403:Y,CPU:BGA,DDR3:ELPIDA_8GB_1866,SSD:Y
639-00917	PCBA_MLB,SA,8GB,23NM,1866,SSD,J94	MLB_CMNPPTS,ALTERNATE,X403:Y,CPU:BGA,DDR3:SAMSUNG_8GB_1866,SSD:Y
639-00918	PCBA_MLB,EL,16GB,25NM,1866,SSD,J94	MLB_CMNPPTS,ALTERNATE,X403:Y,CPU:BGA,DDR3:ELPIDA_16GB_1866,SSD:Y
639-00919	PCBA_MLB,HY,16GB,25NM,1866,SSD,J94	MLB_CMNPPTS,ALTERNATE,X403:Y,CPU:BGA,DDR3:HYNIX_16GB_1866,SSD:Y
639-00920	PCBA_MLB,HY,8GB,29NM,1866,HDD,J94	MLB_CMNPPTS,ALTERNATE,X403:Y,CPU:BGA,DDR3:HYNIX_8GB_1866,SSD:N
639-00921	PCBA_MLB,EL,8GB,25NM,1866,HDD,J94	MLB_CMNPPTS,ALTERNATE,X403:Y,CPU:BGA,DDR3:ELPIDA_8GB_1866,SSD:N
639-00922	PCBA_MLB,SA,8GB,23NM,1866,HDD,J94	MLB_CMNPPTS,ALTERNATE,X403:Y,CPU:BGA,DDR3:SAMSUNG_8GB_1866,SSD:N
639-00923	PCBA_MLB,EL,16GB,25NM,1866,HDD,J94	MLB_CMNPPTS,ALTERNATE,X403:Y,CPU:BGA,DDR3:ELPIDA_16GB_1866,SSD:N
639-00924	PCBA_MLB,HY,16GB,25NM,1866,HDD,J94	MLB_CMNPPTS,ALTERNATE,X403:Y,CPU:BGA,DDR3:HYNIX_16GB_1866,SSD:N
639-00946	PCBA_MLB,X432,HY,8GB,29NM,1866,SSD,J94	MLB_CMNPPTS,ALTERNATE,X403:N,CPU:BGA,DDR3:HYNIX_8GB_1866,SSD:Y
639-00947	PCBA_MLB,X432,EL,16GB,25NM,1866,SSD,J94	MLB_CMNPPTS,ALTERNATE,X403:N,CPU:BGA,DDR3:ELPIDA_16GB_1866,SSD:Y
685-00030	PCBA_MLB,COMMON PARTS,J94	J94_COMMON

BOM Groups

BOM GROUP	BOM OPTIONS
J94_COMMON	COMMON,ALTERNATE,J94_PROGPARTS,SHCKRES:1SUP,XDB,CPUPBRS:KX88,USB_OC_180:Y,AUDIO_DP_SNS:Y,CPU_OUT_CAF:H,FP1V2_S1_CAF:H,FP1V05_S0_CAF:H,FPCPU_VDDQ_CAF:H,RXCRST:Y,CPUVCC:3PBASE,SHIELDS
J94_PROGPARTS	SMC:PROG,BOOTROM:PROG,CAMROM:PROG,TBTROM:PROG,ENETROM:PROG
J94_DEVEL	XDP_CONN,TEMPSNSDEV,SAMCONN

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S00110	1	CPU,BW,QRPD,8S,8S,4/3,2,7,K5W,-8,8GAL364	U0500	CRITICAL	CPU:BGA
998-6122	1	INTERFUSER,8GAL364	U0500	CRITICAL	CPU:SOCKET

ASIC Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S00106	1	IC,LP_FCH_QMR2,9-DESERIO-D,80,8S,FCHQMR95	U1100	CRITICAL	
338S1247	1	IC,TBT_FB-4C,A0,PAQ,C10,SR13C,FCBGA288	U2800	CRITICAL	
343S0616	1	IC,BCH07766A,C1V+,A0,8x8	U3900	CRITICAL	

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
341S00185	1	IC,EFT,V0119,J94	U5210	CRITICAL	BOOTROM:PROG
335S00006	1	IC,SERIAL_FLASH,64MBIT,3V,8P,N80W,QB=1	U5210	CRITICAL	BOOTROM:BLANK
341S00181	1	IC,SMC-B1,EXTERNAL,(VTRD),POC,J94	U5000	CRITICAL	SMC:PROG
338S1159	1	IC,SMC12-AJ,49MB1/50MB2,OCV,PW,157WGA	U5000	CRITICAL	SMC:BLANK
341S3778	1	IC,CAMERA_FLASH,V7229,J16	U4202	CRITICAL	CAMROM:PROG
335S0852	1	IC,FLASH,SPI,1MBIT,3V3	U4202	CRITICAL	CAMROM:BLANK
341S00164	1	IC,EPROM,T29,FALCON RIDGE,(VTRD),POC,J94	U2890	CRITICAL	TBTROM:PROG
335S0915	1	IC,FLASH,SPI,4MBIT,50MHZ	U2890	CRITICAL	TBTROM:BLANK
341S3912	1	IC,ENET_EPI_ROM,N80WTA,V1.15,216/216G/217/2170	U3990	CRITICAL	ENETROM:PROG
335S1025	1	IC,SERIAL_FLASH,2MBIT,2-TV,REV F	U3990	CRITICAL	ENETROM:BLANK

CPU DRAM SPD Straps

BOM GROUP	BOM OPTIONS
DDR3:HYNIX_8GB_1600	RAMCFG0:L, RAMCFG1:L, RAMCFG2:L, RAMCFG3:H, RAMCFG4:L, HYNIX_8GB_1600
DDR3:HYNIX_16GB_1600	RAMCFG0:L, RAMCFG1:L, RAMCFG2:H, RAMCFG3:H, RAMCFG4:L, HYNIX_16GB_1600
DDR3:HYNIX_8GB_1866	RAMCFG0:L, RAMCFG1:L, RAMCFG2:L, RAMCFG3:H, RAMCFG4:H, HYNIX_8GB_1866
DDR3:HYNIX_16GB_1866	RAMCFG0:L, RAMCFG1:L, RAMCFG2:H, RAMCFG3:H, RAMCFG4:H, HYNIX_16GB_1866
DDR3:ELPIDA_8GB_1600	RAMCFG0:H, RAMCFG1:H, RAMCFG2:L, RAMCFG3:H, RAMCFG4:L, ELPIDA_8GB_1600
DDR3:ELPIDA_16GB_1600	RAMCFG0:H, RAMCFG1:H, RAMCFG2:H, RAMCFG3:H, RAMCFG4:L, ELPIDA_16GB_1600
DDR3:ELPIDA_8GB_1866	RAMCFG0:H, RAMCFG1:L, RAMCFG2:L, RAMCFG3:H, RAMCFG4:H, ELPIDA_8GB_1866
DDR3:ELPIDA_16GB_1866	RAMCFG0:H, RAMCFG1:H, RAMCFG2:H, RAMCFG3:H, RAMCFG4:H, ELPIDA_16GB_1866
DDR3:SAMSUNG_8GB_1600	RAMCFG0:H, RAMCFG1:L, RAMCFG2:L, RAMCFG3:L, RAMCFG4:L, SAMSUNG_8GB_1600
DDR3:SAMSUNG_8GB_1866	RAMCFG0:H, RAMCFG1:L, RAMCFG2:L, RAMCFG3:L, RAMCFG4:H, SAMSUNG_8GB_1866

Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0155	377S00011		ALL	USB3 diodes
377S0184	377S00011		ALL	USB3 diodes
377S0124	377S0057		ALL	TVS
155S0578	155S0367		ALL	120OHM EMI BEAD
128S0368	128S0365		ALL	150UF AL POLY
138S0681	138S0638		ALL	Taiyo 10uf 805 alt
197S0479	197S0478		ALL	12 MHz Cam. Xtal
197S0486	197S0478		ALL	12 MHz Cam. Xtal
107S0251	107S0249		ALL	2mOhm Sense resistor
107S0251	107S0249	SSD:Y	ALL	2mOhm Sense resistor
107S0255	107S0240		ALL	1mOhm Sense resistor
197S0481	197S0480		ALL	25MHz Xtal
197S0343	197S0480		ALL	25MHz Xtal
197S0369	197S0392		ALL	32 KHz PCH Xtal
197S0399	197S0392		ALL	32 KHz PCH Xtal
138S0860	138S0775		ALL	Single-source 1uF 402
138S0860	138S0775	SSD:Y	ALL	Single-source 1uF 402
378S0391	378S0390		ALL	Debug LEDs
341S00016	341S3912		ALL	ENET ROM,ADESTO,V1.15
138S0747	138S0773		ALL	1uF,X6S,402
376S0572	376S0659		ALL	Single P-Ch FET
376S0572	376S0659	SSD:Y	ALL	Single P-Ch FET
376S00001	376S0659		ALL	Single P-Ch FET
376S00001	376S0659	SSD:Y	ALL	Single P-Ch FET
376S0972	376S00075		ALL	Single N-Ch FET
138S1103	138S0719		ALL	CAP,4.7uF,20V,10V,X5R,0402
155S0830	155S0316		ALL	FER BD,600 OHM,0.5A,0603
155S00076	155S0546		ALL	FER BD,600 OHM,300mA,0402
152S1373	152S1876		ALL	IND,PWR,7.8 mOhm,13A,7.1uV,9x3mm

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1089	376S1128		ALL	Single N-Ch FET
155S0660	155S0513		ALL	220HM EMI BEAD
155S0694	155S0387		ALL	4700HM EMI BEAD
127S0164	127S0162		ALL	1UF 25V TANT
376S1129	376S0855		ALL	Dual N-Ch FET
376S00074	376S0855		ALL	Dual N-Ch FET
376S1129	376S0855	SSD:Y	ALL	Dual N-Ch FET
376S00074	376S0855	SSD:Y	ALL	Dual N-Ch FET
311S0649	311S0541		ALL	Single AND Gate
107S0375	107S00039		ALL	DDR Sense Res
353S3814	353S3812		ALL	TBT mux
353S4376	353S3384		ALL	HDD OOBv1 comparator
311S00013	311S0508		ALL	Single Driver Buffer OD
371S00019	371S0463		ALL	Rohn Schottky Barrier Diode
138S0746	138S0705		ALL	CAP,CER,X5R,10uF,20V,10V,402
371S00017	371S0749		ALL	DIODE,SCHOTTKY,30V,1A,SOD-323
353S00095	353S3328		ALL	SOC Crossbar
353S00440	353S3052		ALL	USB Power Switch

Strategic Silicon

PART#	STRATEGIC VALUE	COMMENT
998-01027	08	CPU,BRW,4C+3E,65W
338S1247	02	TBT,Falcon Ridge-4c
335S00006	07	IC,SERIAL_FLASH,Quad-10
333S0784	07	HYNIX,32Gb,25nm LPDDR3-1866
333S0786	07	HYNIX,16Gb,29nm LPDDR3-1866
333S0790	07	ELPIDA,32Gb,25nm LPDDR3-1866
333S0792	07	ELPIDA,16Gb,25nm LPDDR3-1866
333S00004	07	SAMSUNG,16Gb,23nm LPDDR3-1866

DRAM Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
333S0783	4	IC,SDRAM,25nm 32Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	HYNIX_16GB_1600
333S0784	4	IC,SDRAM,25nm 32Gb,LPDDR3-1866,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	HYNIX_16GB_1866
333S0785	4	IC,SDRAM,29nm 16Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	HYNIX_8GB_1600
333S0786	4	IC,SDRAM,29nm 16Gb,LPDDR3-1866,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	HYNIX_8GB_1866
333S0789	4	IC,SDRAM,25nm 32Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	ELPIDA_16GB_1600
333S0790	4	IC,SDRAM,25nm 32Gb,LPDDR3-1866,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	ELPIDA_16GB_1866
333S0791	4	IC,SDRAM,25nm 16Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	ELPIDA_8GB_1600
333S0792	4	IC,SDRAM,25nm 16Gb,LPDDR3-1866,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	ELPIDA_8GB_1866
333S00003	4	IC,SDRAM,23nm 16Gb,LPDDR3-1600,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	SAMSUNG_8GB_1600
333S00004	4	IC,SDRAM,23nm 16Gb,LPDDR3-1866,178P FBGA	U2300,U2400,U2500,U2600	CRITICAL	SAMSUNG_8GB_1866

CPU DRAM CFG Chart

VENDOR	CFG 1	CFG 0	DIE REV	CFG 3
HYNIX	0	0	A	0
N/A	0	1	B	1
SAMSUNG	1	0		
ELPIDA	1	1		

SIZE	CFG 2	SPEED	CFG 4
8GB	0	1600	0
16GB	1	1866	1

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		SHEET	2 OF 73

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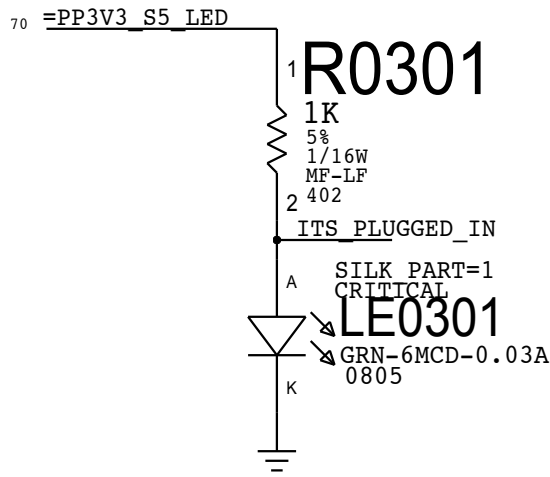
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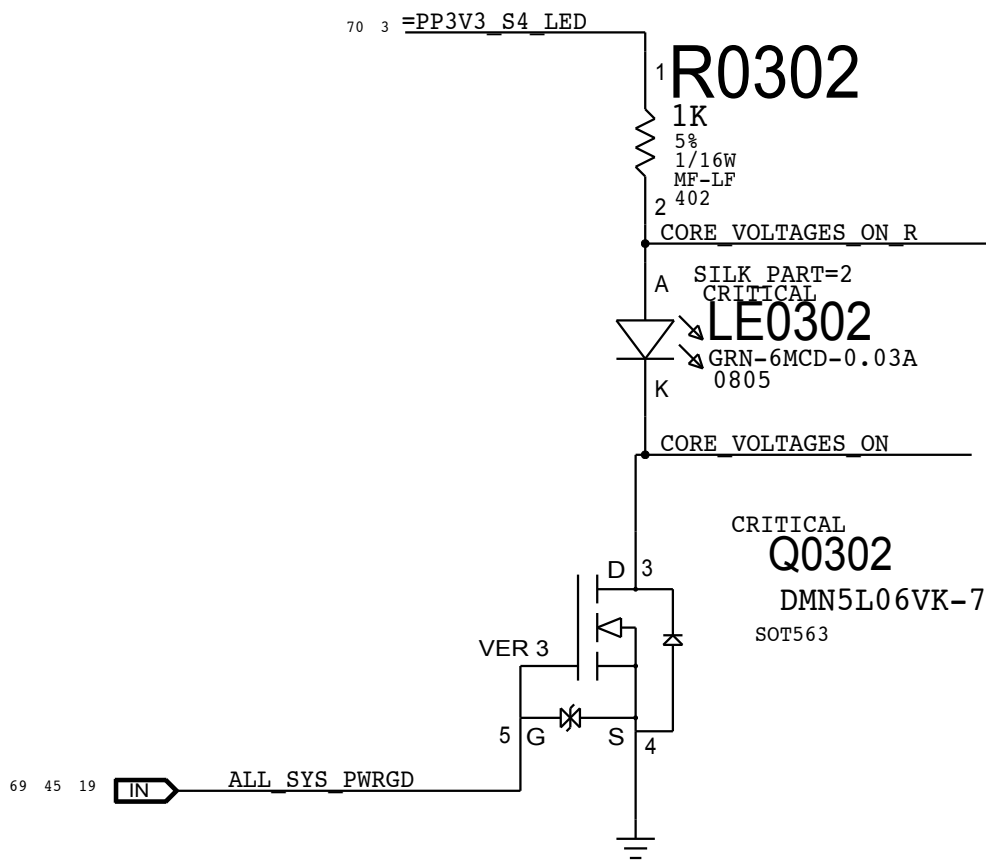
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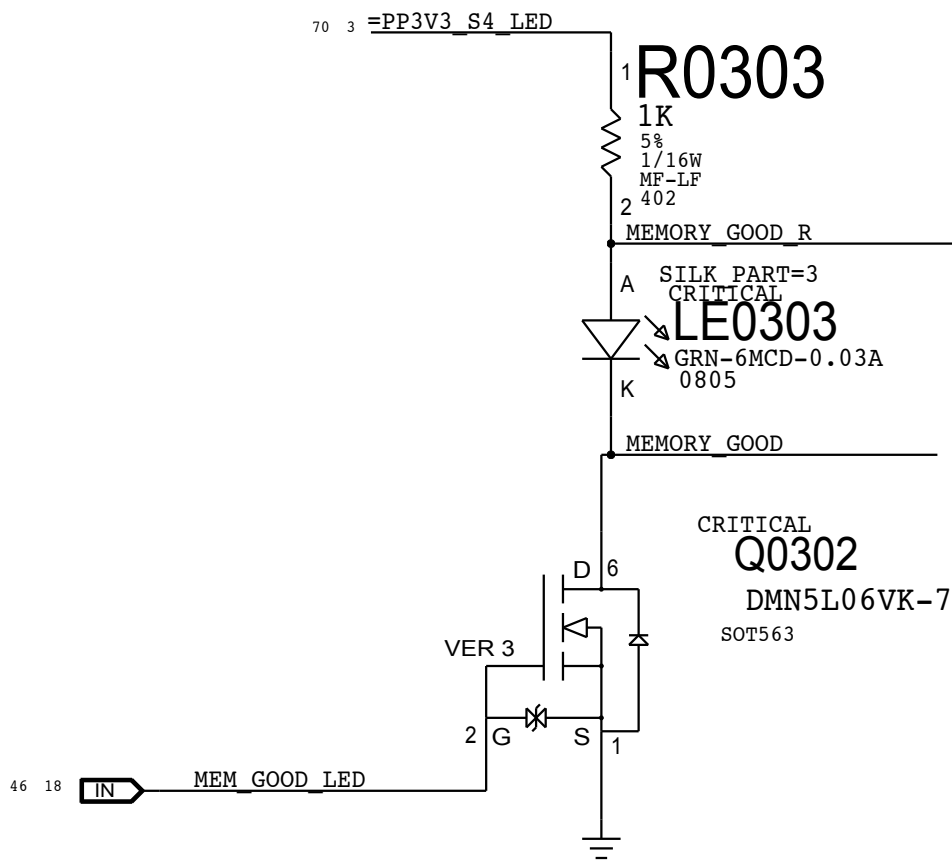
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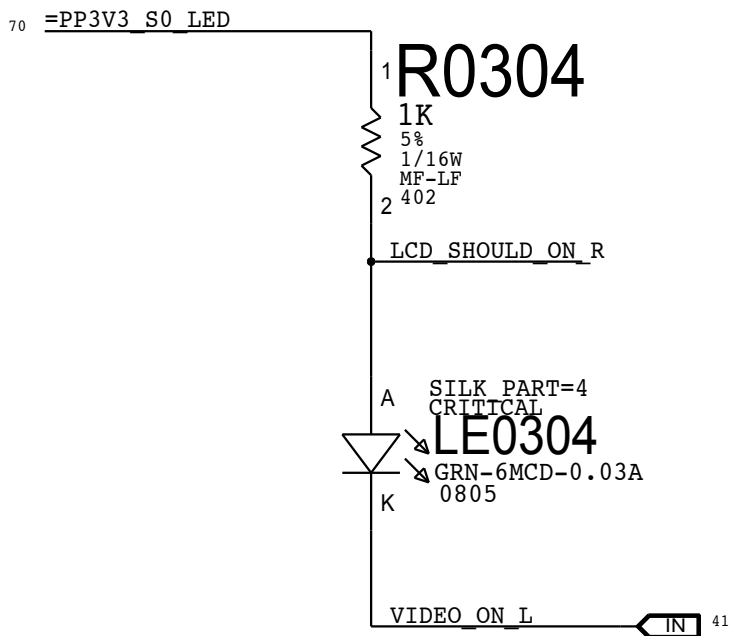
ALL_SYS_PWRGD Led



MEM_GOOD Led



VIDEO ON Led



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			PAGE	3 OF 105	
			SHEET	3 OF 73	

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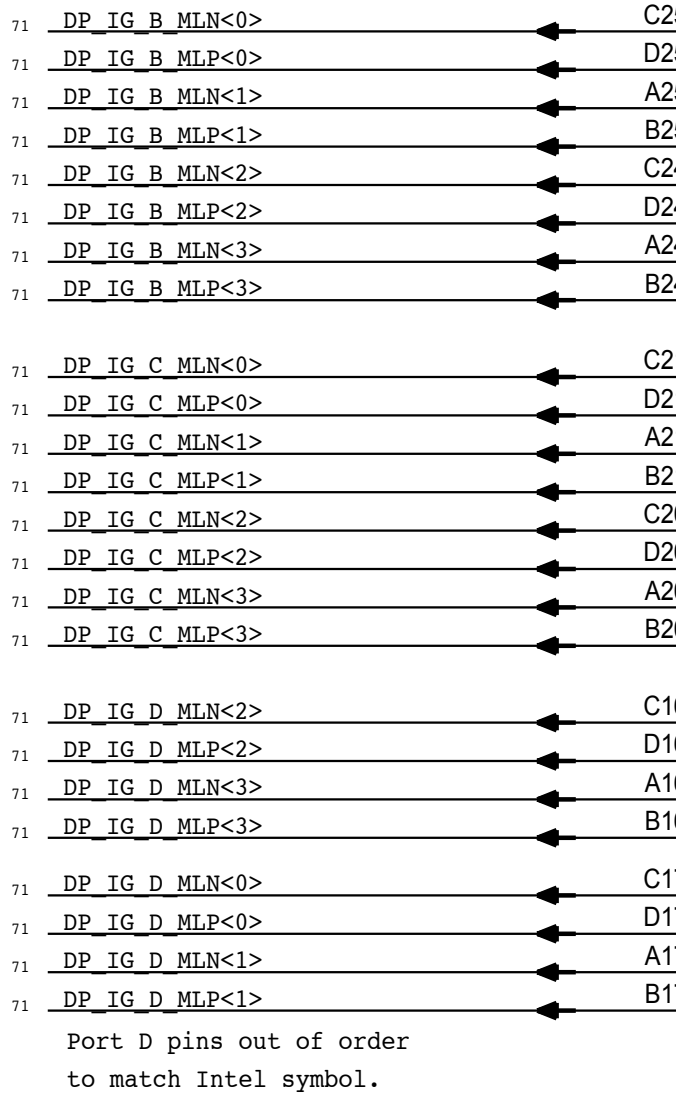
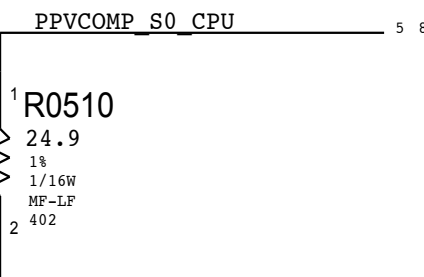
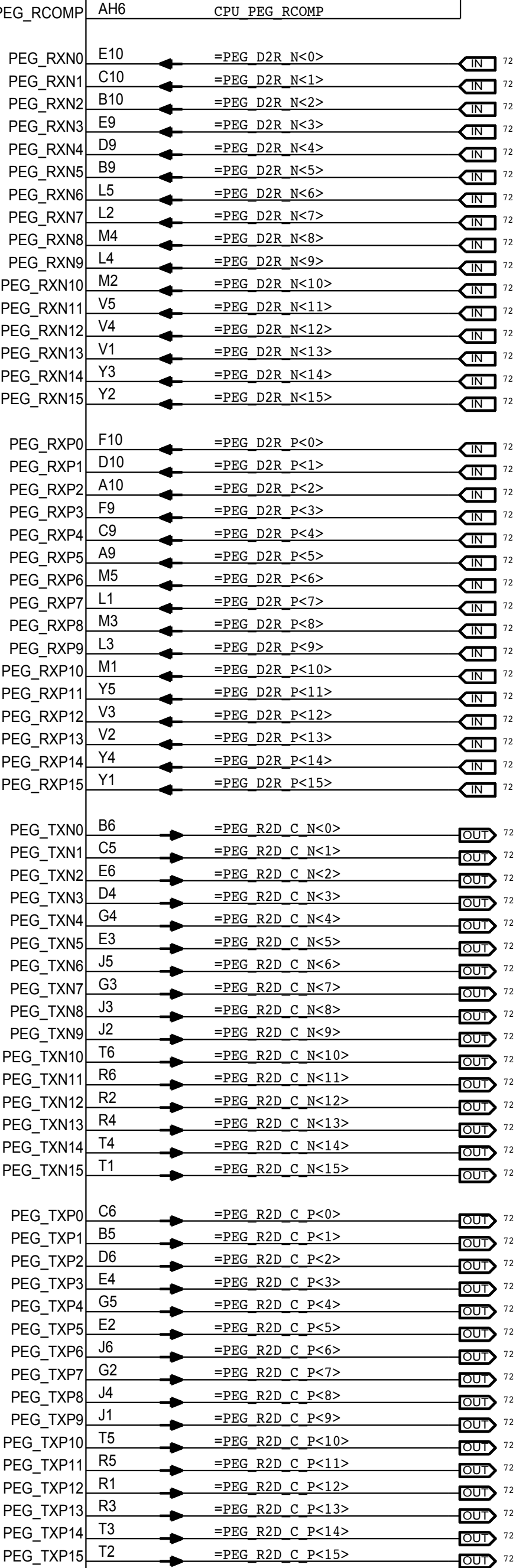
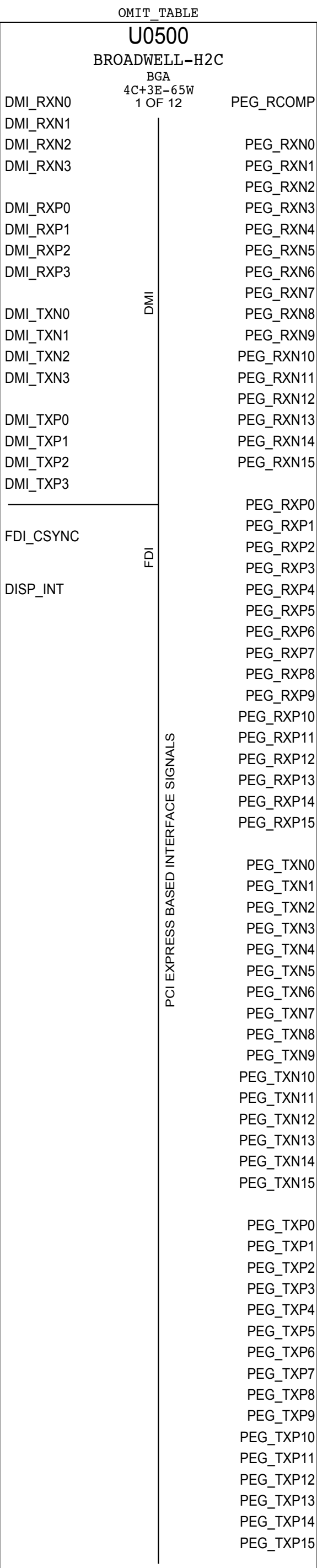
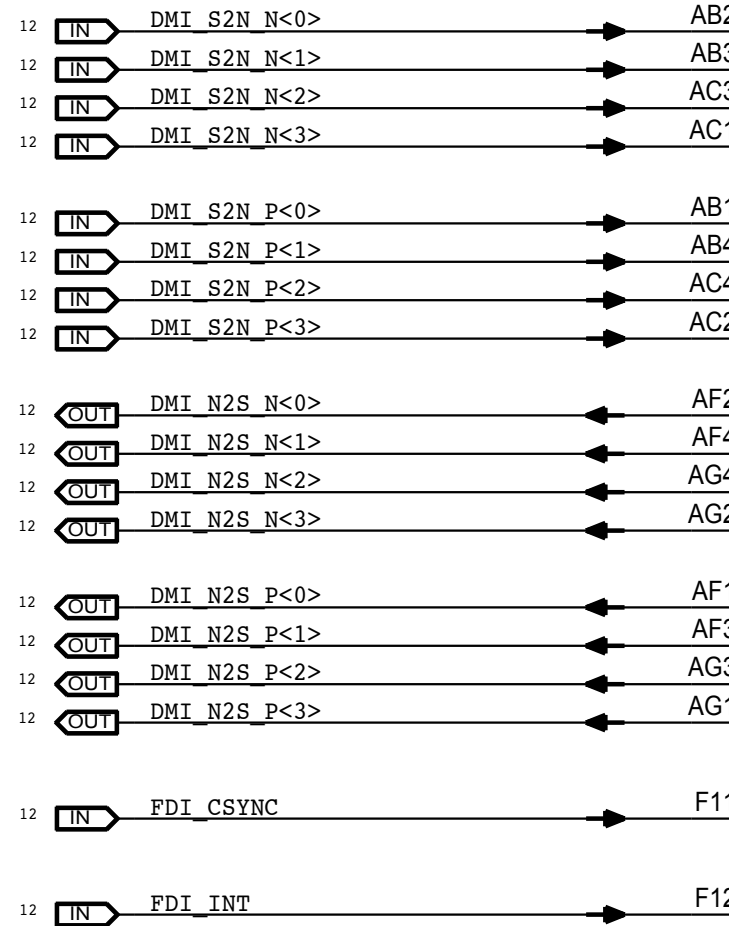
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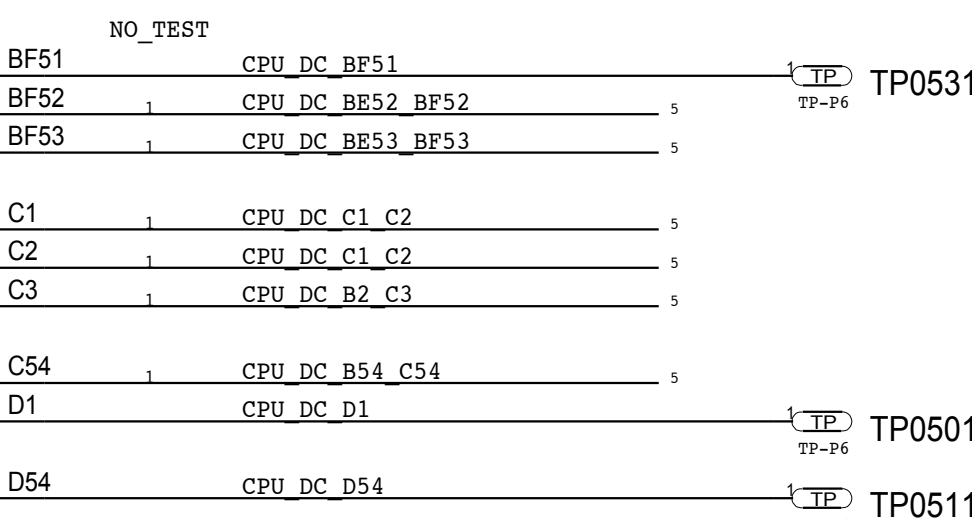
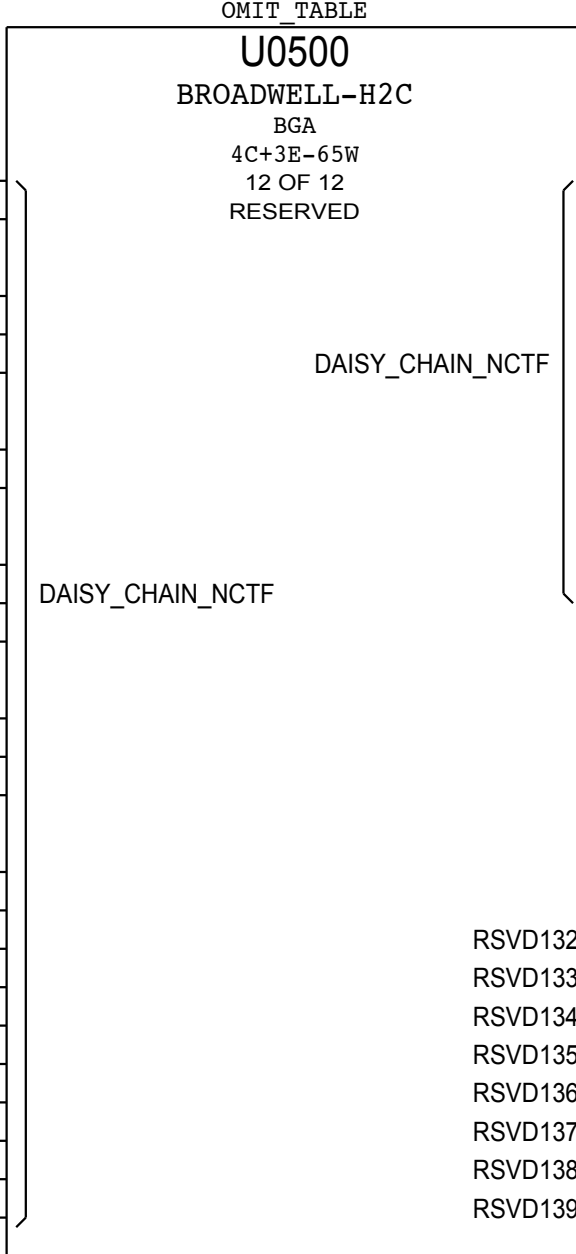
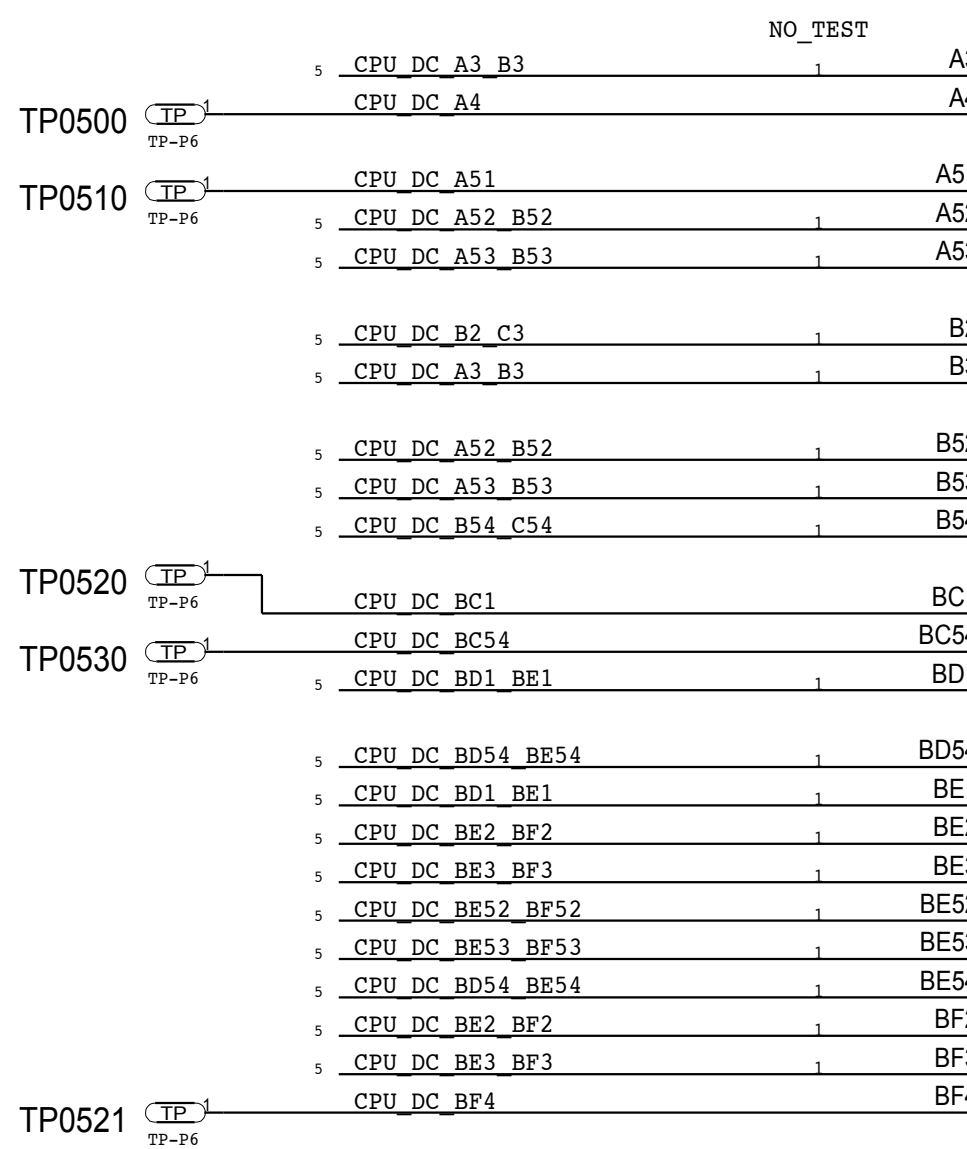
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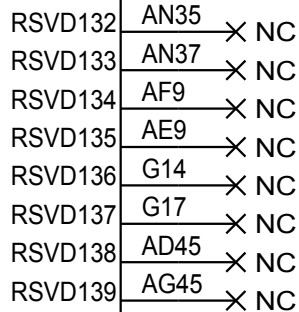
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Port D pins out of order to match Intel symbol.

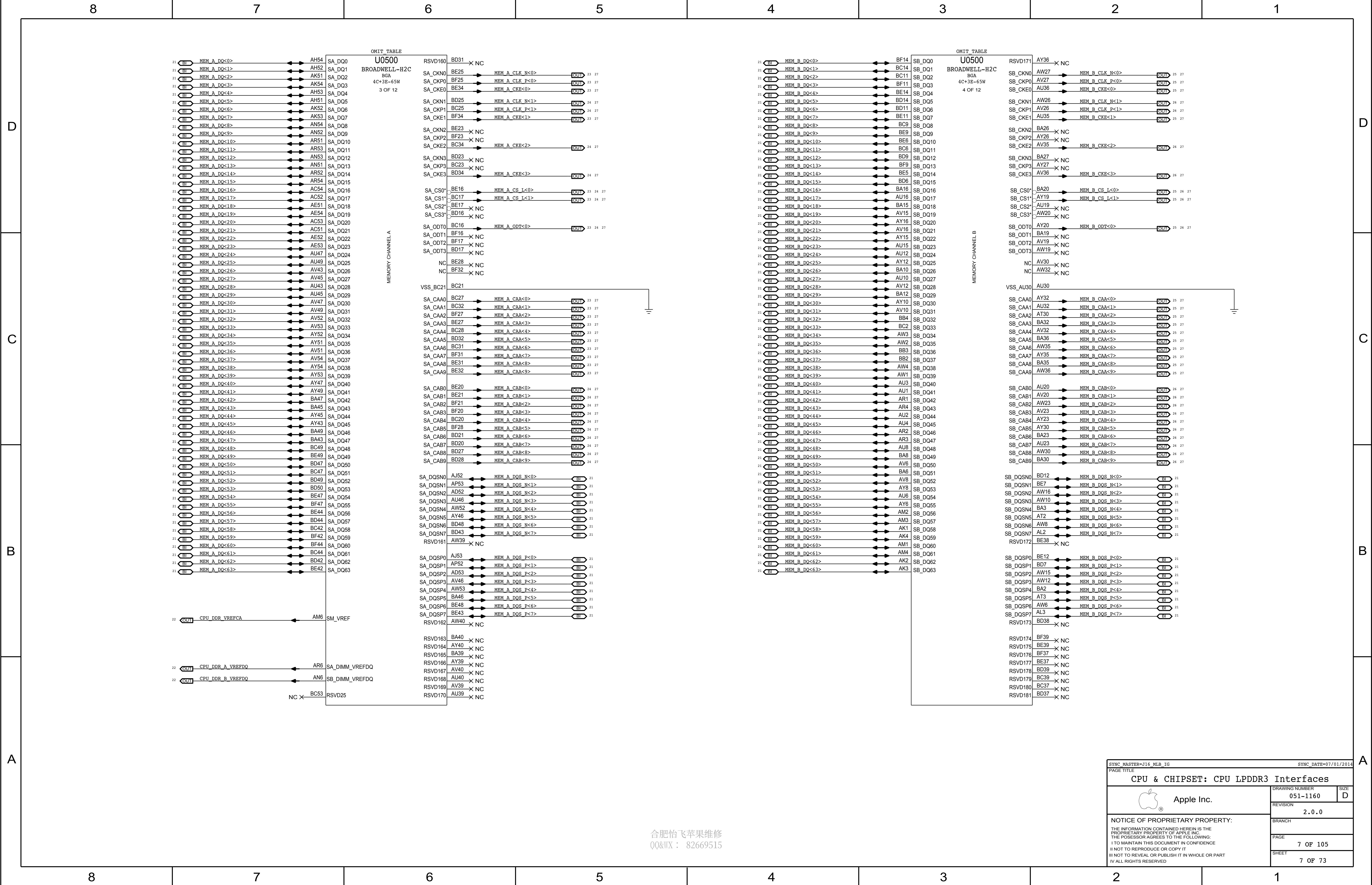


CPU Daisy-Chain Strategy:
Each corner of CPU has two testpoints. Other corner test signals connected in daisy-chain fashion. Continuity should exist between both TP's on each corner.



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CPU & CHIPSET: CPU DMI/PEG/FDI/RSVD			
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		REVISION	2.0.0
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		SHEET	5 OF 73

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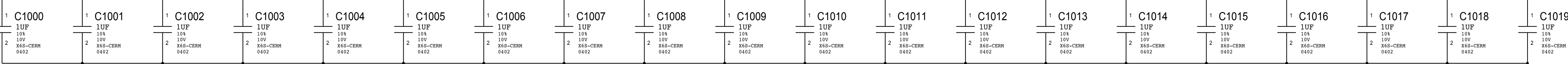


CPU VCORE Decoupling

Intel recommendation: 4x 470uF 4mOhm (3 CPU-side, 1 opposite), 20x 22uF 0805 (10 CPU-side, 10 opposite near edge, 4x 10uF 0603 (2 CPU-side, 2 opposite), 20x 1uF 0402 (under CPU)
Apple Implementation: 9x 210uF 6mOhm, 44x 10uF 0402, 4x 10uF 0402, 20x 1uF 0402

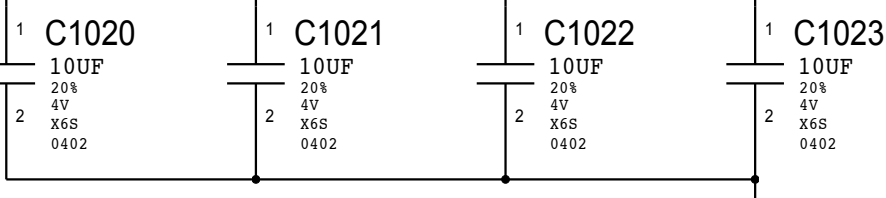
PLACEMENT_NOTE (C1000-C1019):

Place on bottom side of U0500



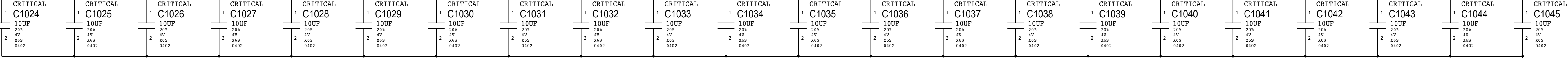
PLACEMENT_NOTE (C1020-C1023):

Place near U0500 on bottom side



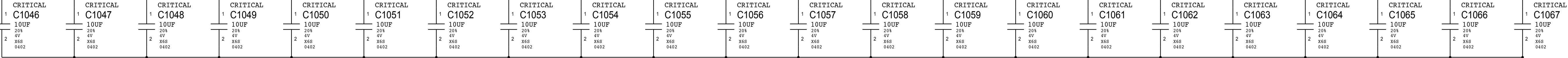
PLACEMENT_NOTE (C1024-C1045):

Place near inductors on bottom side.



PLACEMENT_NOTE (C1046-C1067):

Place near inductors on bottom side.



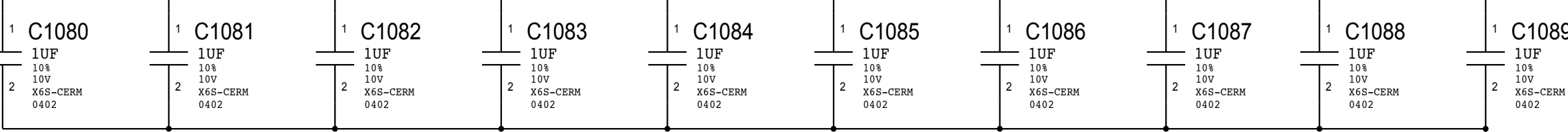
BULK CAPS ON REGULATOR PAGE

CPU VDDQ Decoupling

Intel recommendation: 2x 330uF, 8x 10uF 0603, 10x 1uF 0402
Apple Implementation: 2x 330uF, 8x 10uF 0603, 10x 1uF 0402

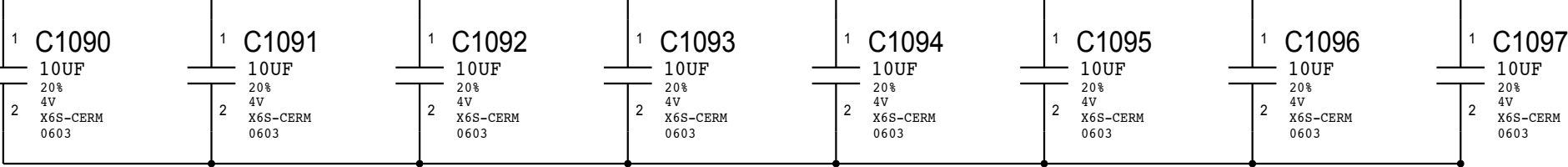
PLACEMENT_NOTE (C1080-C1089):

Place on bottom side of U0500



PLACEMENT_NOTE (C1090-C1097):

Place near U0500 on bottom side



OMIT TABLE

CRITICAL

C1078

470UF-0.00450HM

20V

POLY-TANT

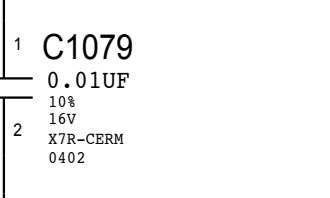
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CPU VCCIO Decoupling

Intel recommendation: 2x 0.01uF 0402 (1 near CPU, 1 near SVID pull-ups)
Apple Implementation: 2x 0.01uF 0402 (second cap is on CPU VR page)

PLACEMENT_NOTE (C1079):

Place near U0500 on bottom side



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
128S0381	1	CAP,AL,POLY,470UF	C1078	CRITICAL	PPCPU_VDDQ_CAP:H
128S0391	1	CAP,AL,POLY,270UF	C1078	CRITICAL	PPCPU_VDDQ_CAP:L

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NOTE: Intel decoupling recommendations from Shark Bay Mobile Platform Power Delivery Design Guide (doc #487822, Rev 0.8 dated January 2012), Section 5.

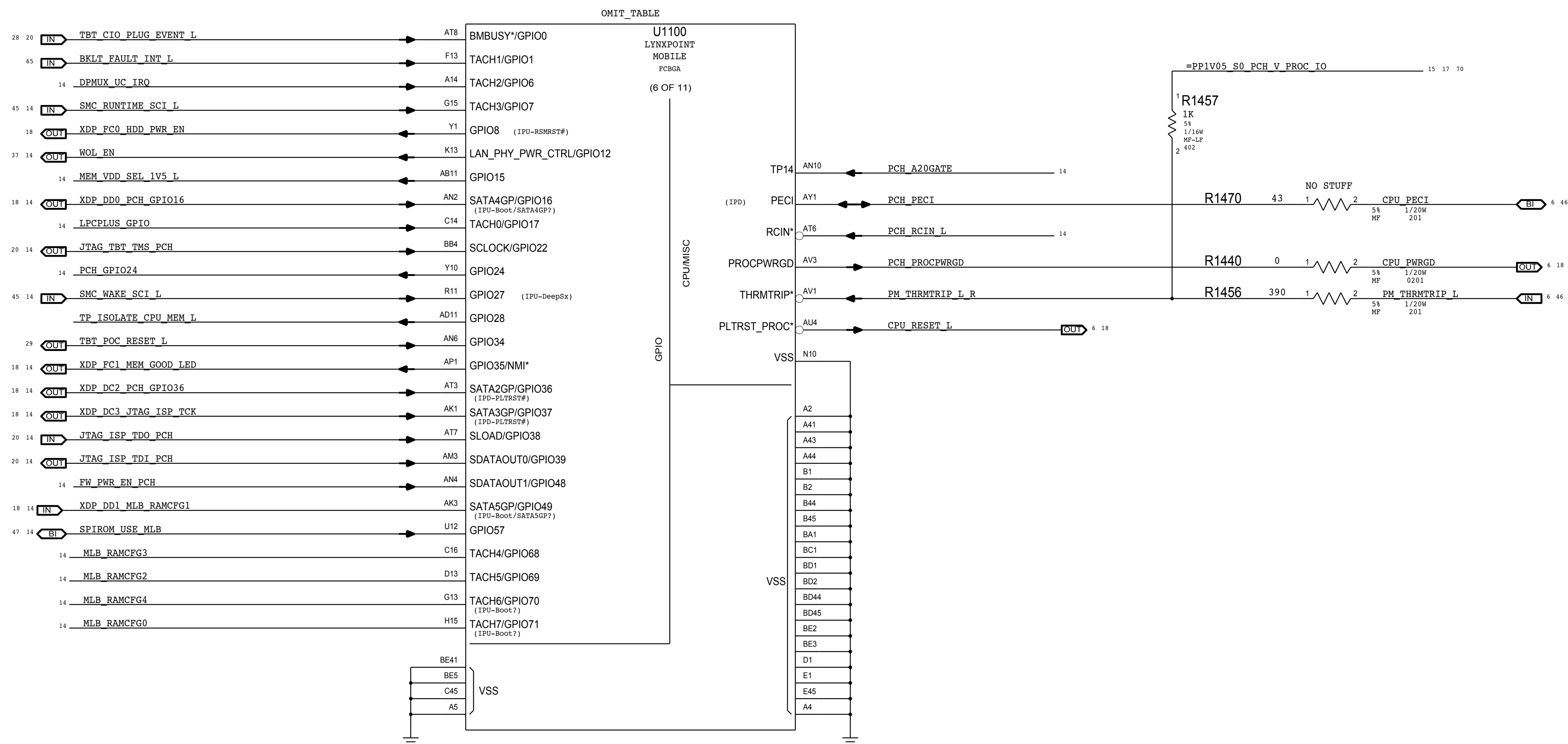
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	BRANCH		
	PAGE		
	10 OF 105		
	SHEET		
	10 OF 73		



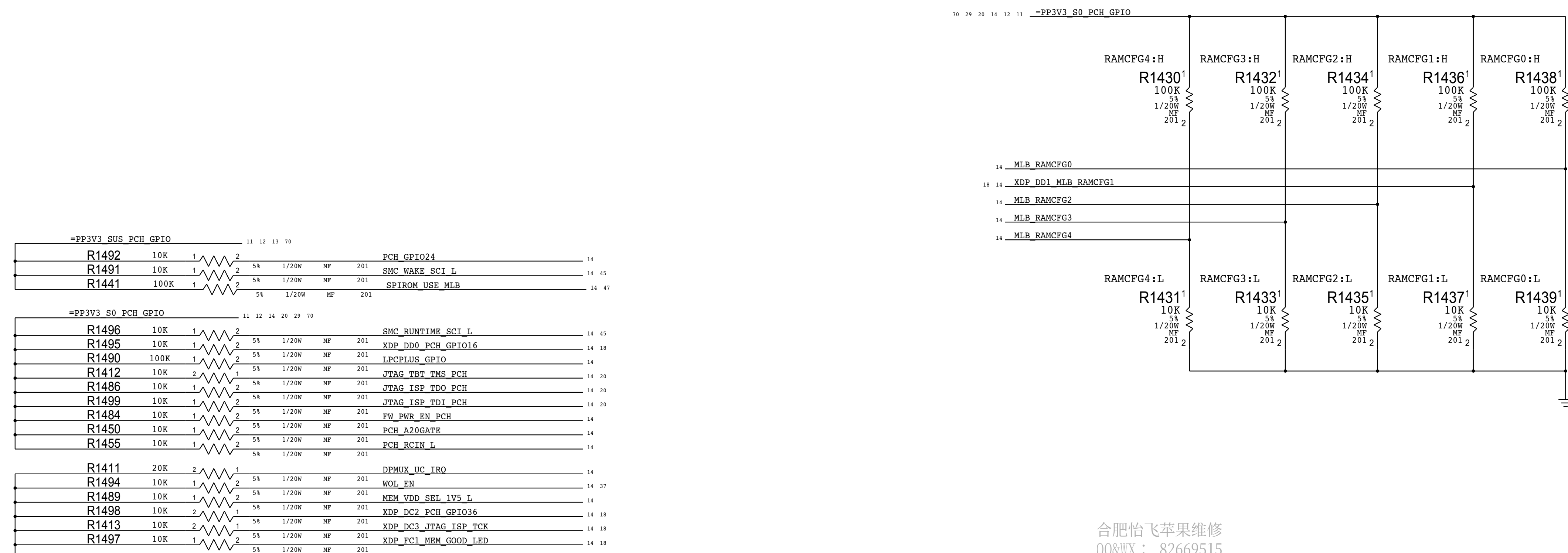


SYNC_MASTER=J16_MLB_IG		SYNC_DATE=07/01/2014	
PAGE TITLE			
CPU & CHIPSET: PCH DMI/FDI/PM/GFX/PCI			
	Apple Inc.	DRAWING NUMBER	D SIZE
		051-1160	
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	PAGE	12 OF 105	
	SHEET	12 OF 73	



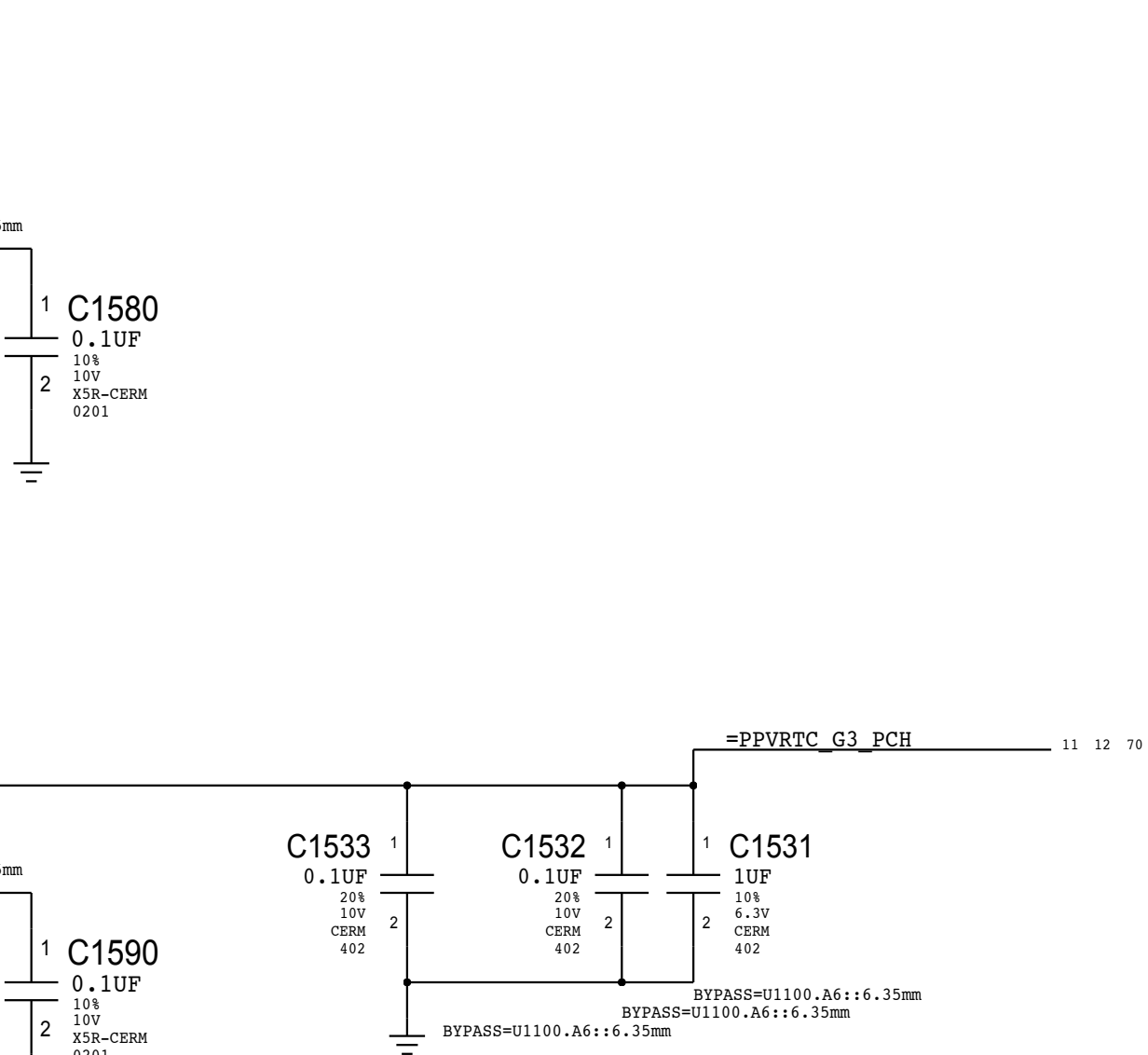
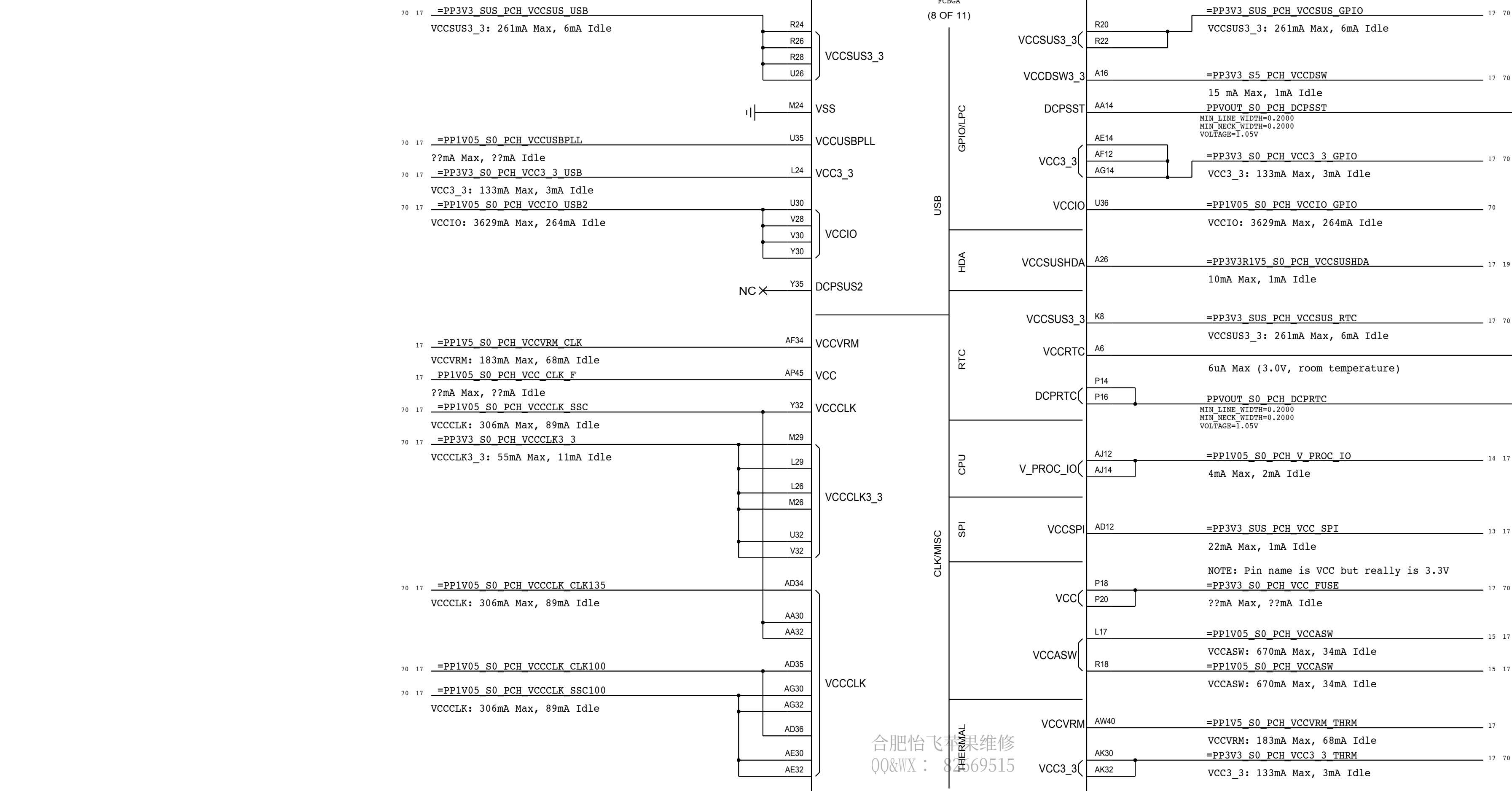


RAM Configuration Straps

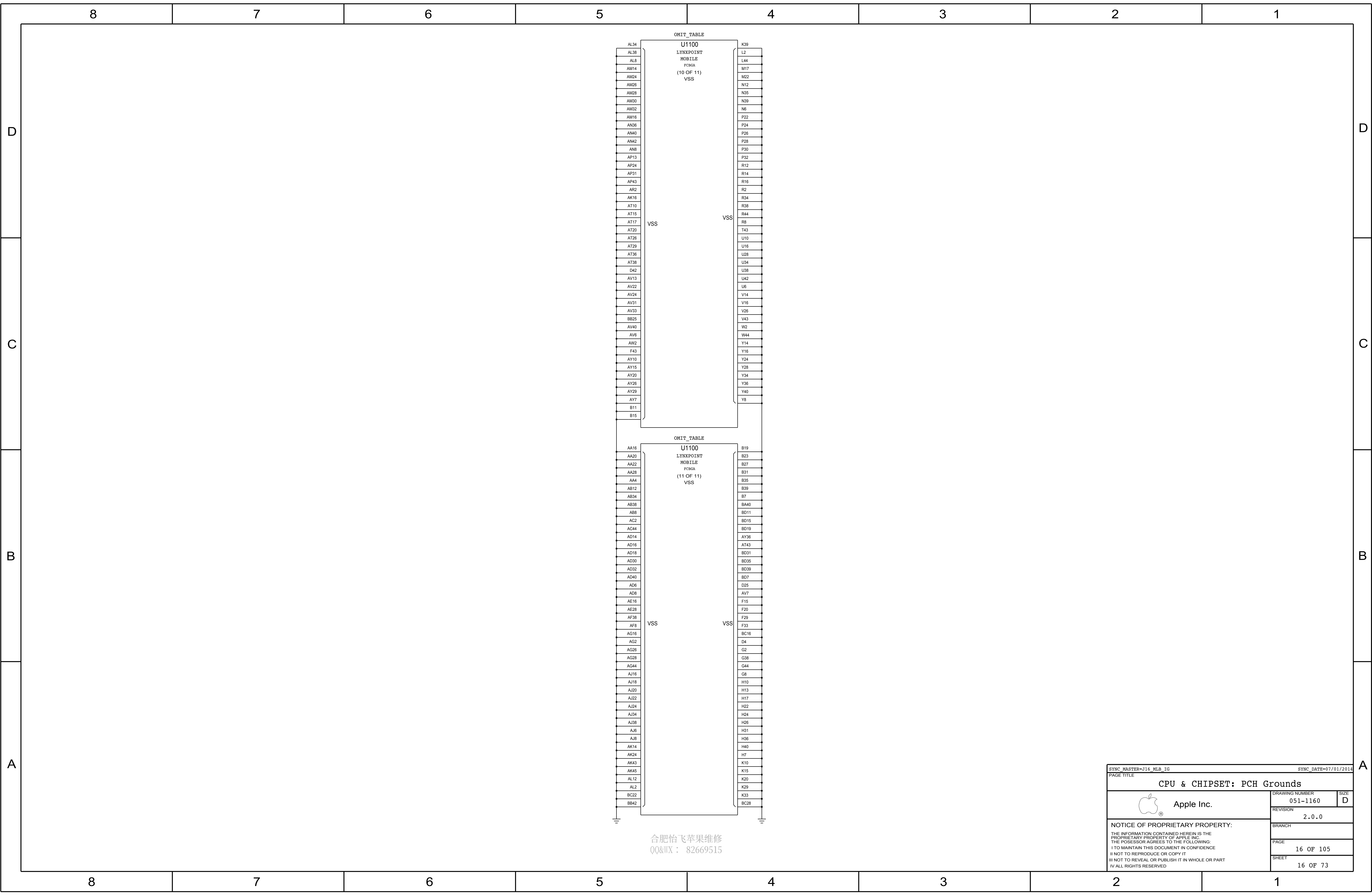


SYNC MASTER=J94 ANDRES		SYNC_DATE=07/09/2014	
PAGE TITLE			
CPU & CHIPSET: PCH GPIO/MISC/NCTF			
 Apple Inc.	DRAWING NUMBER 051-1160		SIZE D
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		SHEET 14 OF 73	

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SYNC_MASTER=J16_MLB_IG		SYNC_DATE=07/01/2014	
PAGE TITLE			
CPU & CHIPSET: PCH Power			
 Apple Inc.	DRAWING NUMBER 051-1160		SIZE D
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	BRANCH		
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PAGE		15 OF 105	
SHEET		15 OF 73	



SYNC_MASTER=J16_MLB_IG		SYNC_DATE=07/01/2014	
PAGE TITLE			
CPU & CHIPSET: PCH Grounds			
 Apple Inc.	DRAWING NUMBER	051-1160	SIZE D
	REVISION	2.0.0	
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	PAGE	16 OF 105	
	SHEET	16 OF 73	

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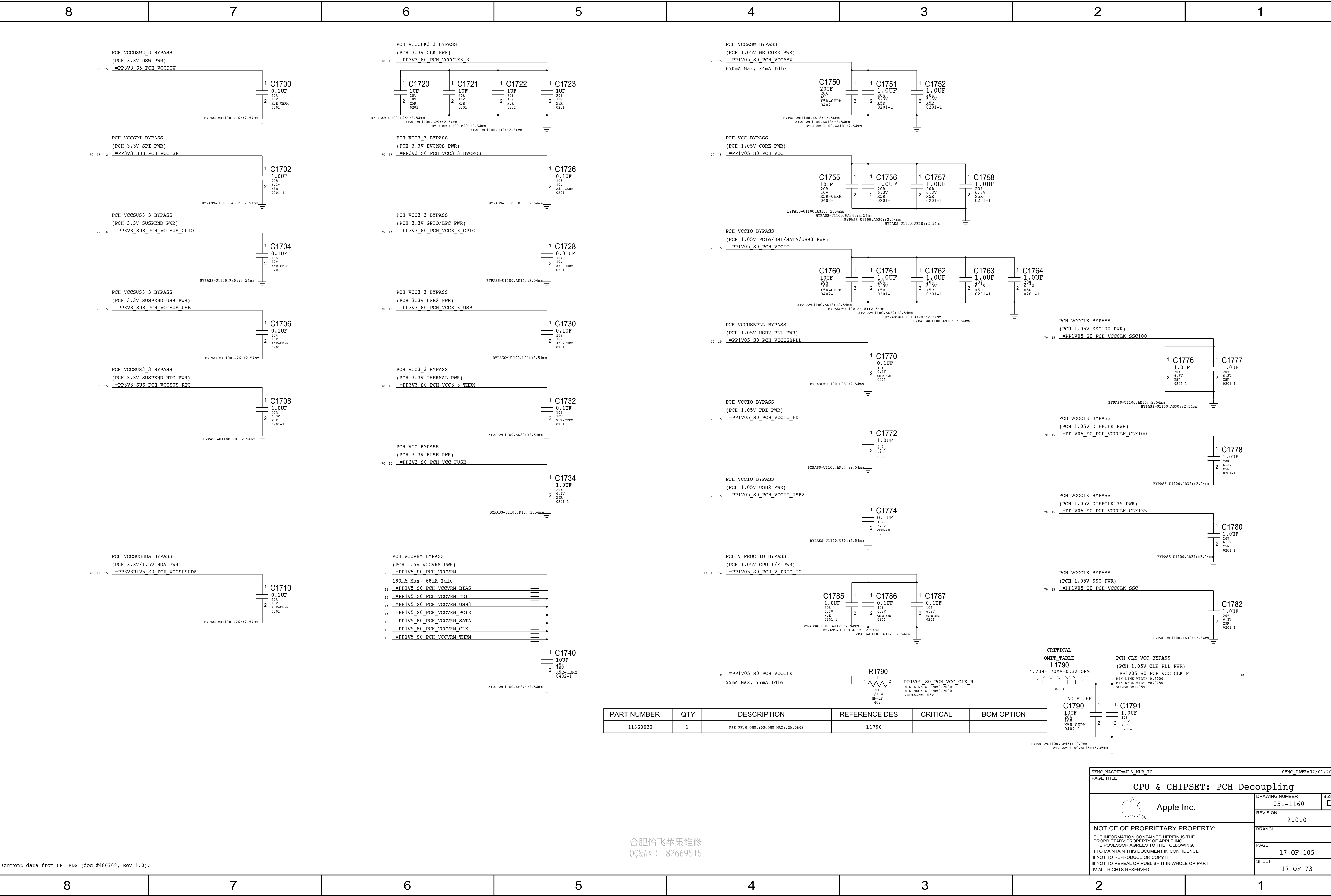
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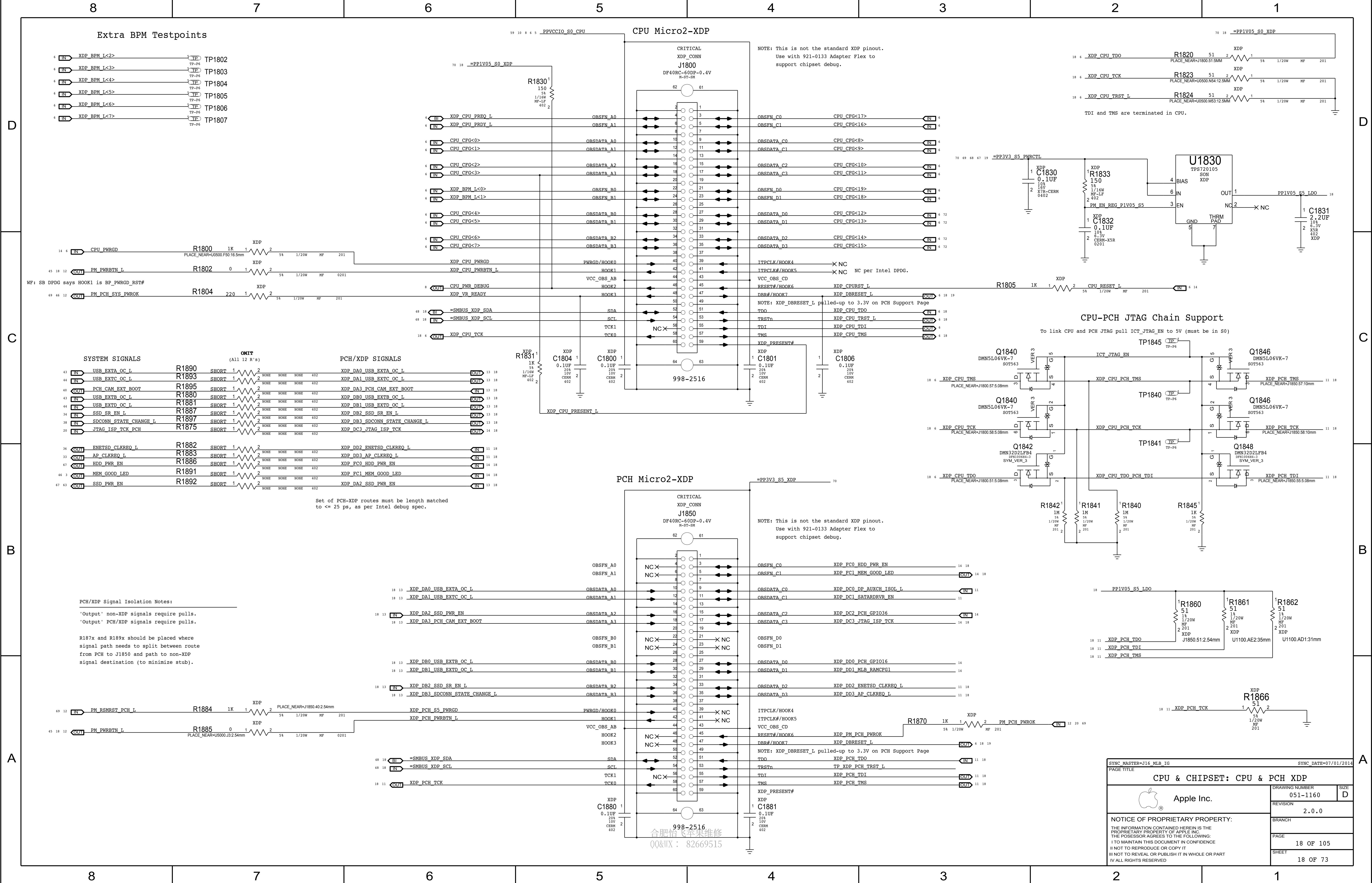
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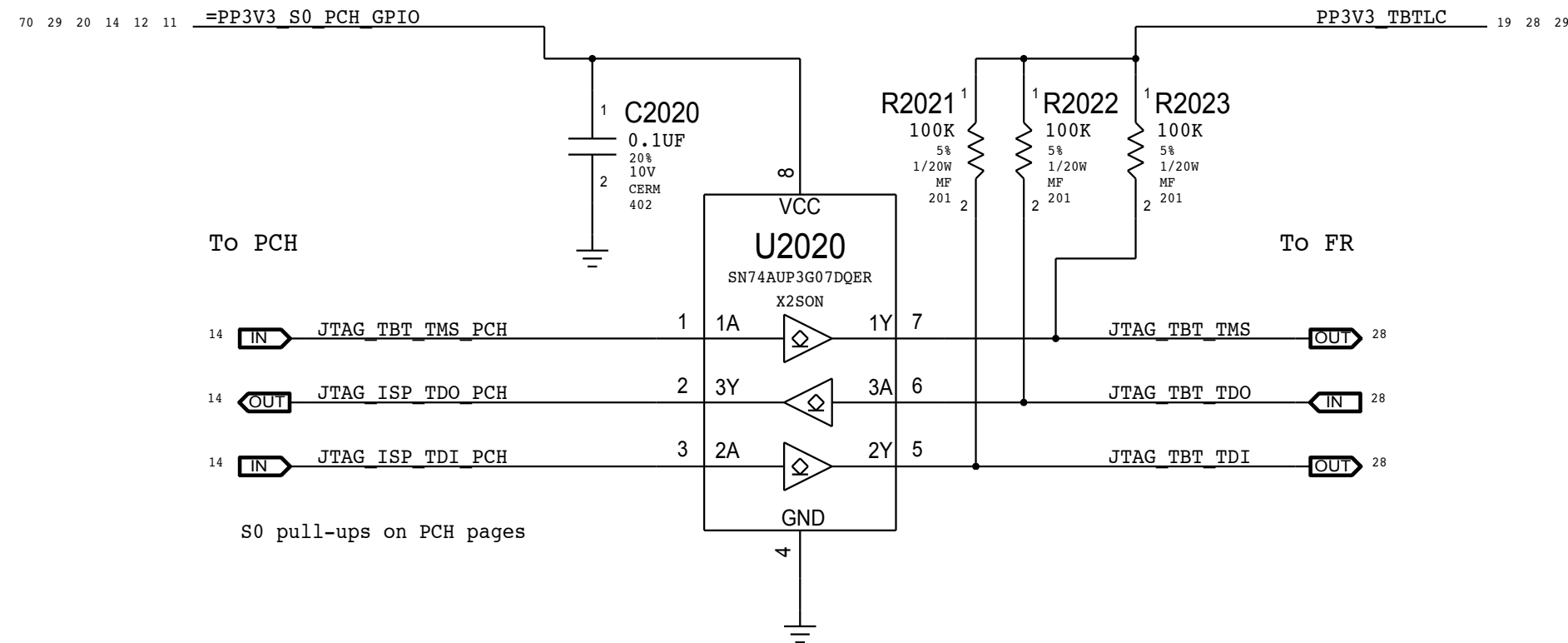
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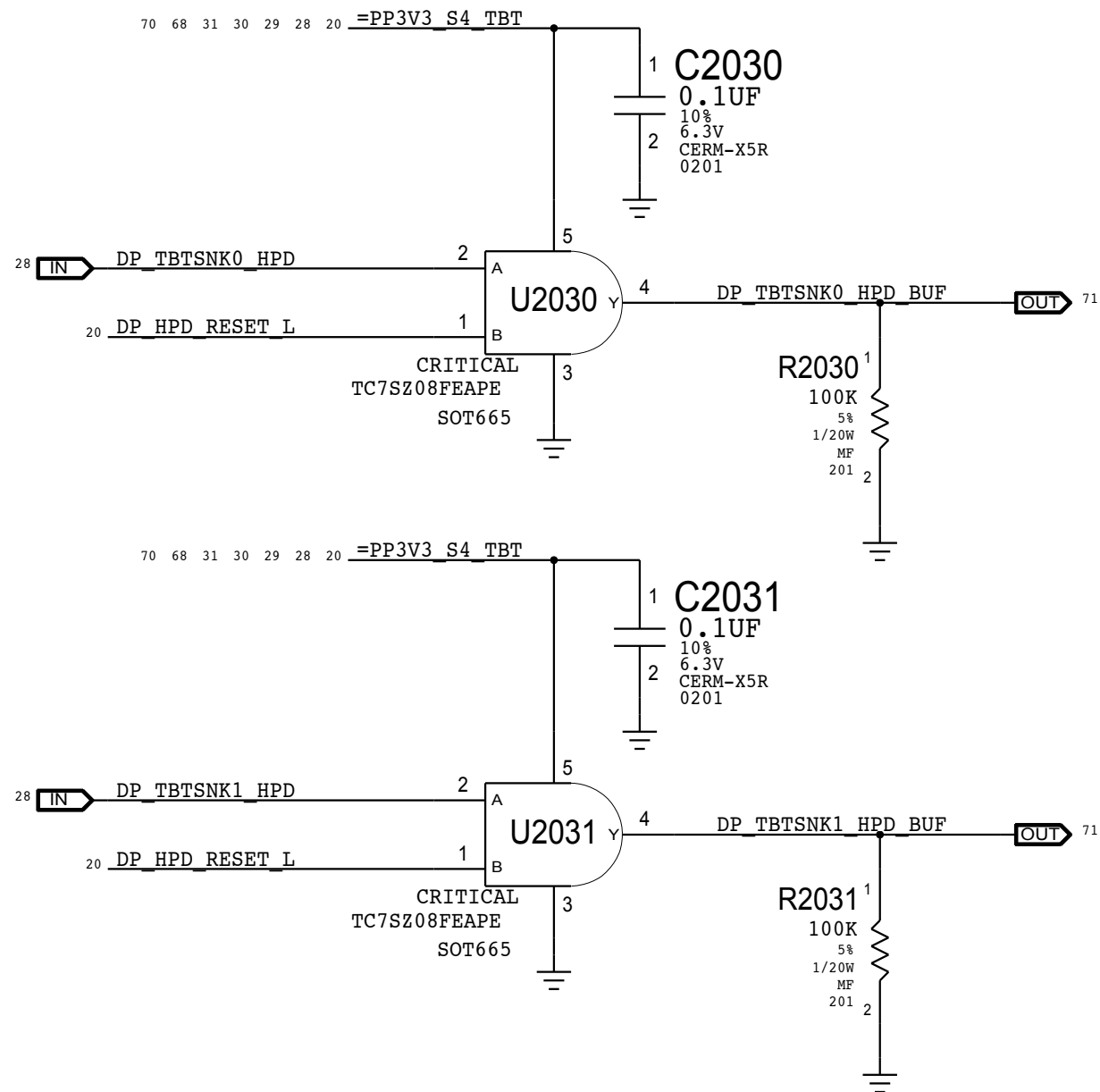
Falcon Ridge JTAG Isolation

TBTLIC can be on when S0 is off, and vice-versa
Isolation ensures no leakage to RR or PCH



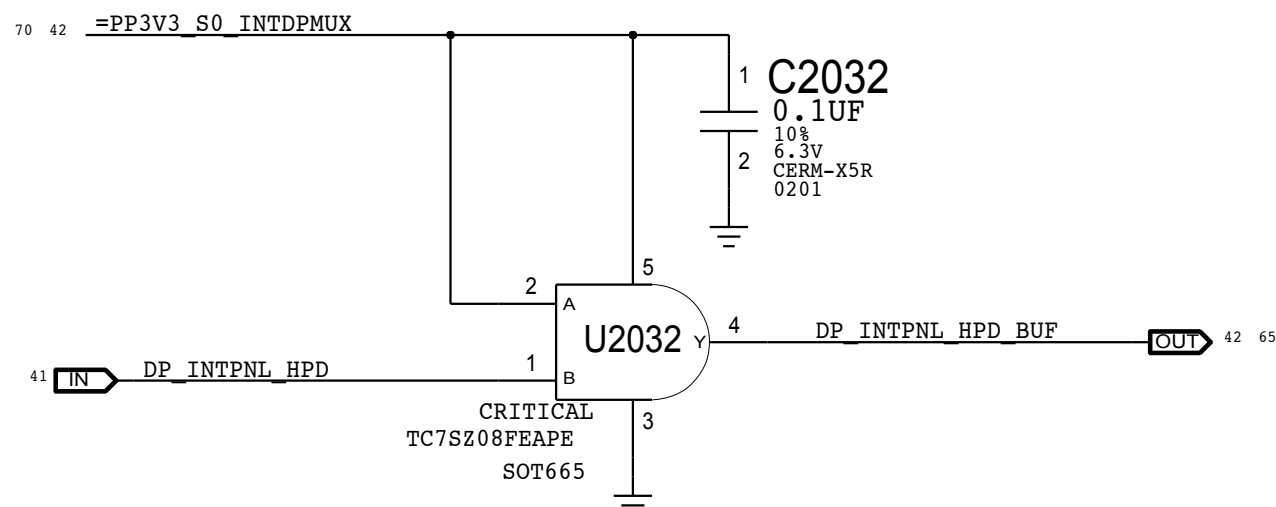
Falcon Ridge HPD Isolation

NOTE: PLT_RESET_L used as other input to the AND gate so that HPD is only driven high to the PCH in S0.

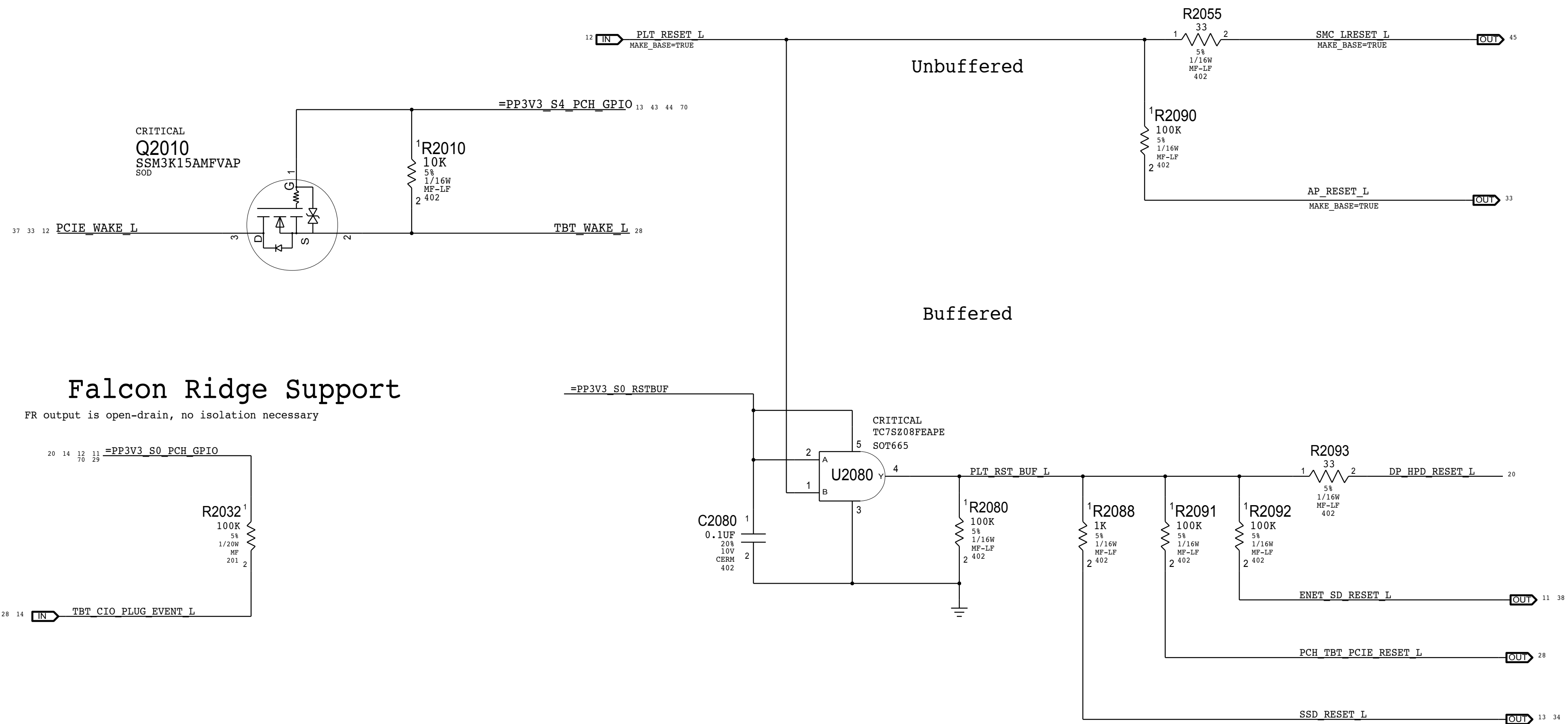


Internal Display HPD Isolation

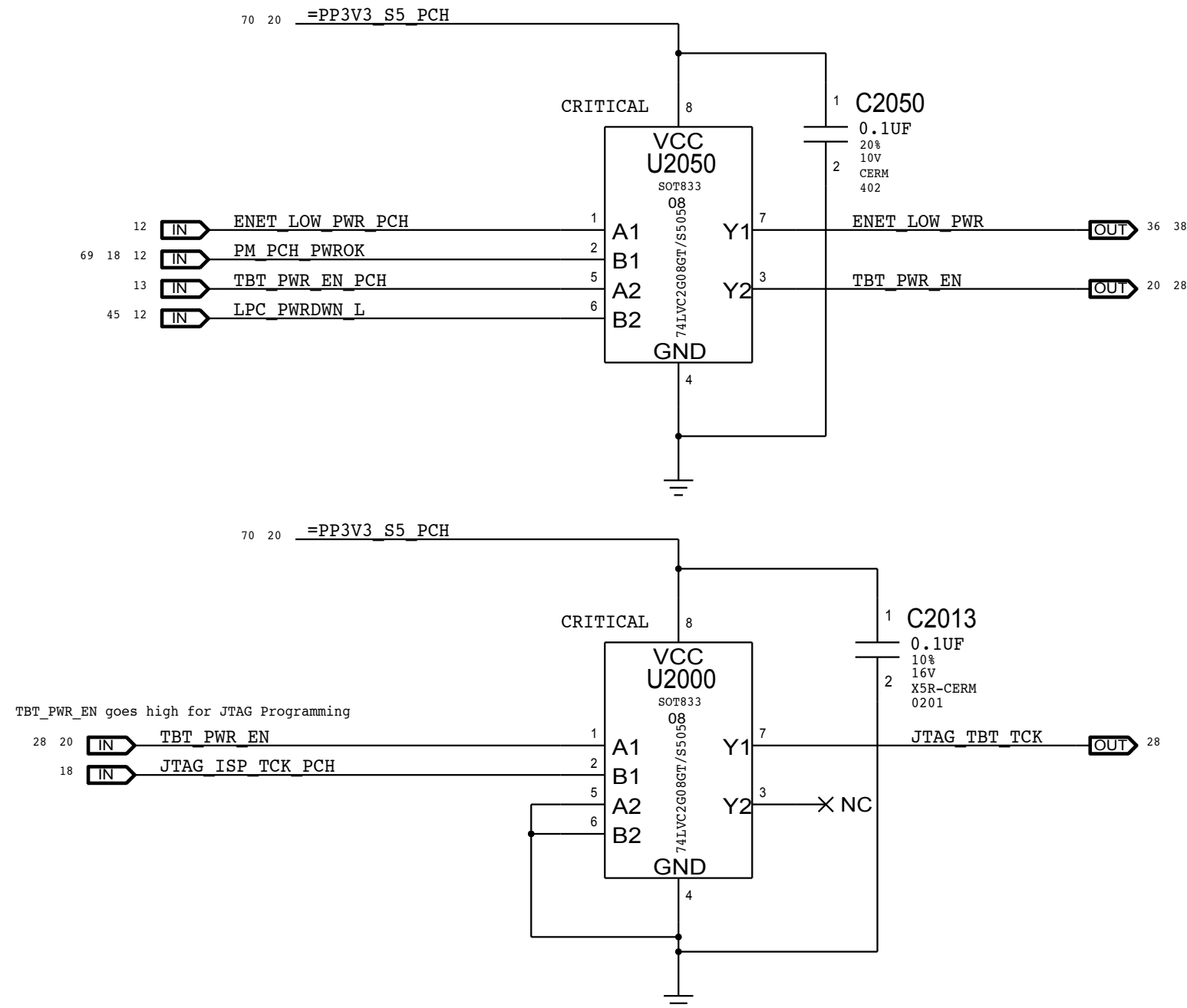
NOTE: PP3V3_S0 used as other input to the AND gate so that HPD is only driven high when PCH 3V3_S0 is up.



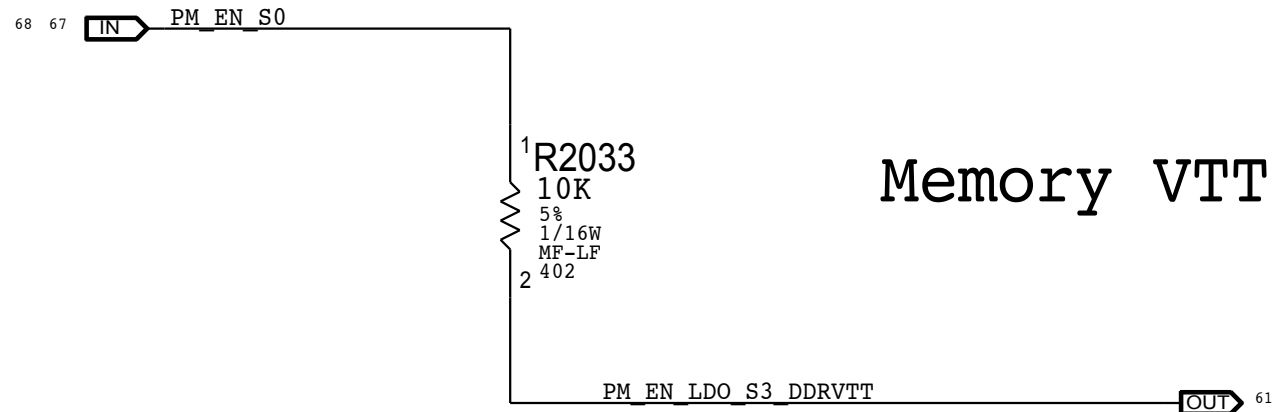
Platform Reset Connections

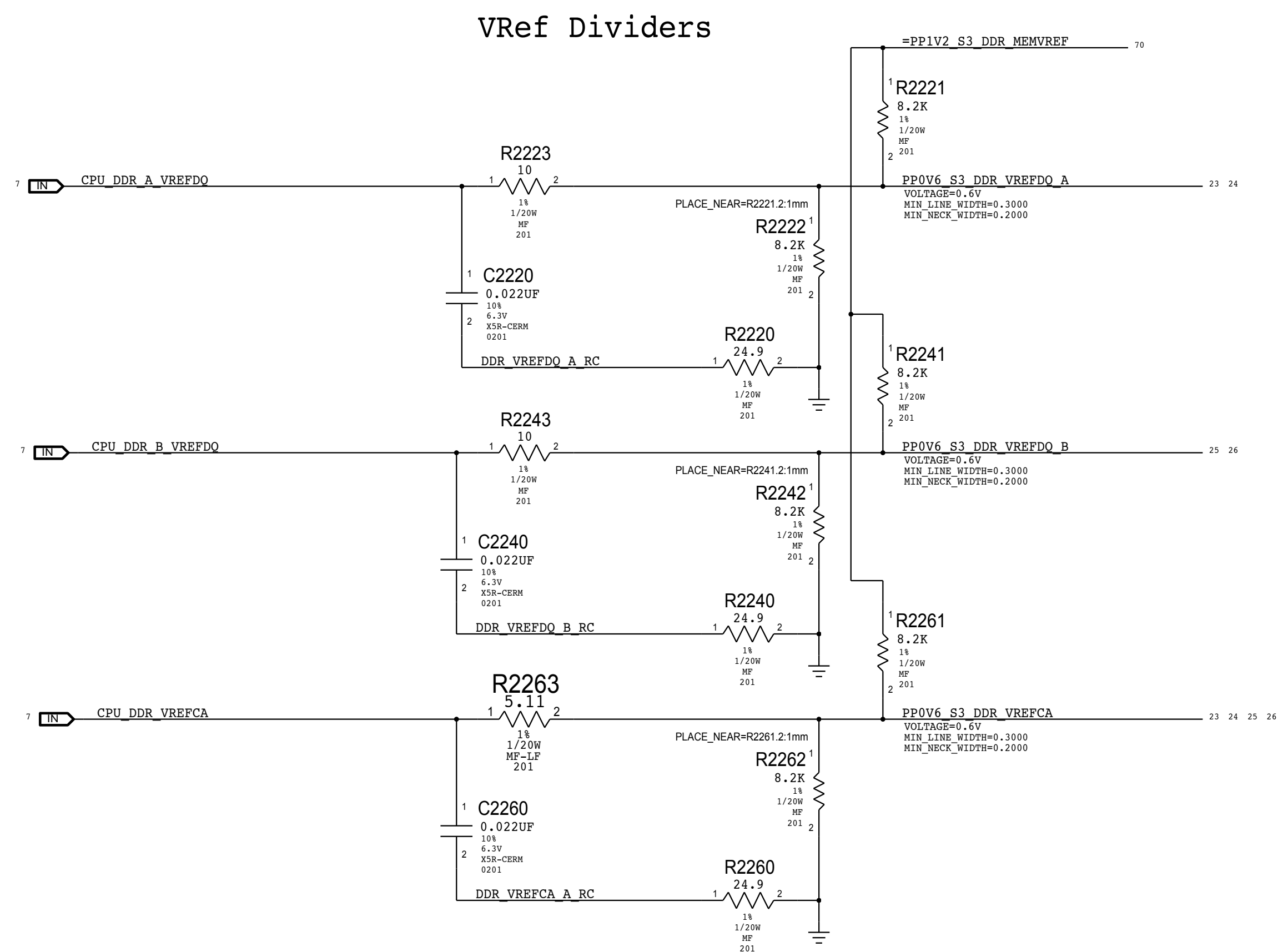


GPIO Glitch Prevention



Memory VTT Enable

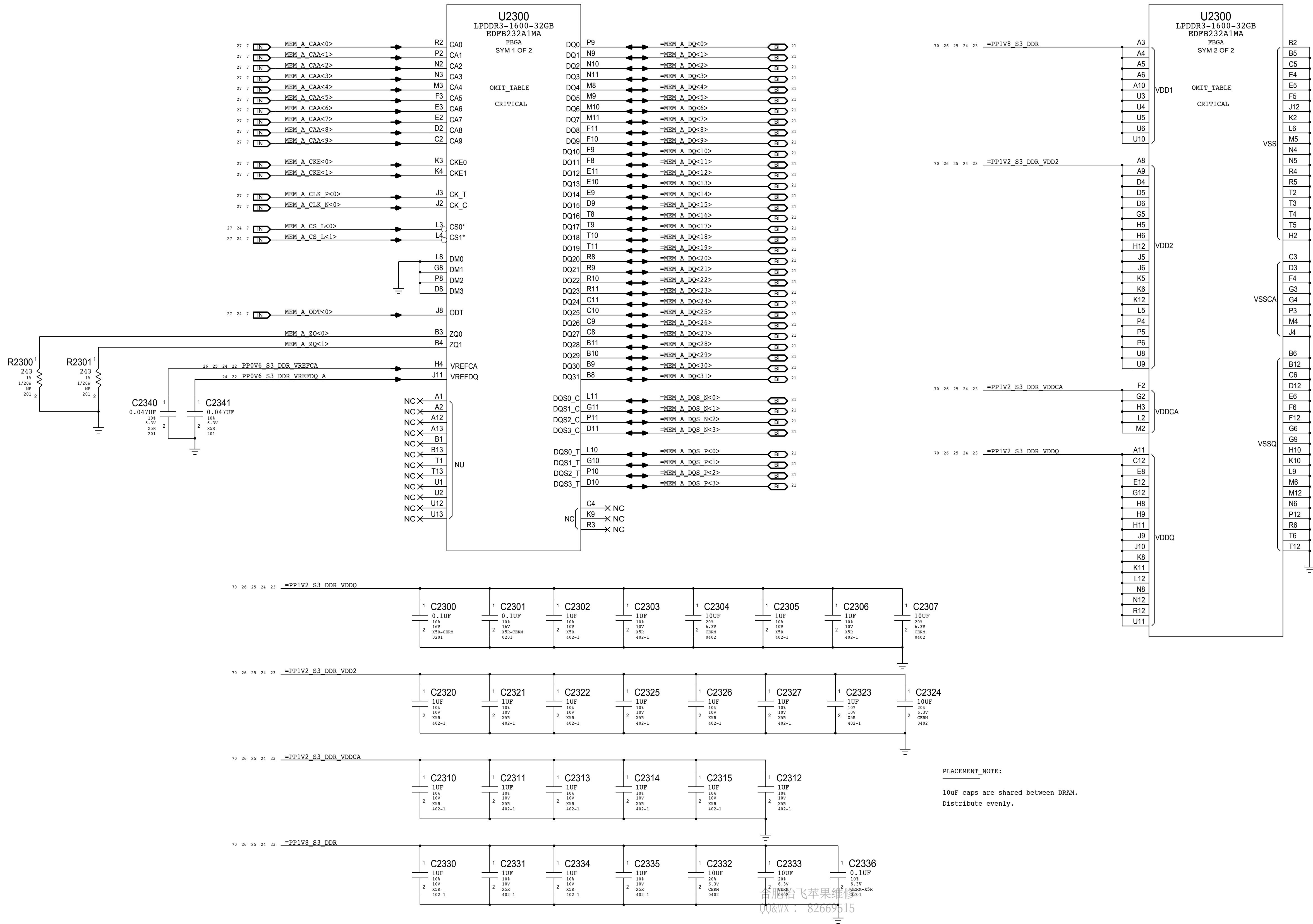




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SYNC MASTER=J117 ANDRES		SYNC DATE=02/27/2014	
PAGE TITLE			
DRAM: DDR3 VREF MARGINING			
	Apple Inc.		DRAWING NUMBER 051-1160
			SIZE D
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		BRANCH	
		PAGE 22 OF 105	
		SHEET 22 OF 73	

LPDDR3 CHANNEL A (0-31)



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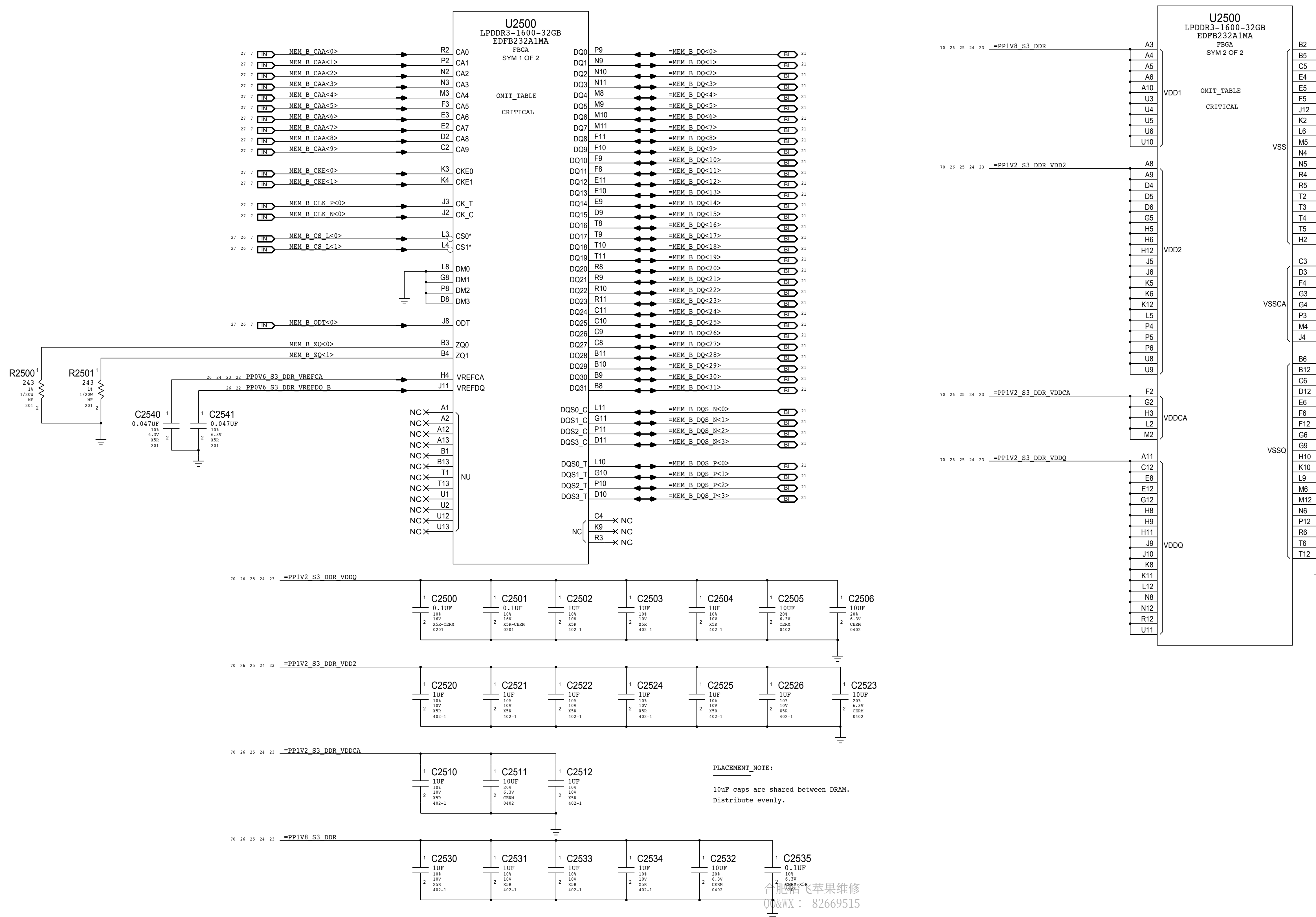


10uF caps are shared between DRAM.
Distribute evenly.

A

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LPDDR3 CHANNEL B (0-31)



SYNC MASTER=J41_MLB PAGE TITLE		SYNC DATE=09/03/2013	
DRAM: LPDDR3 Channel B (0-31)			
 Apple Inc.	DRAWING NUMBER 051-1160		SIZE D
	REVISION 2.0.0		
	BRANCH		
	PAGE 25 OF 105		
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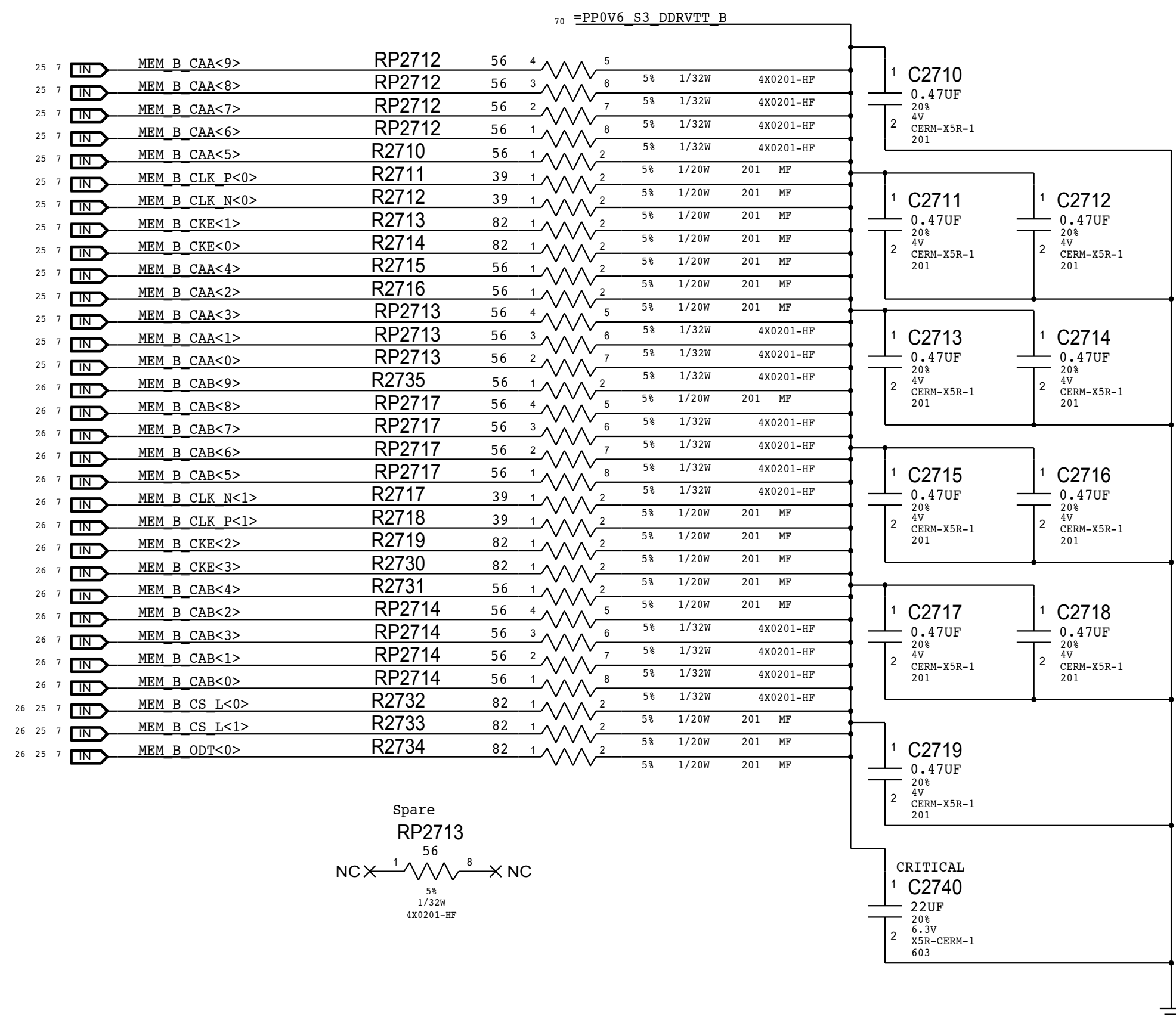
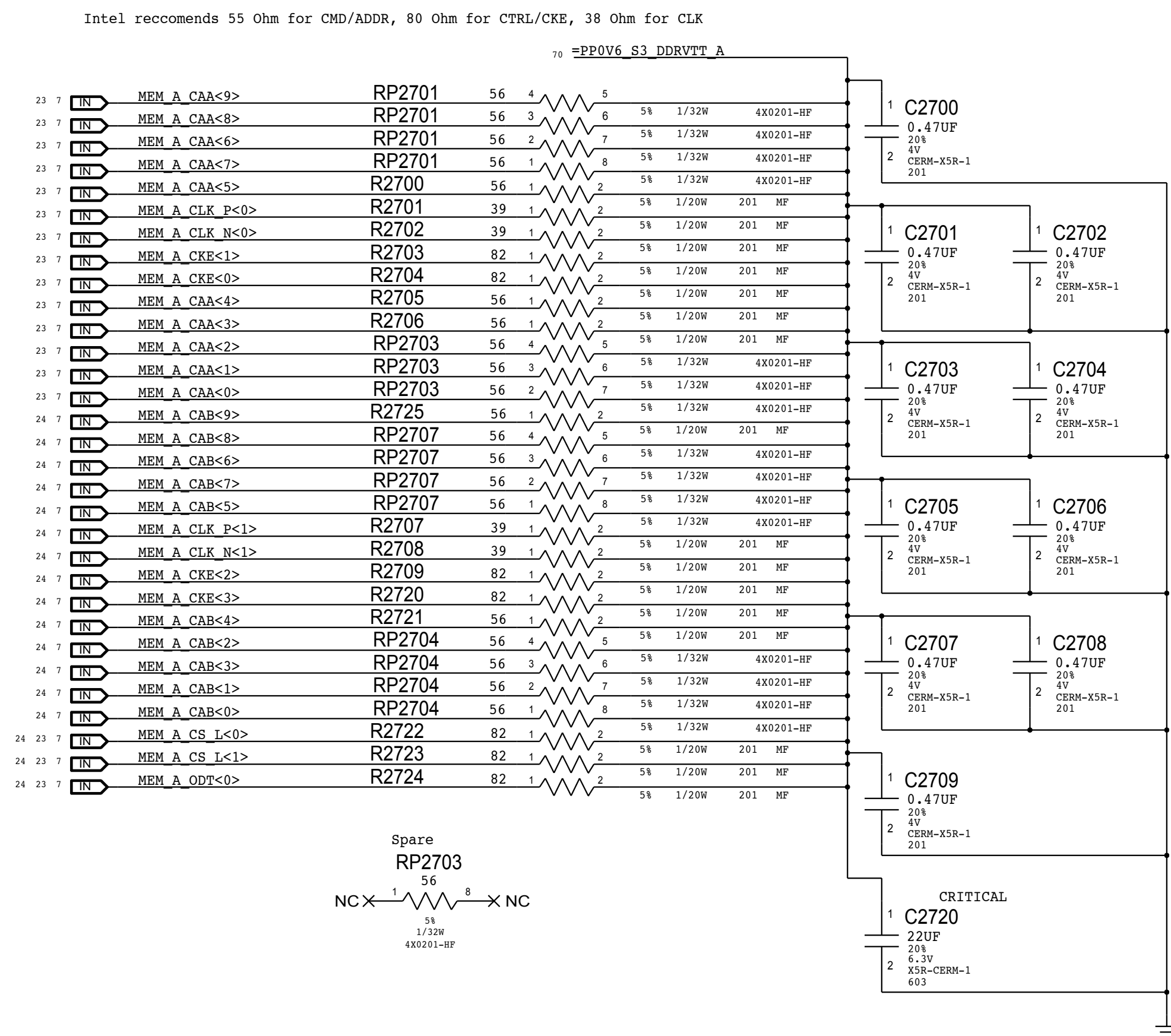
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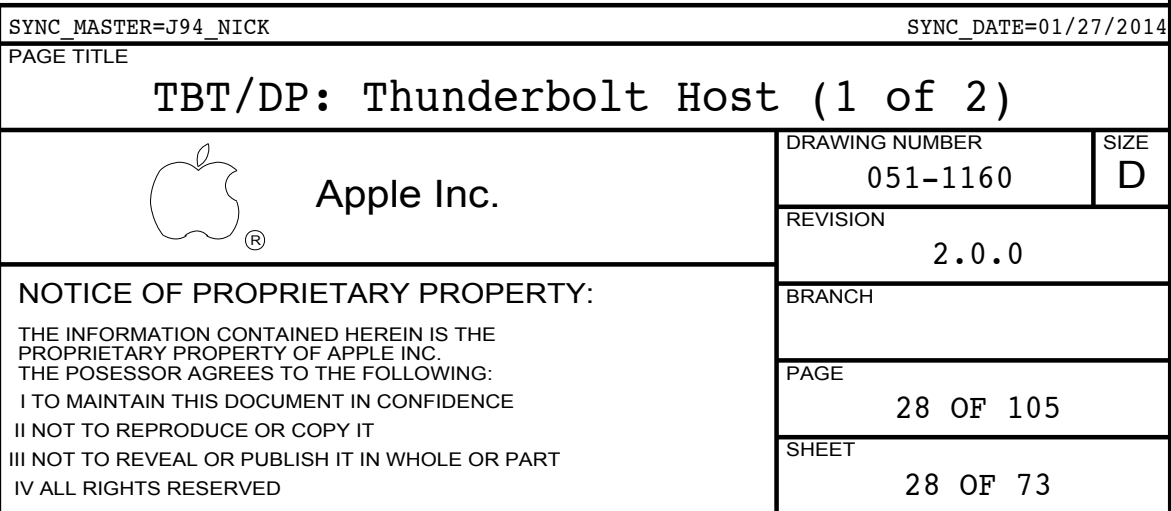
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SYNC MASTER~J41_MLB PAGE TITLE		SYNC DATE=09/03/2013	
DRAM: LPDDR3 Channel B (32-63)			
	DRAWING NUMBER 051-1160		SIZE D
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	PAGE 26 OF 105		
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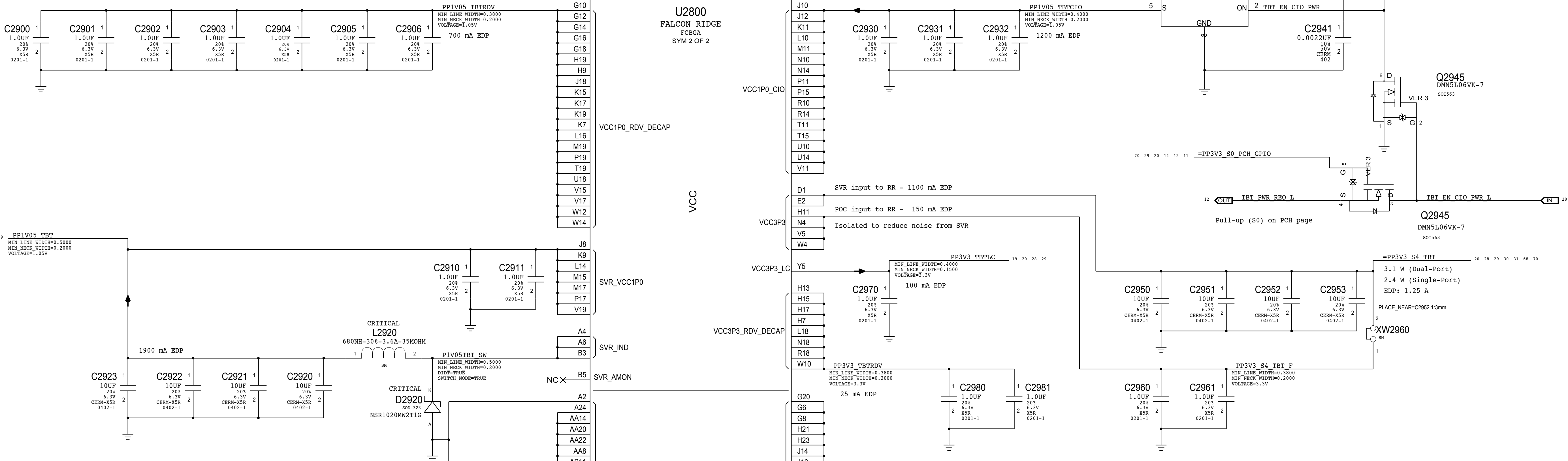
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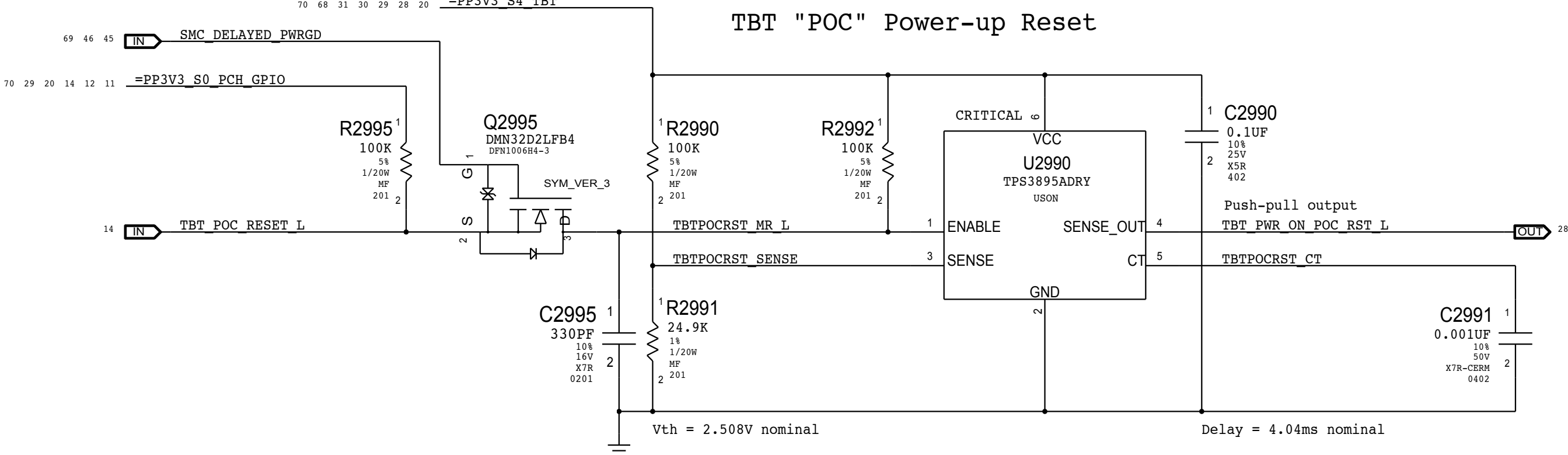
U2950	
Part	SLG5AP304V
Type	Load Switch
R(on)	7.8 mOhm Typ
@ 1.05V	9.6 mOhm Max
Max Current = 4A (85C)	

1.05V TBT "CIO" Switch

Internal switch not functional on RR.



A2, B1, and C2 are the ground current pins for A4, A6, and B3. They should be routed as a shape to the anode of D2920.



SYNC_MASTER=J70_NICK		SYNC_DATE=01/12/2014	
PAGE TITLE			
TBT/DP: Thunderbolt Host (2 of 2)			
 Apple Inc.	DRAWING NUMBER		SIZE
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	REVISION		
	2.0.0		
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PAGE		29 OF 105	
SHEET		29 OF 73	

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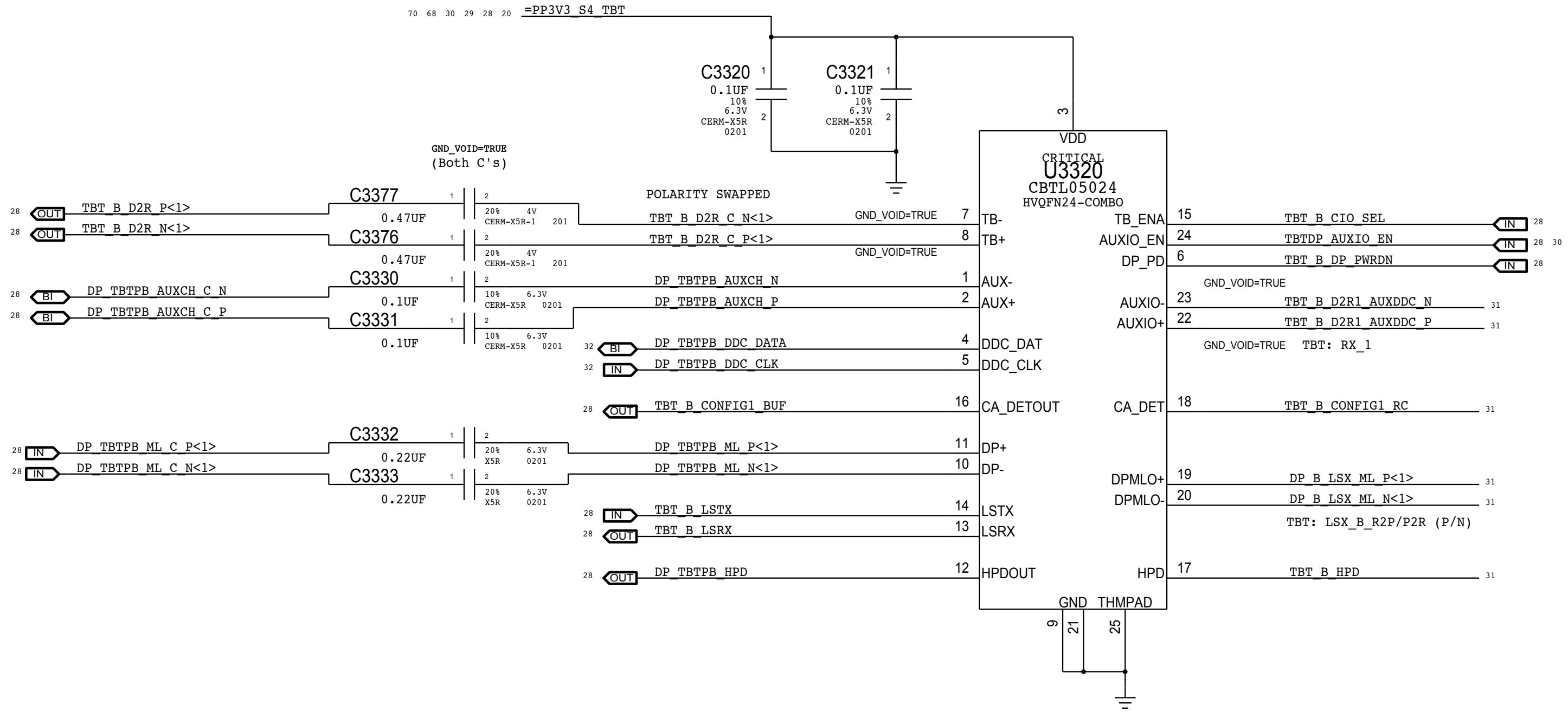
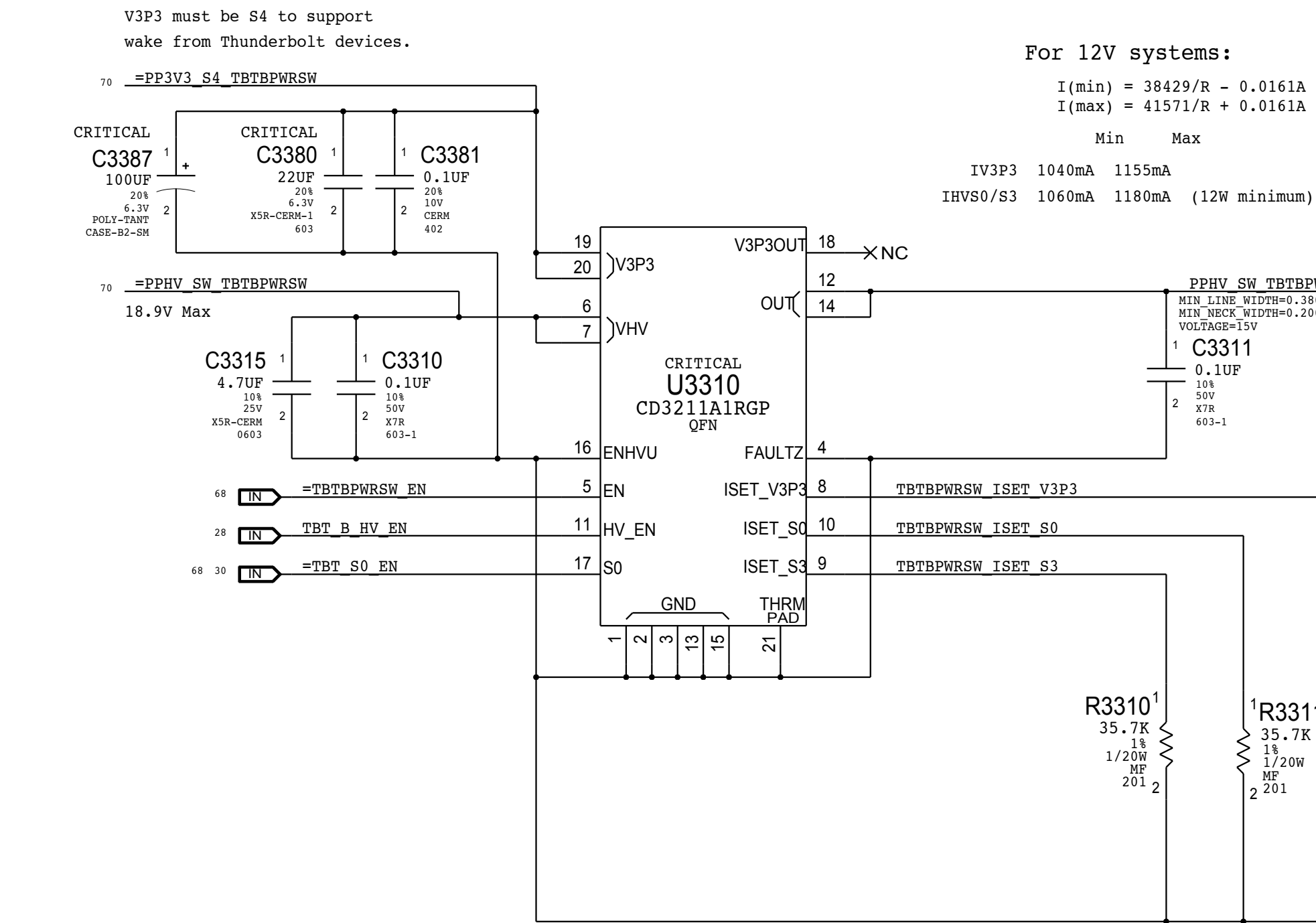


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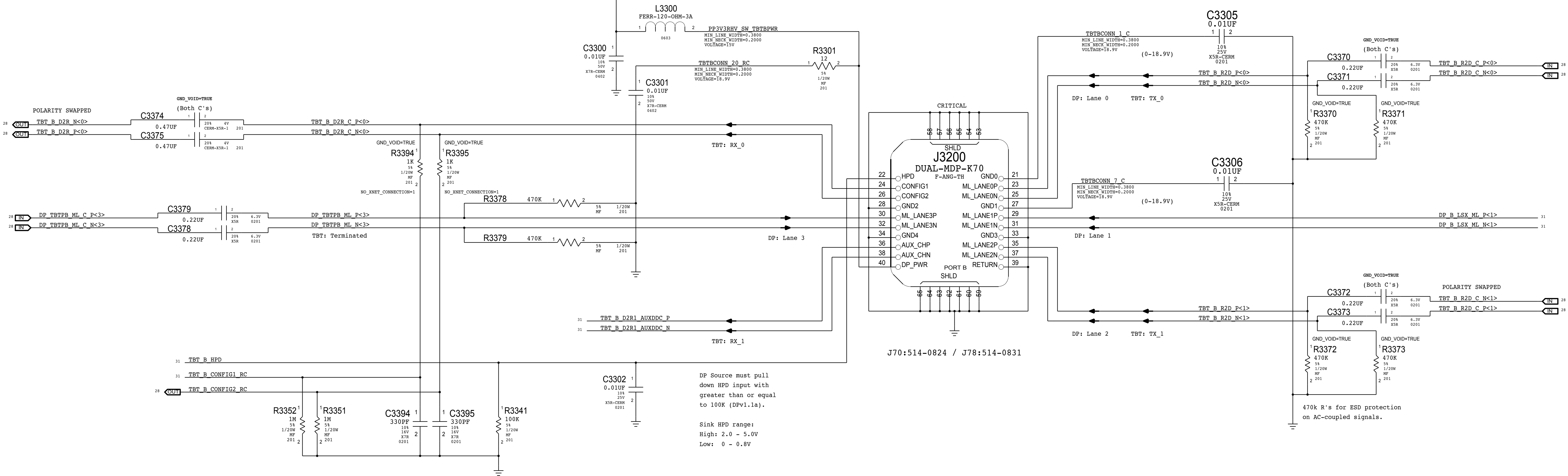
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SYNC MASTER=J70 NICK		SYNC DATE=10/16/2013	
PAGE TITLE			
TBT/DP: Thunderbolt Connector A			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-1160		D
	REVISION		
2.0.0			
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		32 OF 105	
		SHEET	
		30 OF 73	

3.3V/HV Power MUX



Thunderbolt Connector B



DP Source must pull down HPD input with greater than or equal to 100K (DPv1.1a).

Sink HPD range:
High: 2.0 - 5.0V
Low: 0 - 0.8V

SYNC_MASTER=J70 NICK		SYNC_DATE=10/16/2013	
PAGE TITLE			
TBT/DP: Thunderbolt Connector B			
	Apple Inc.	DRAWING NUMBER	051-1160
		REVISION	2.0.0
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		PAGE	33 OF 105
		SHEET	31 OF 73

D

C

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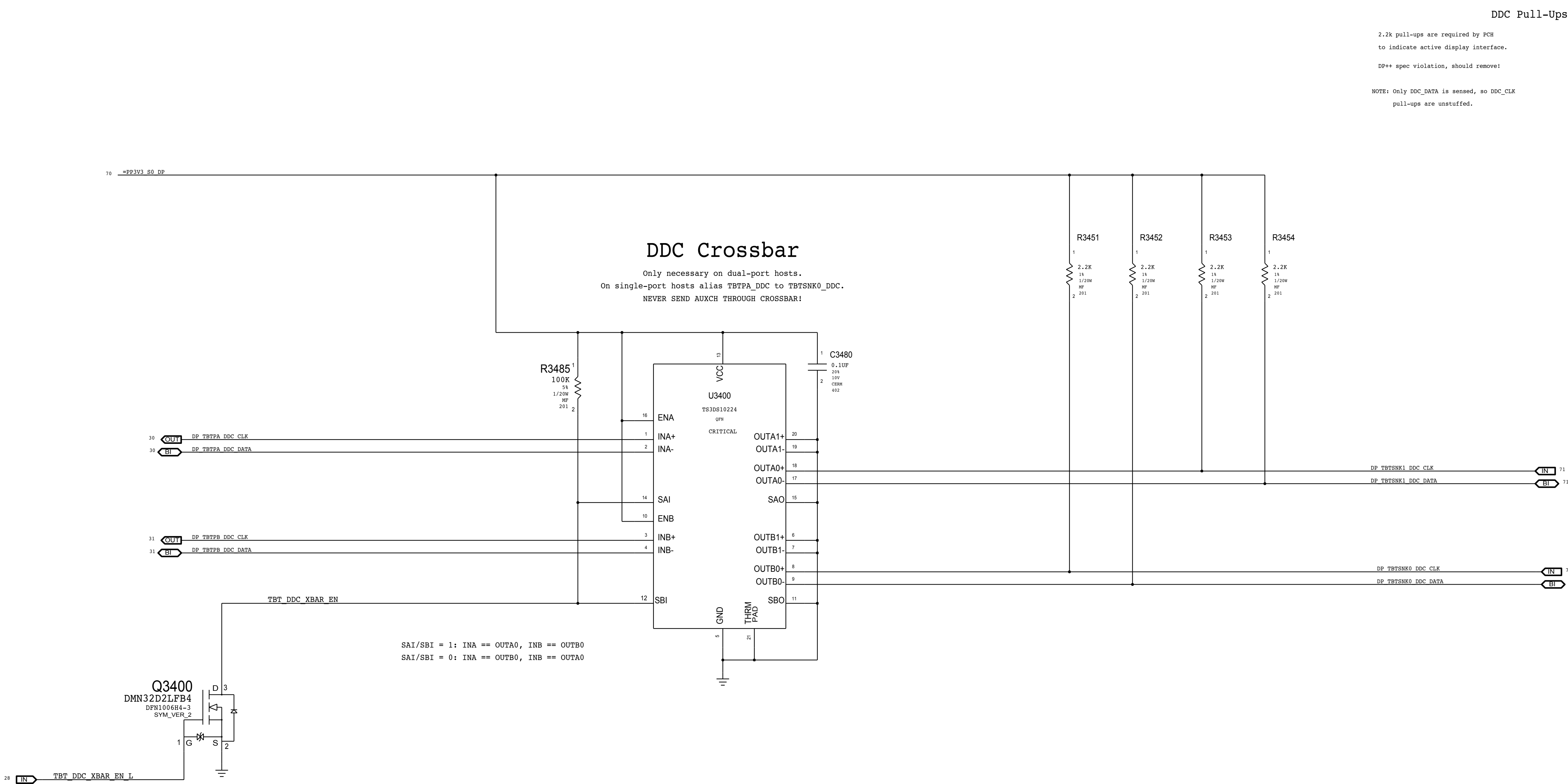
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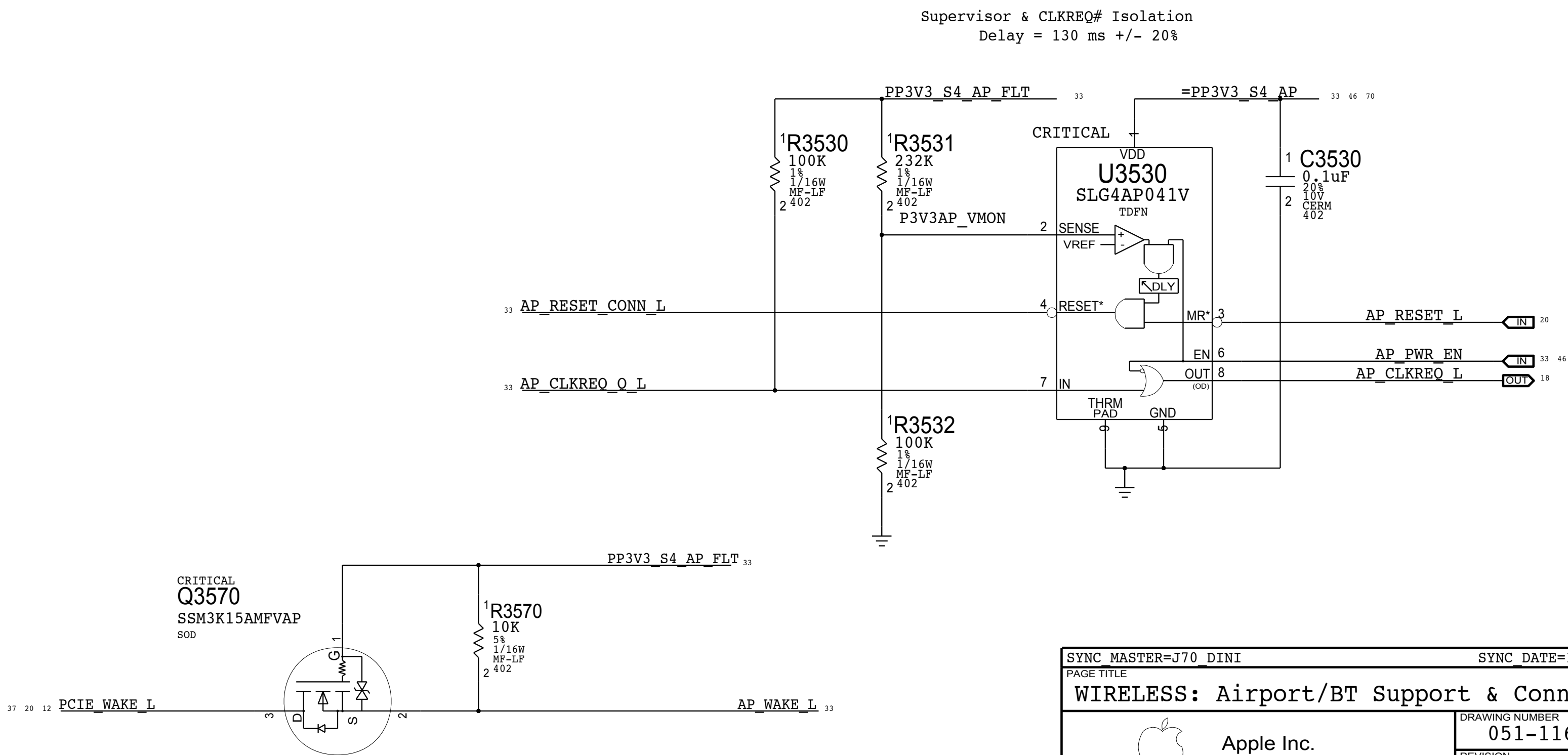
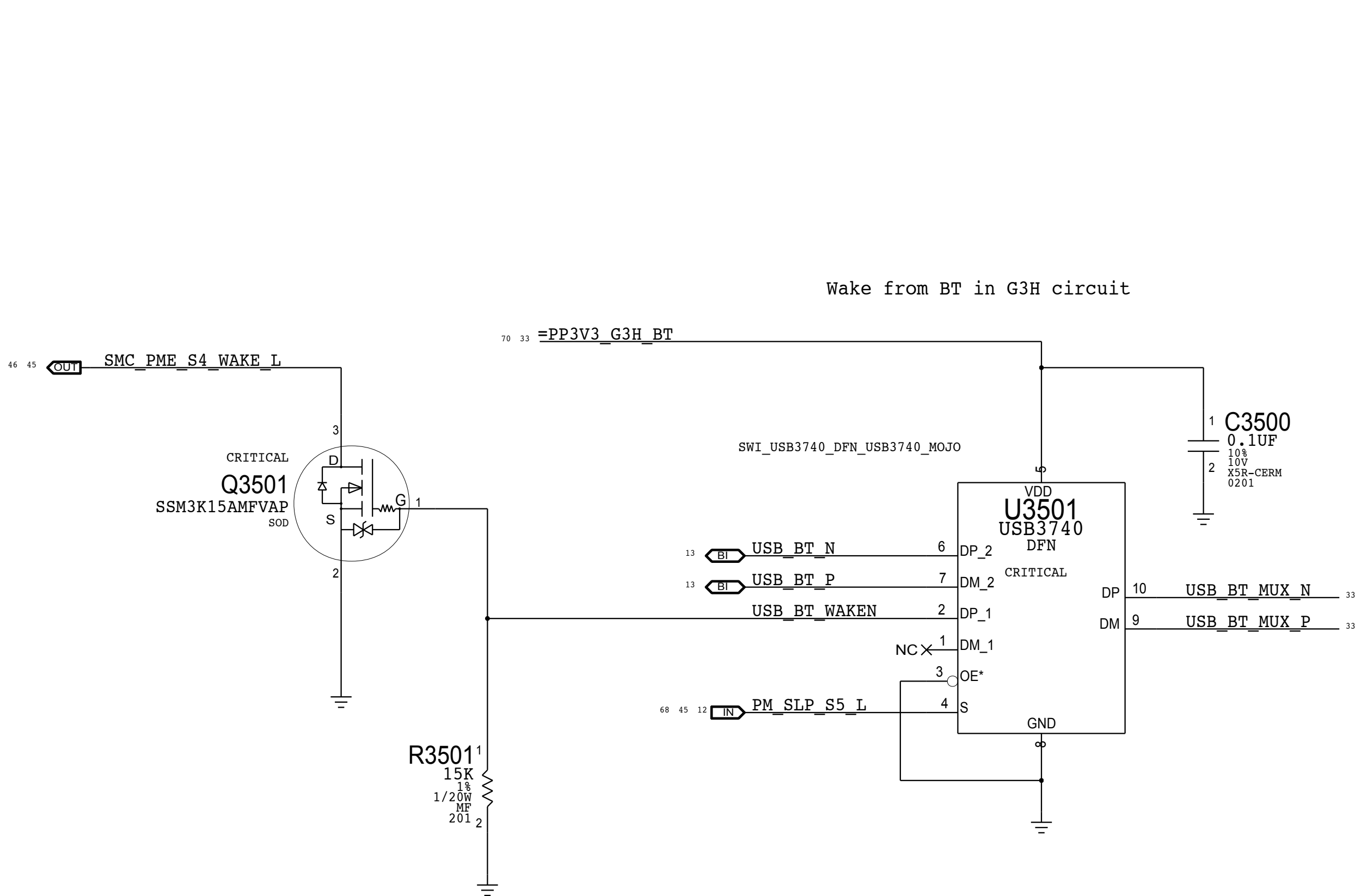
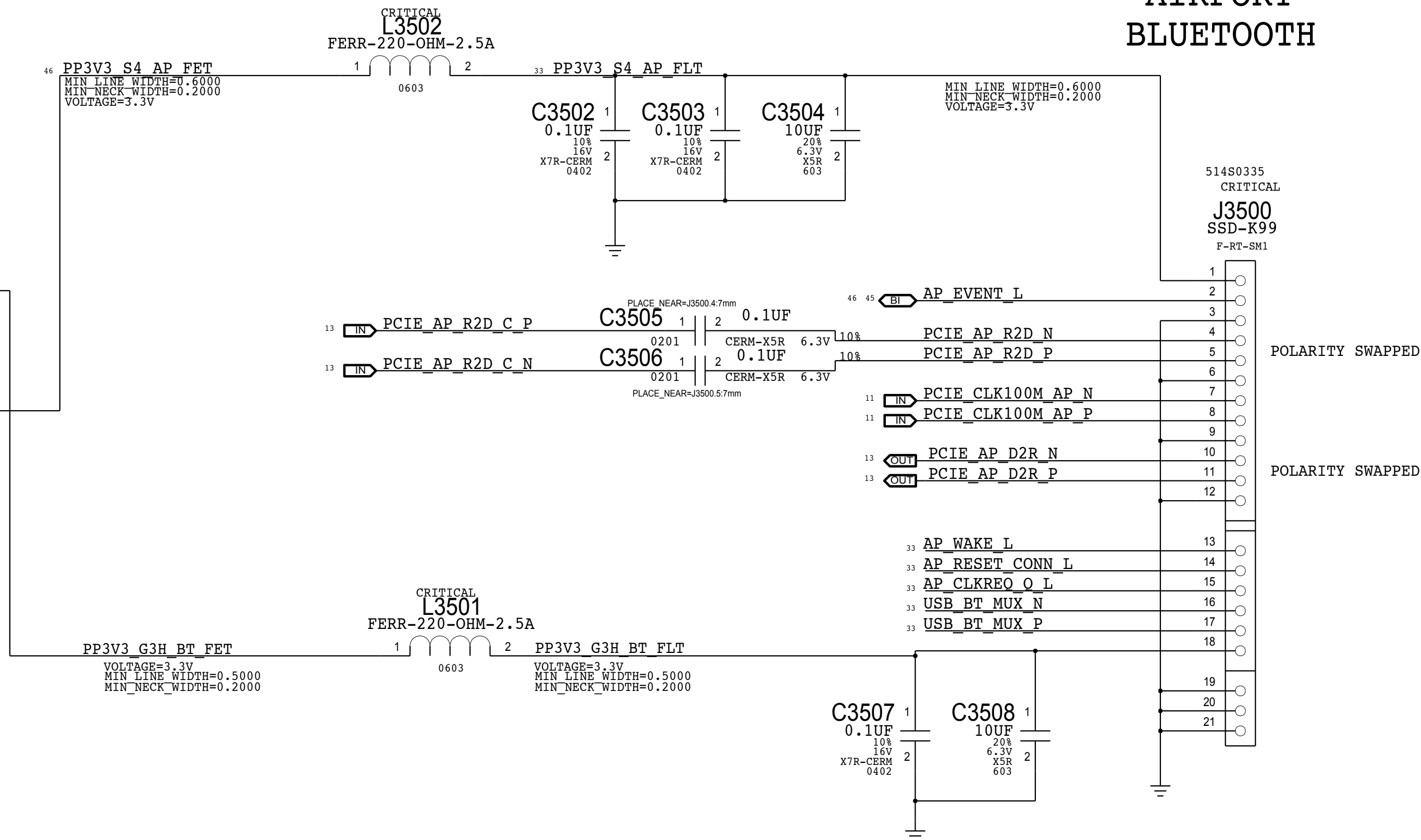
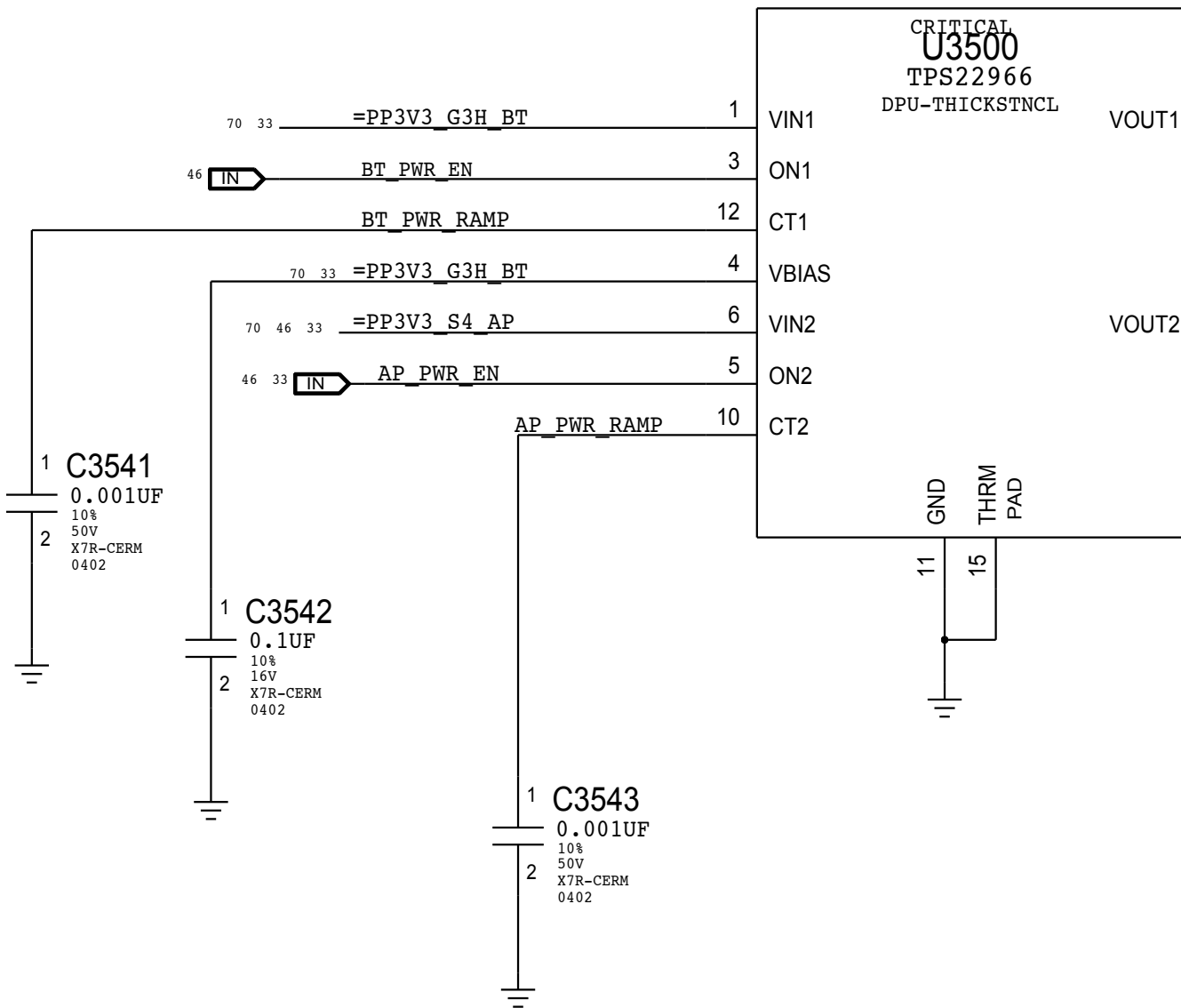
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A



SYNC_MASTER~J70_TONY		SYNC_DATE~09/13/2013	
PAGE TITLE			
TBT/DP: DDC Crossbar			
 Apple Inc.	DRAWING NUMBER	051-1160	SIZE D
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	SHEET	32 OF 73	

AP & BT Load Switch	
SWITCH	TPS22966
AP SLEW RATE	1185 us
BT SLEW RATE	1185 us
Equation	$0.32 * C_L + 13.7$



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SYNC_MASTER=J70 DINI		SYNC_DATE=10/08/2013	
PAGE TITLE			
WIRELESS: Airport/BT Support & Connector			
	Apple Inc.	DRAWING NUMBER	051-1160
		REVISION	2.0.0
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		SHEET	33 OF 73

D

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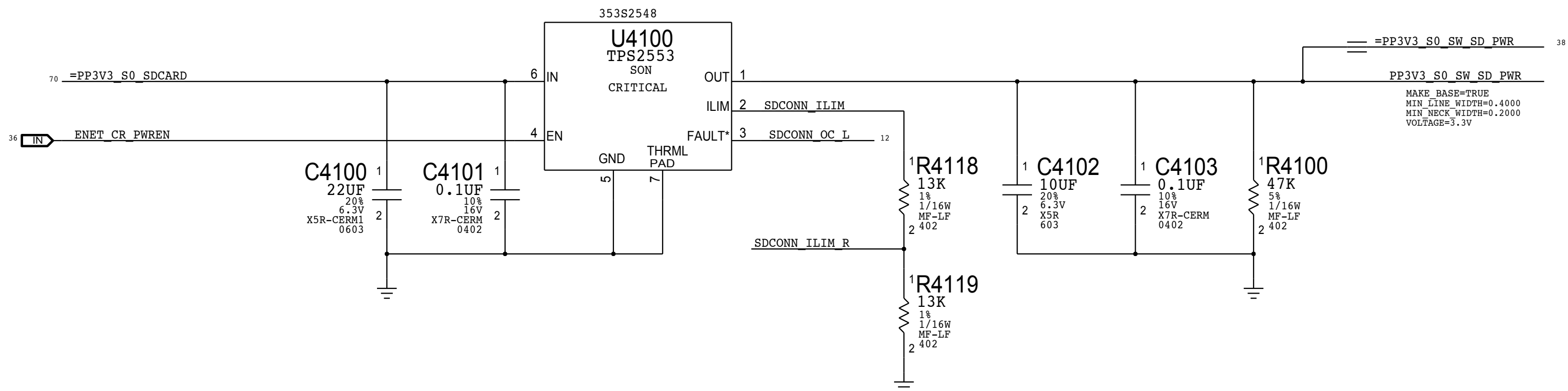


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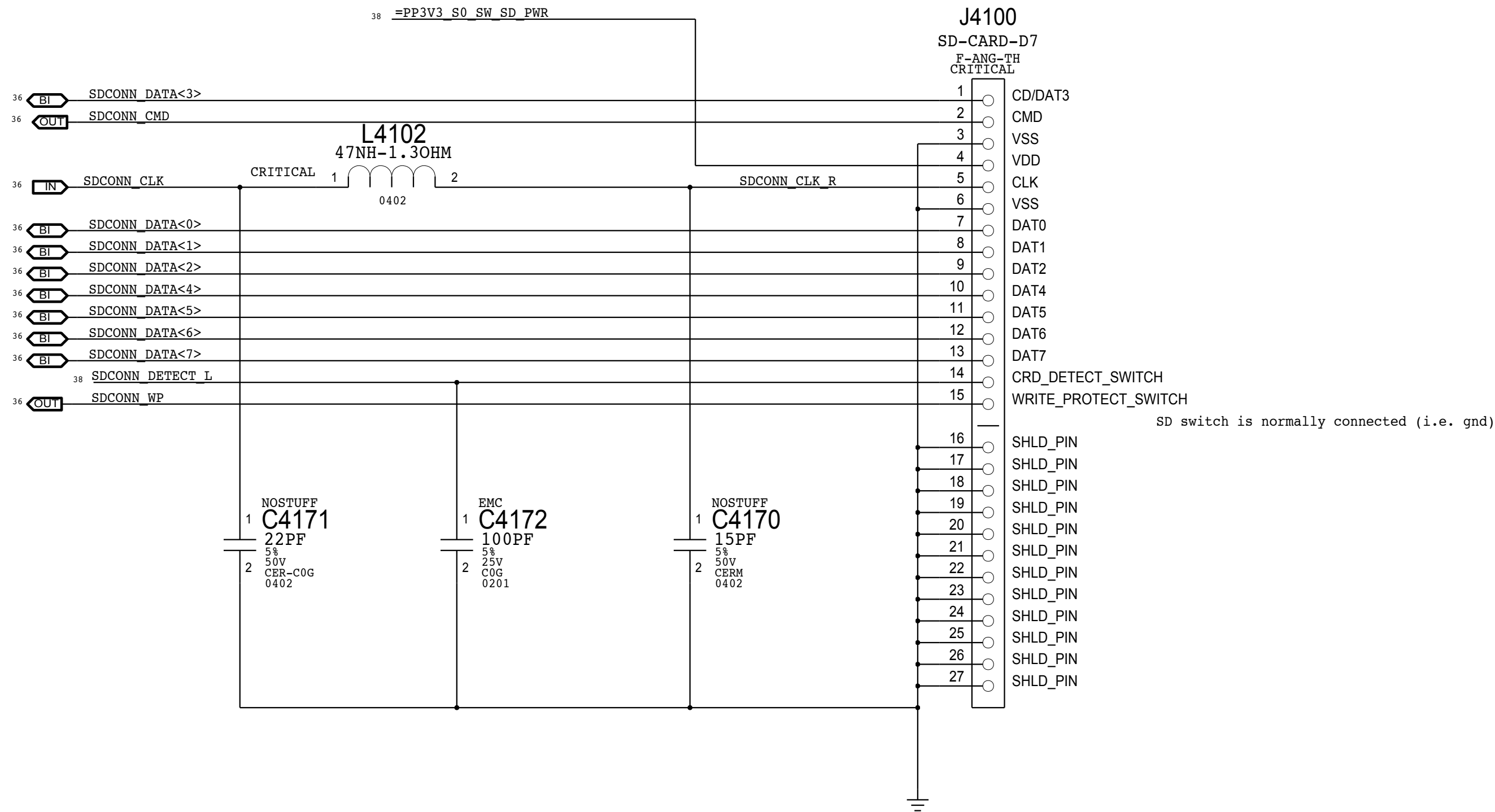
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SD CARD 3.3V OVERCURRENT PROTECTION CHIP

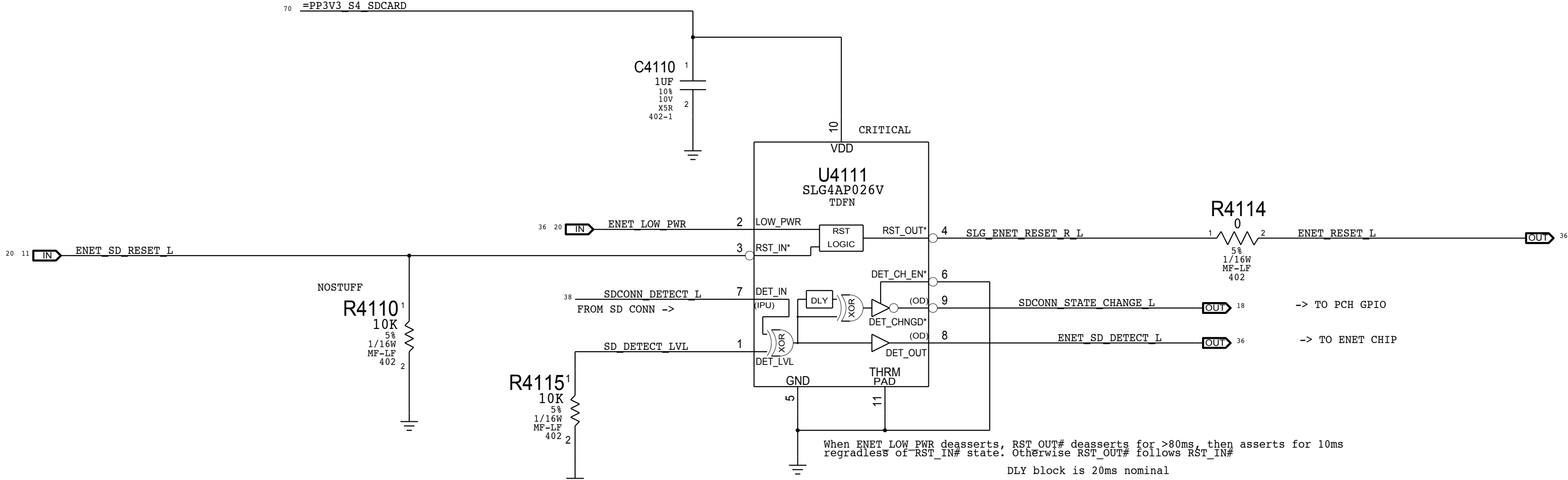


J16:516-0249 / J17:512-0038

SD CARD CONNECTOR



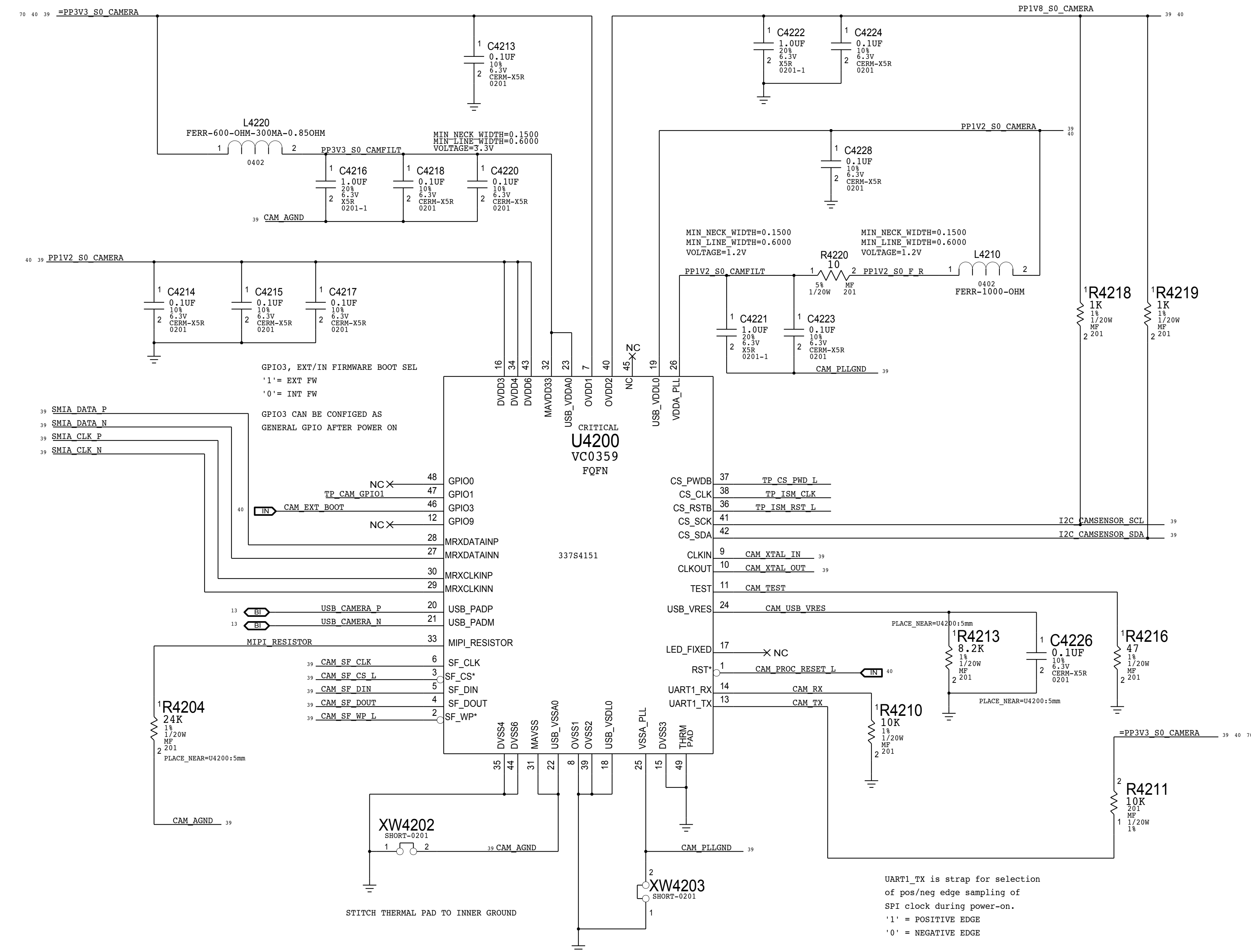
SDCONN DETECT DEBOUNCE. ENET_RESET AND DETECT-CHANGED PCH GPIO PULSE GENERATION.



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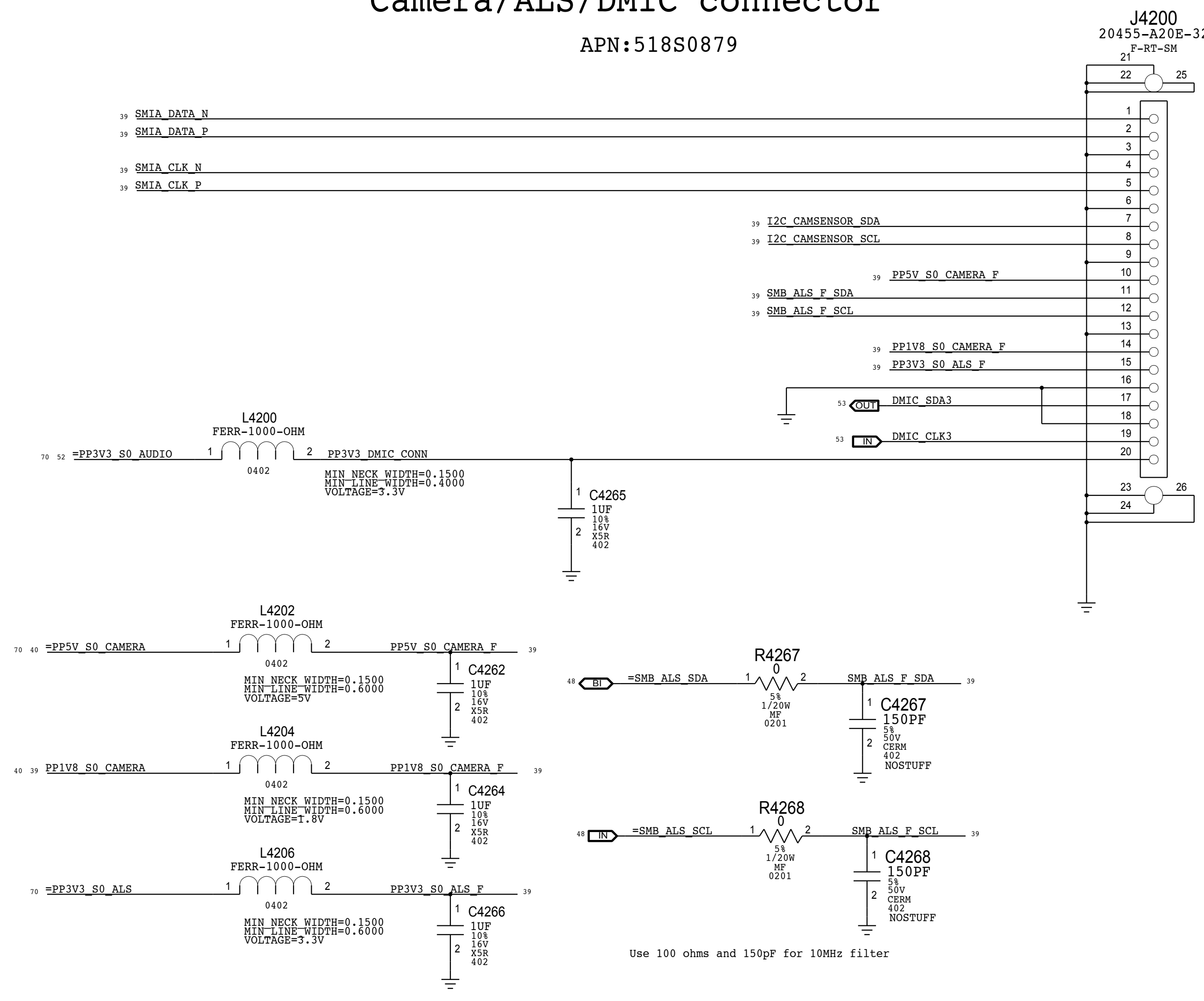
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PAGE TITLE			
SD CARD: SD Reader Connector			
	DRAWING NUMBER		SIZE
	051-1160		D
REVISION		2.0.0	
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USB CAMERA CONTROLLER

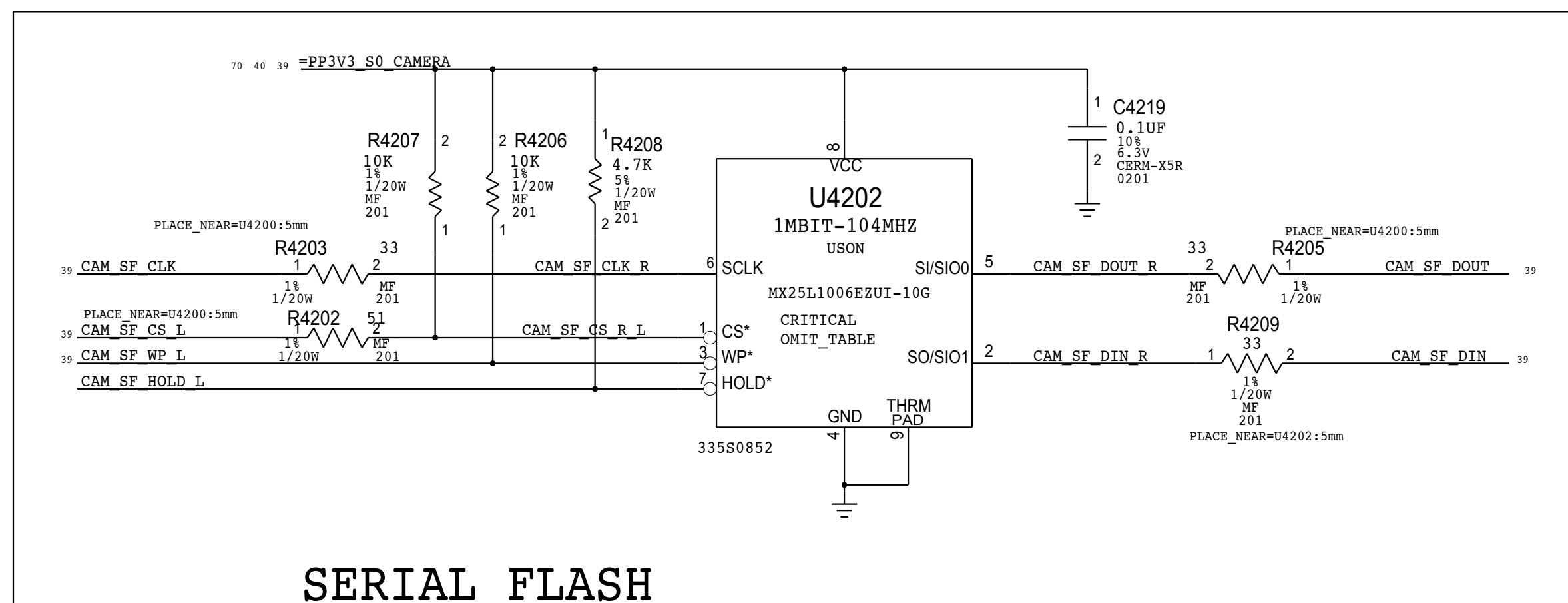


Camera/ALS/DMIC connector

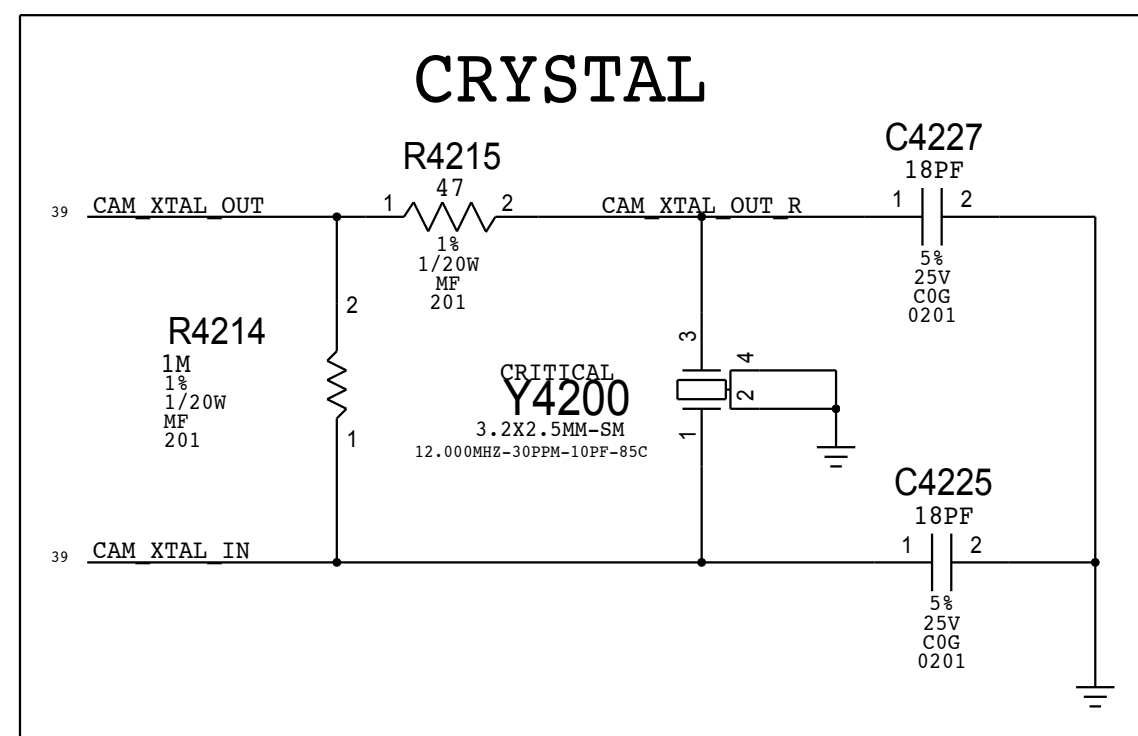
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SERIAL FLASH

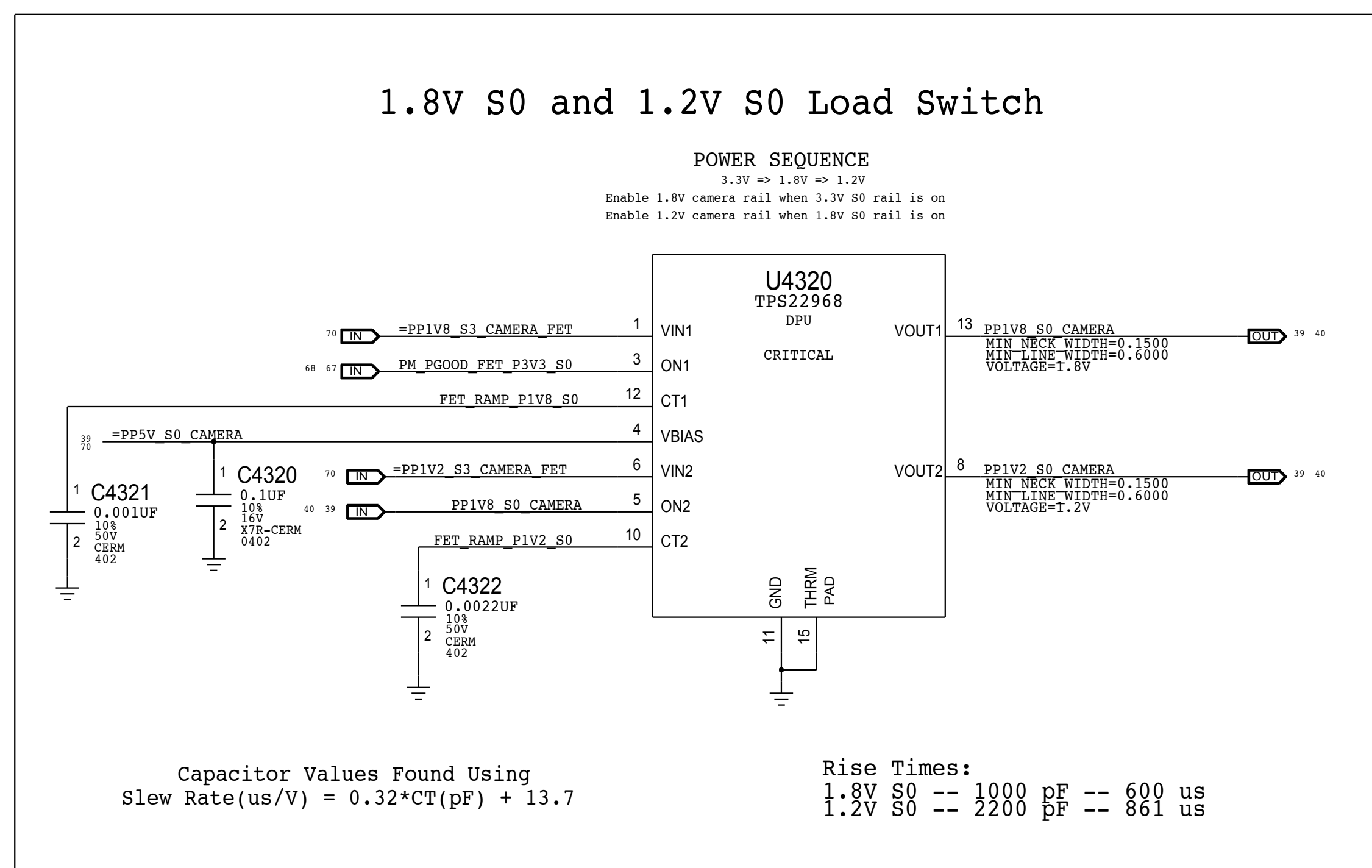
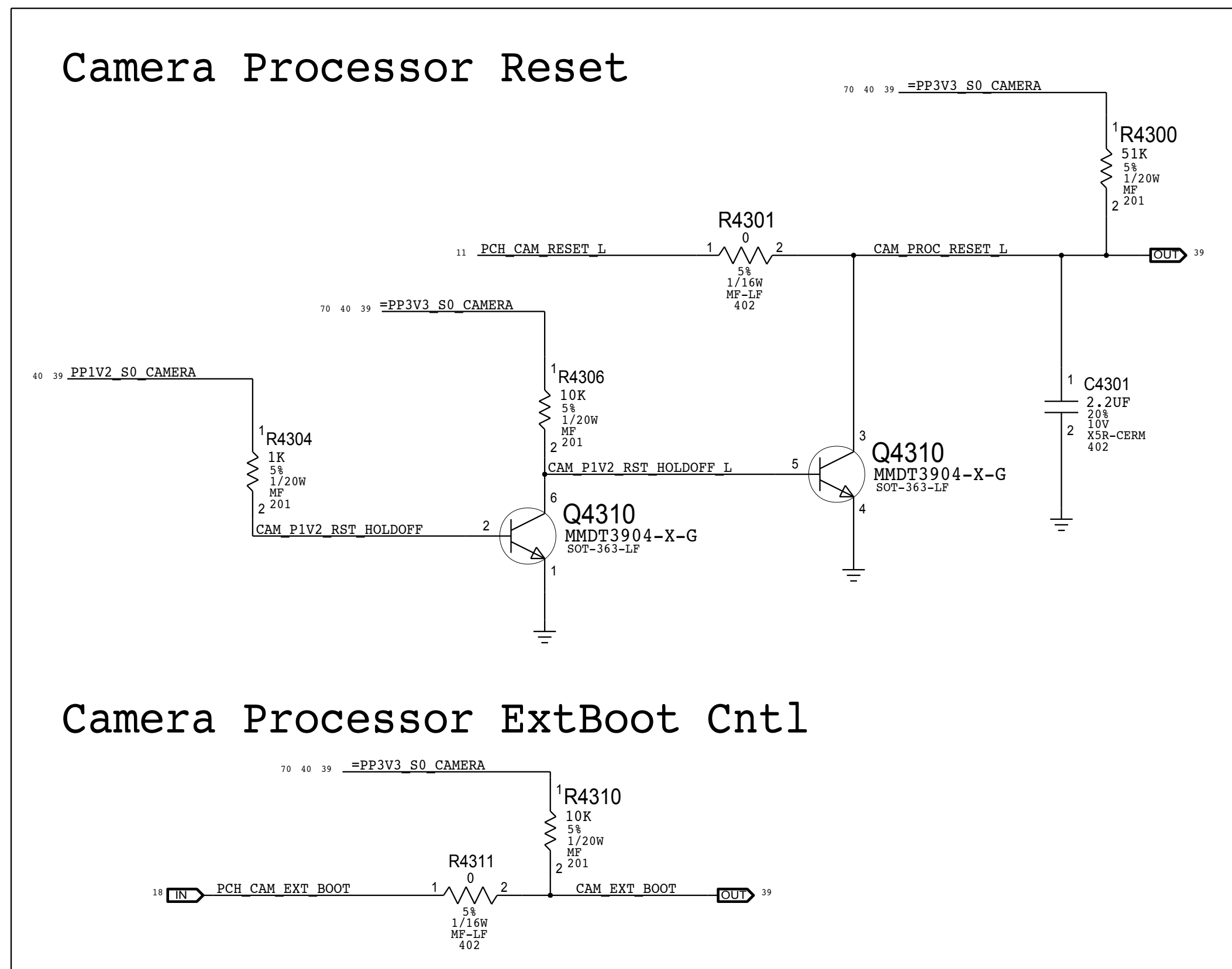


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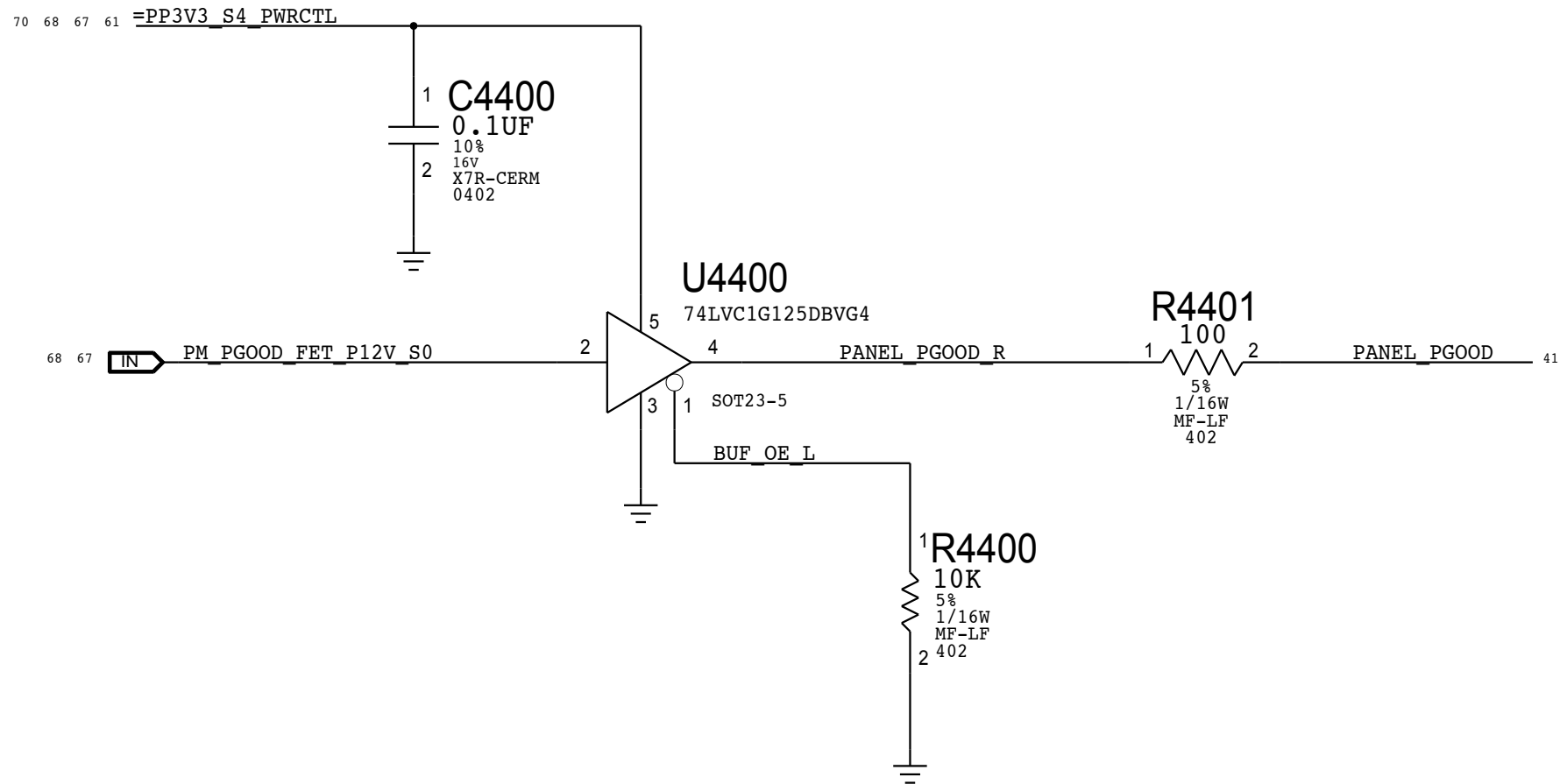
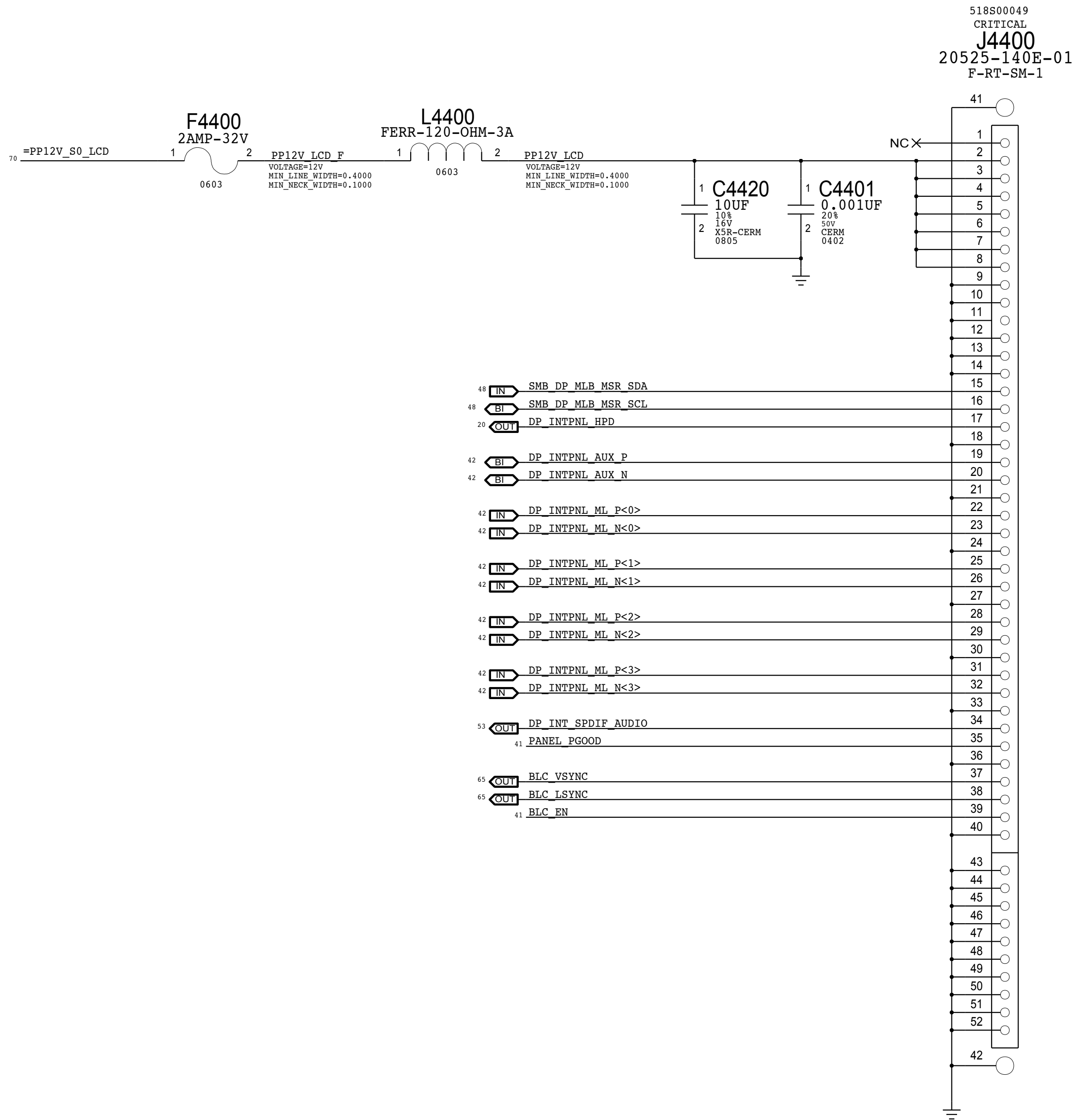


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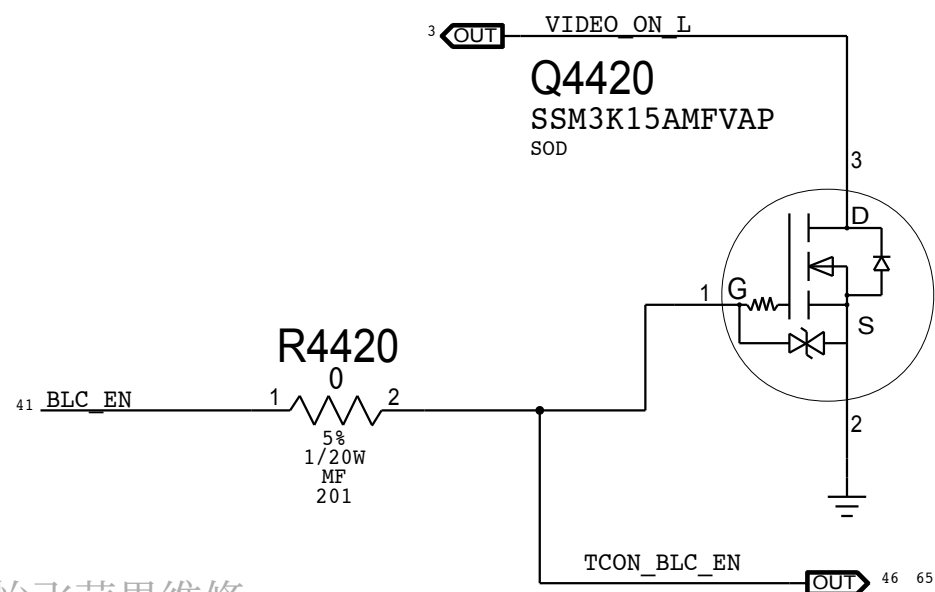
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PAGE TITLE			
CAMERA: Camera Controller			
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Internal DP Connector



X403 Config



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SYNC_MASTER=J94_NICK		SYNC_DATE=06/23/2014	
PAGE TITLE			
DISPLAY: Internal DP Support			
	DRAWING NUMBER		SIZE
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	PAGE		
	44 OF 105		
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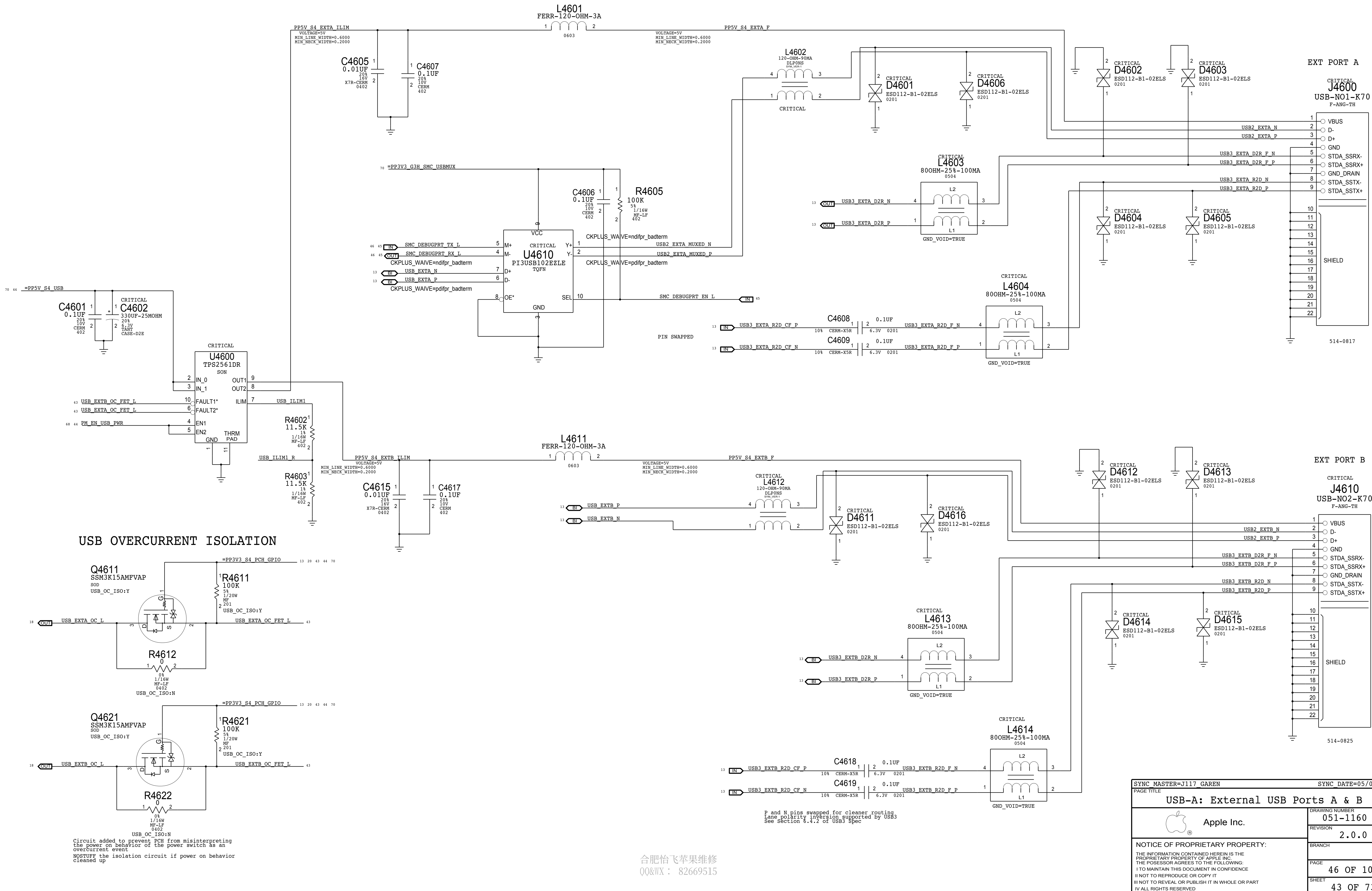
A

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P and N pins swapped for cleaner routing
Lane polarity inversion supported by USB3
See Section 6.4.2 of USB3 Spec

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SYNC_MASTER=J117_GAREN		SYNC_DATE=05/05/2014	
PAGE TITLE			
USB-A: External USB Ports A & B			
	DRAWING NUMBER	051-1160	SIZE
	REVISION	2.0.0	D
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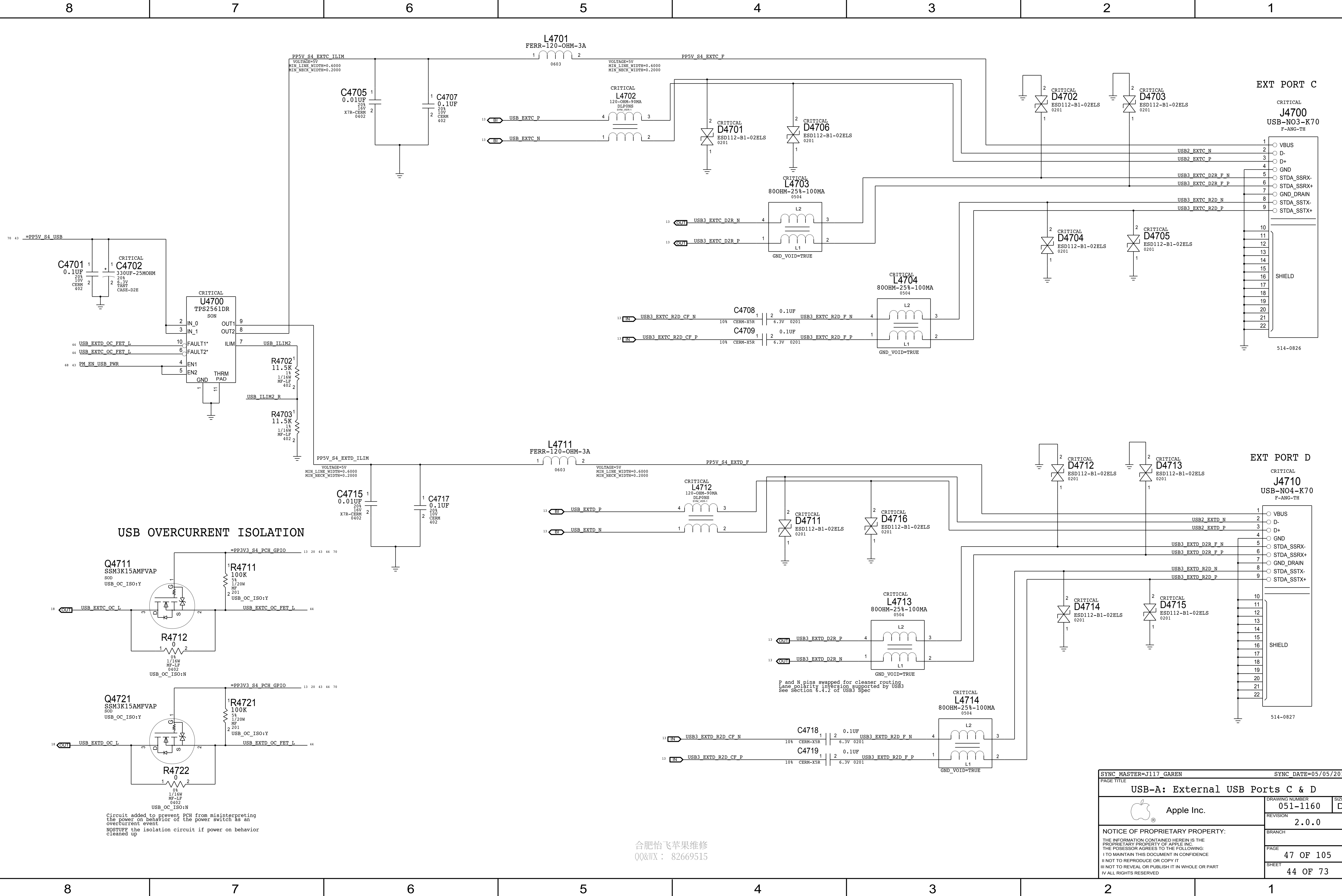
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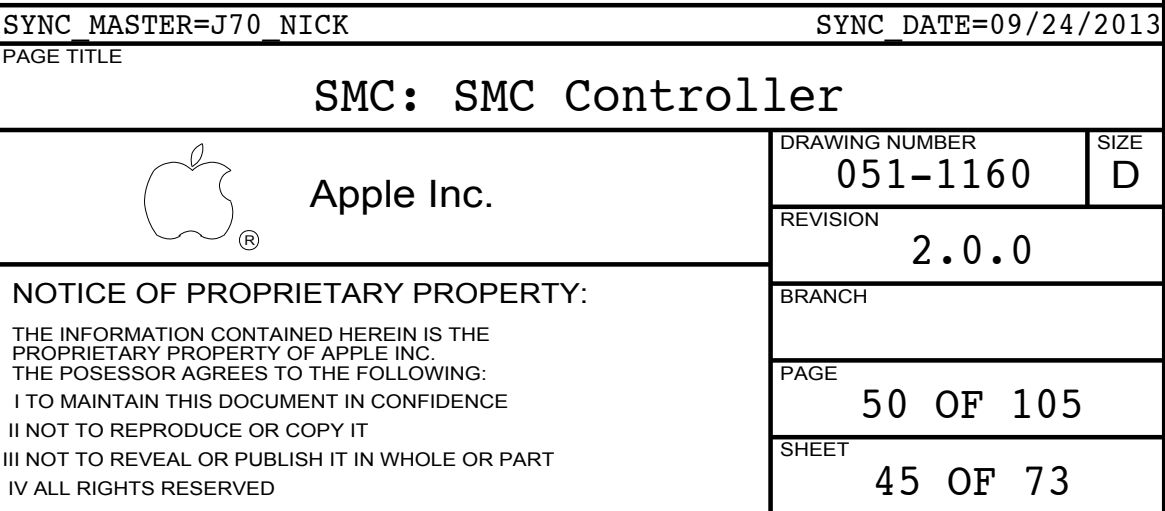
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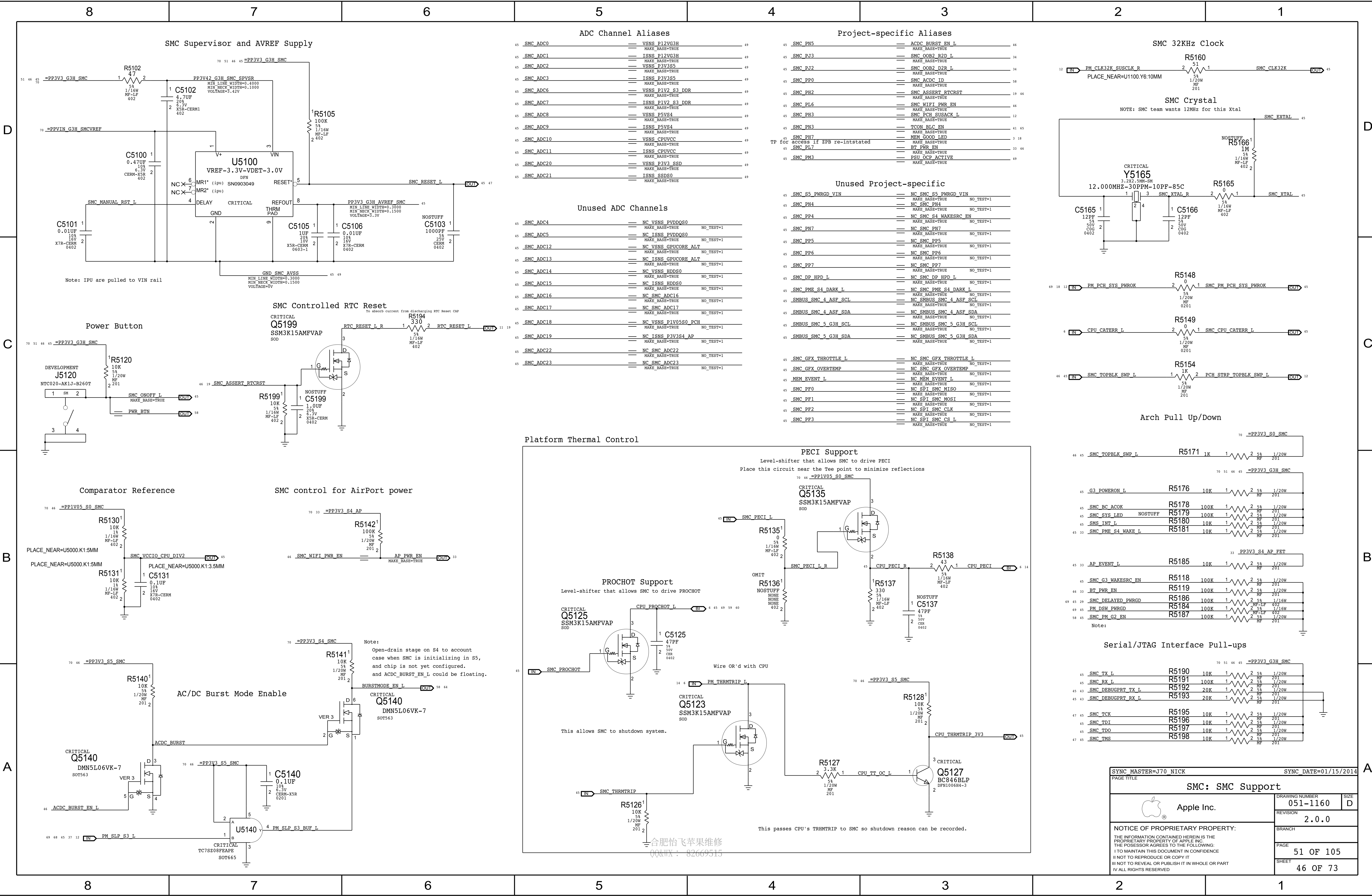
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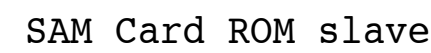
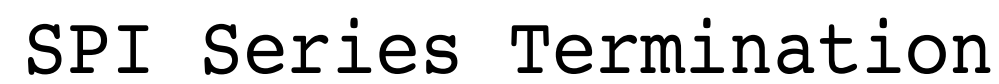
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PAGE TITLE			
USB-A: External USB Ports C & D			
 Apple Inc.	DRAWING NUMBER	051-1160	SIZE
	REVISION	2.0.0	
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		PAGE	
		47 OF 105	
		SHEET	
		44 OF 73	

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QQ&WX : 82669515

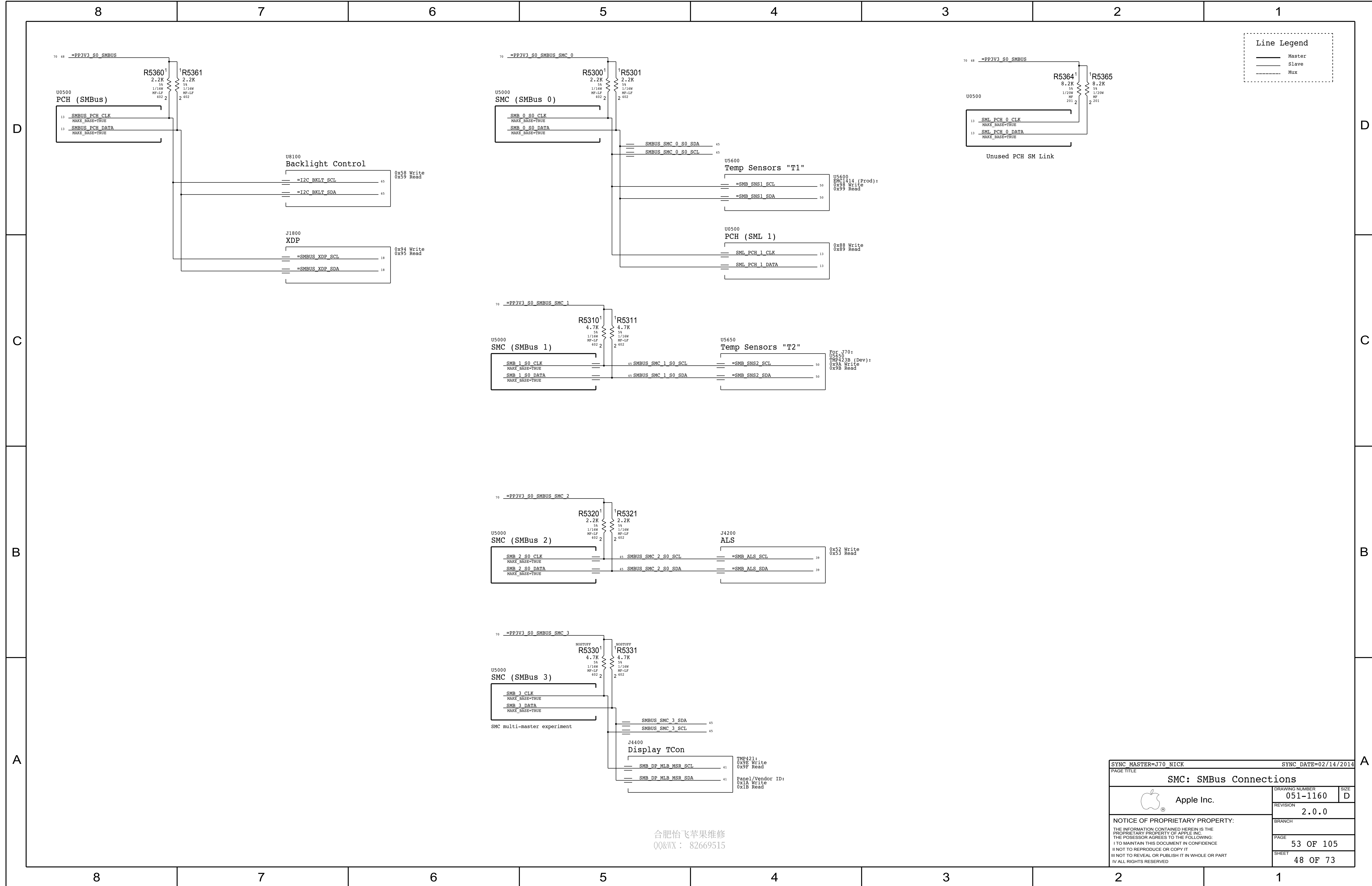




Quad_IO Mode (Mode 0 & 3) supported
SPI Freq: 50MHz for PCH, 20 MHz for SMC

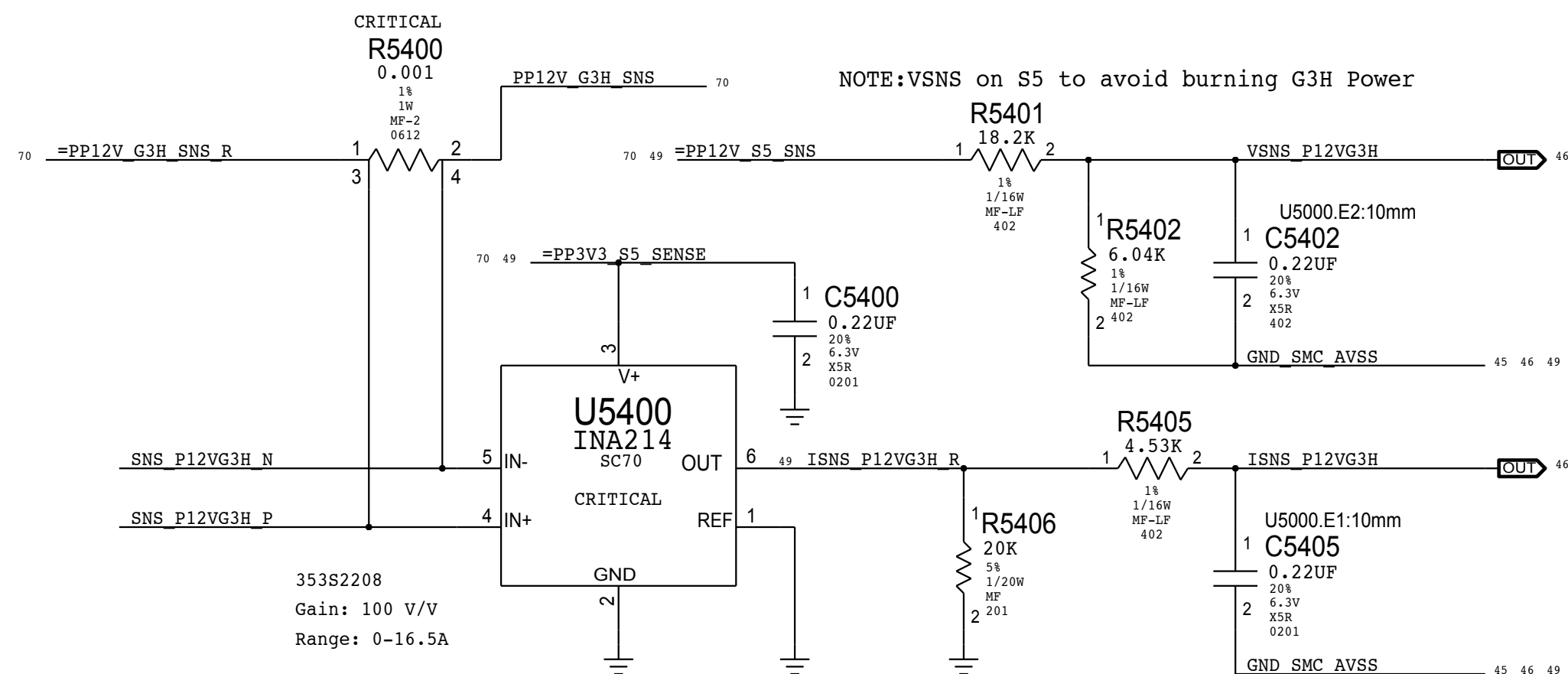


SYNC MASTER=J117 TONY		SYNC DATE=05/05/2014	
PAGE TITLE			
CPU & CHIPSET: SPI and Debug Connector			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-1160	D
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		2.0.0	
		BRANCH	
		PAGE	52 OF 105
		SHEET	
			47 OF 73



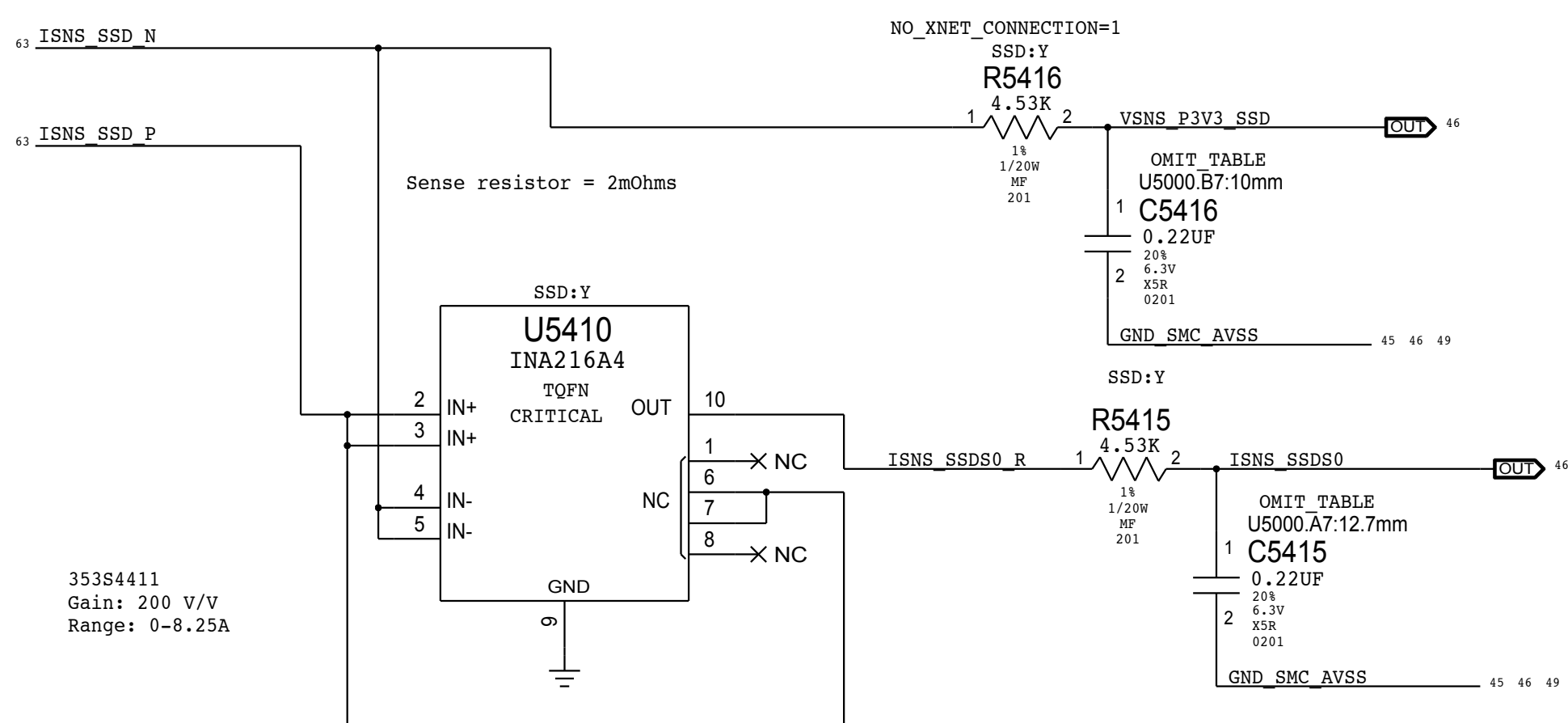
12V G3H (VD2R:ADC0/ID2R:ADC1)

AC/DC lowside sense (System total)



SSD S0 (IH1R:ADC21/VH1R:ADC20)

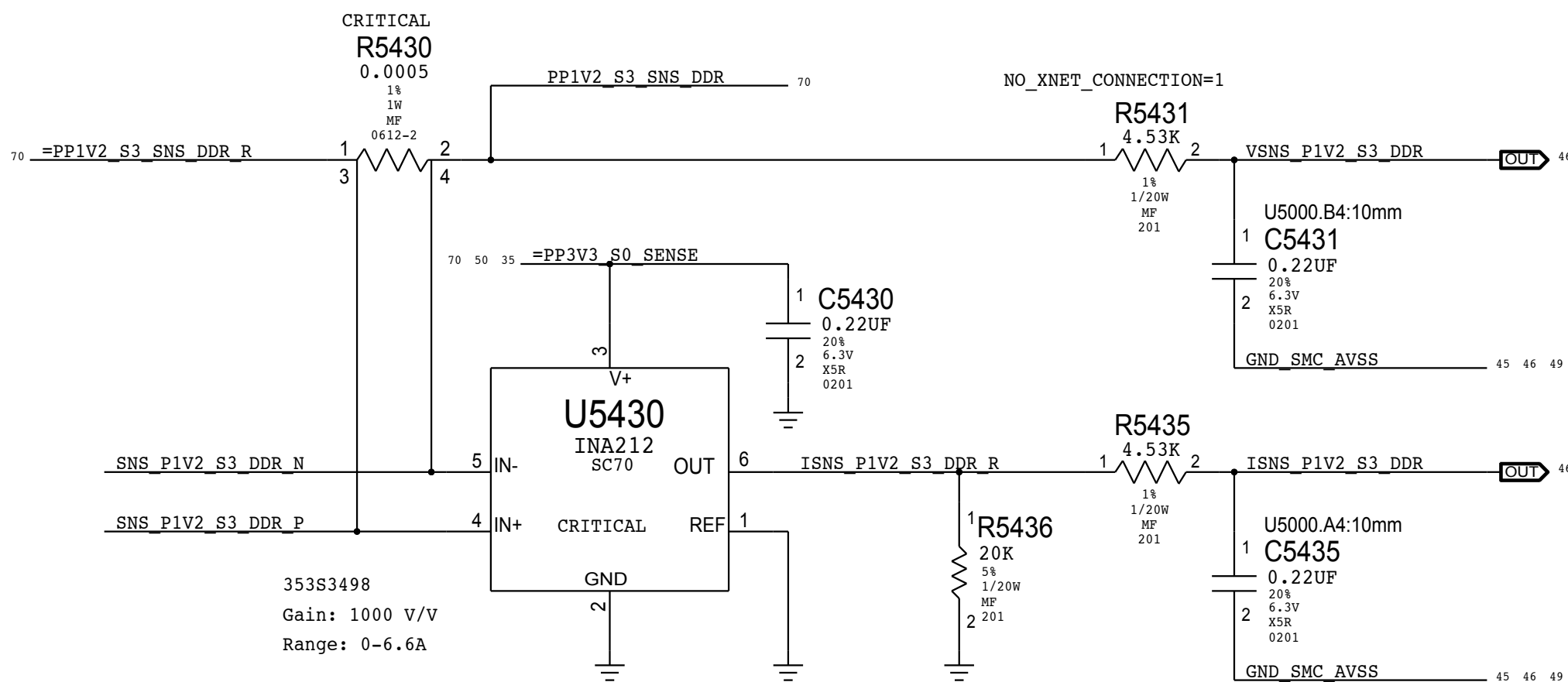
I-sense / V-sense for SSD



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
13280304	2	CAP,0.22UF,201	C5415,C5416	SSD:Y
11780201	2	RES,0 OHM,201	C5415,C5416	SSD:N

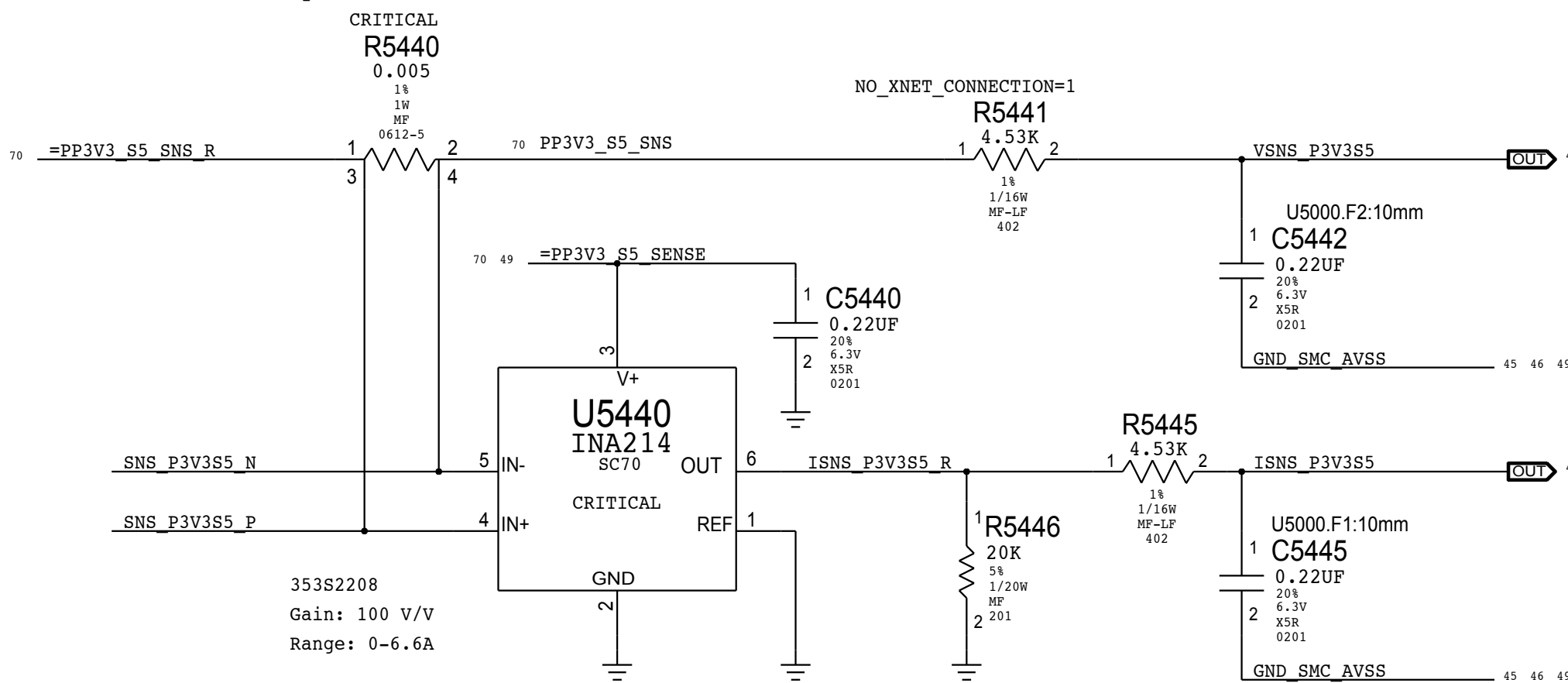
VDDQ S3 (VM0R:ADC6/IM0R:ADC7)

VDDQ lowside sense for DDR



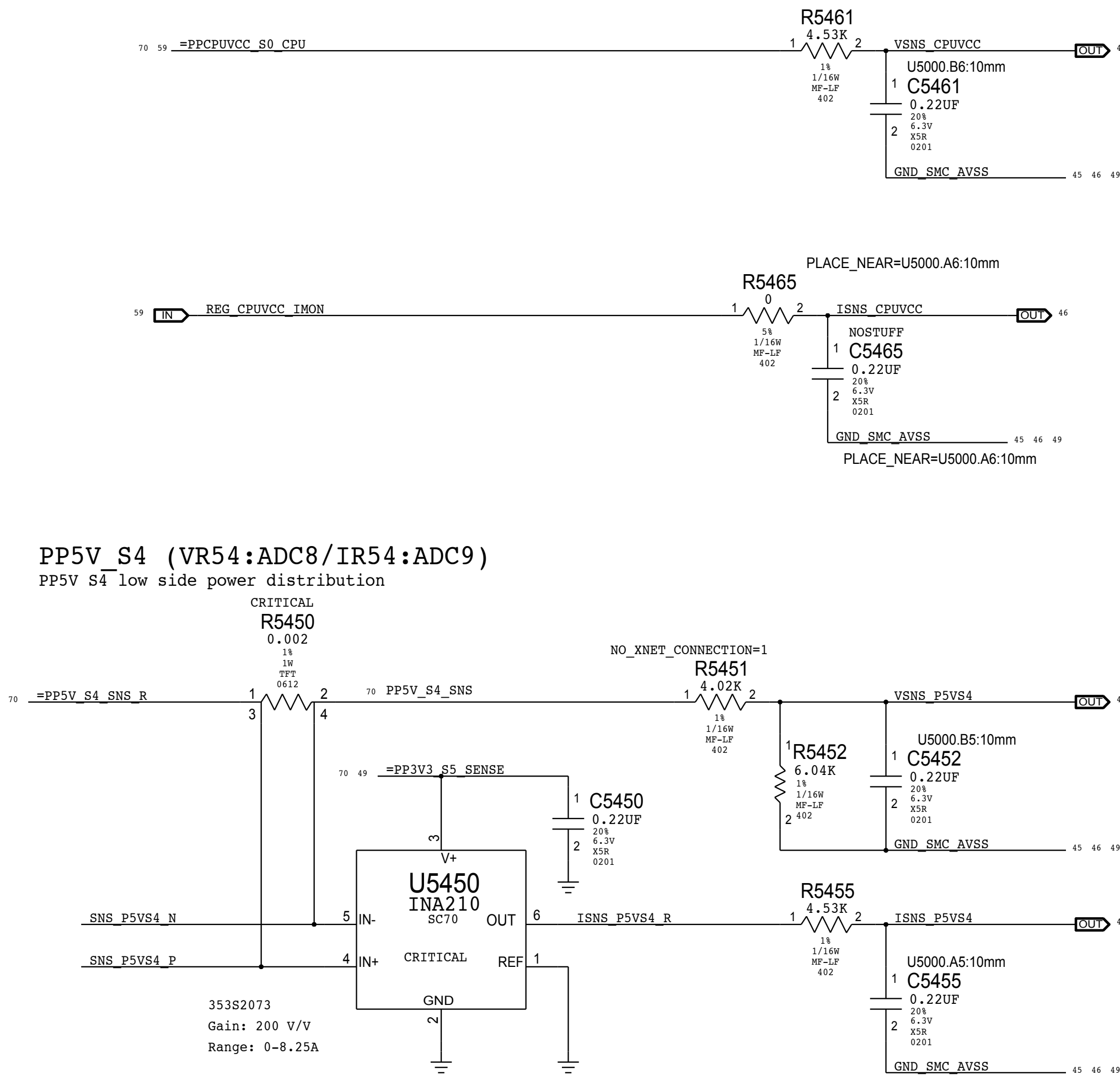
PP3V3 S5 (VR35:ADC2/IR35:ADC3)

PP3V3 S5 low side power distribution

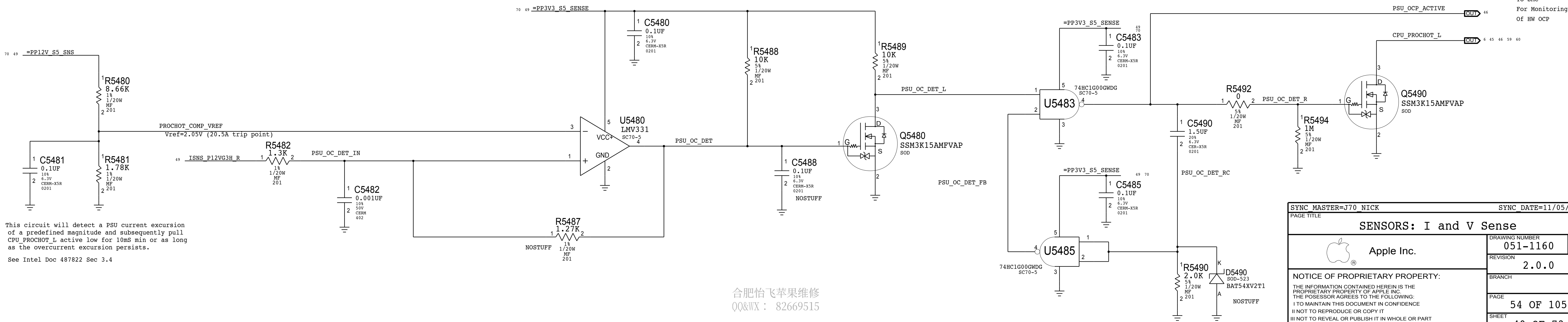


CPU Core (VC0C:ADC10/IC0C:ADC11)

Voltage sense and IMON amp (VC0C, IC0C)

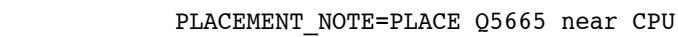
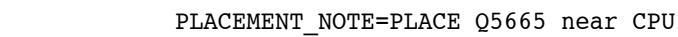
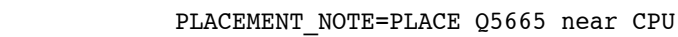
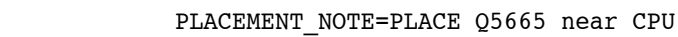
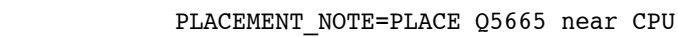


Fast Response PSU Overcurrent Protection

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SYNC_MASTER=J70_NICK		SYNC_DATE=11/05/2013	
PAGE TITLE			
SENSORS: I and V Sense			
 Apple Inc.	DRAWING NUMBER 051-1160		SIZE D
	REVISION 2.0.0		
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		PAGE	54 OF 105
		SHEET	49 OF 73

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2
1

D

C

B

A

D

C

B

A

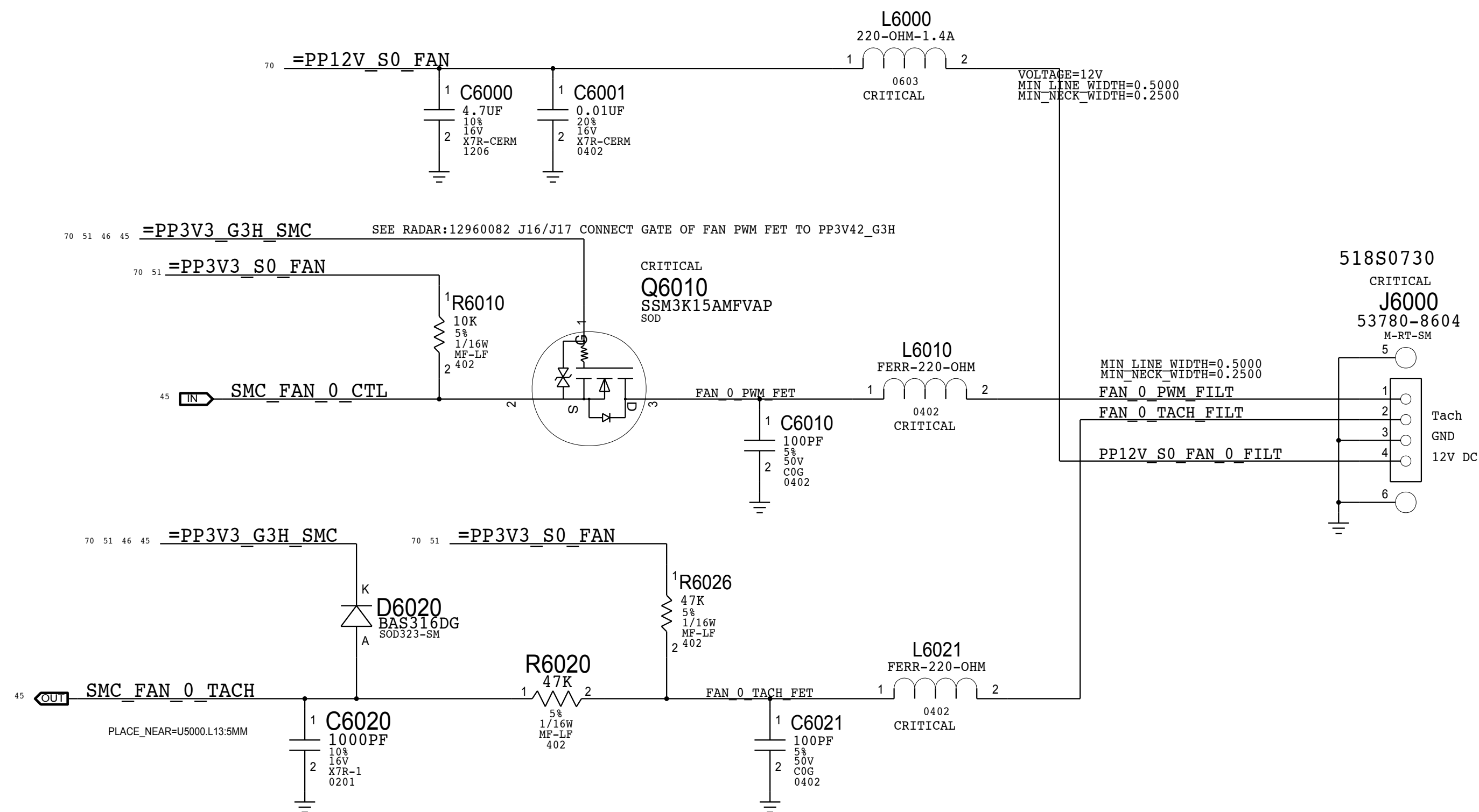
SMC Fan 0 (System)

Note:

The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q6010 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q6010 is at common and the SMC sinks current when Q6010 is on.

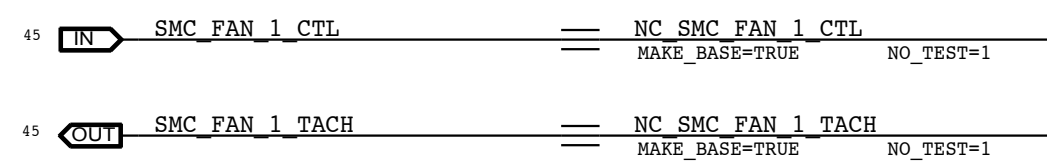
This resembles an open-drain if there is a pull-up, going to a Hi-Z FET input.

Otherwise, this is simply a pass-FET.
See RADAR: 10565825- D7: Need scematic and PCB file of fan(All Vendors).



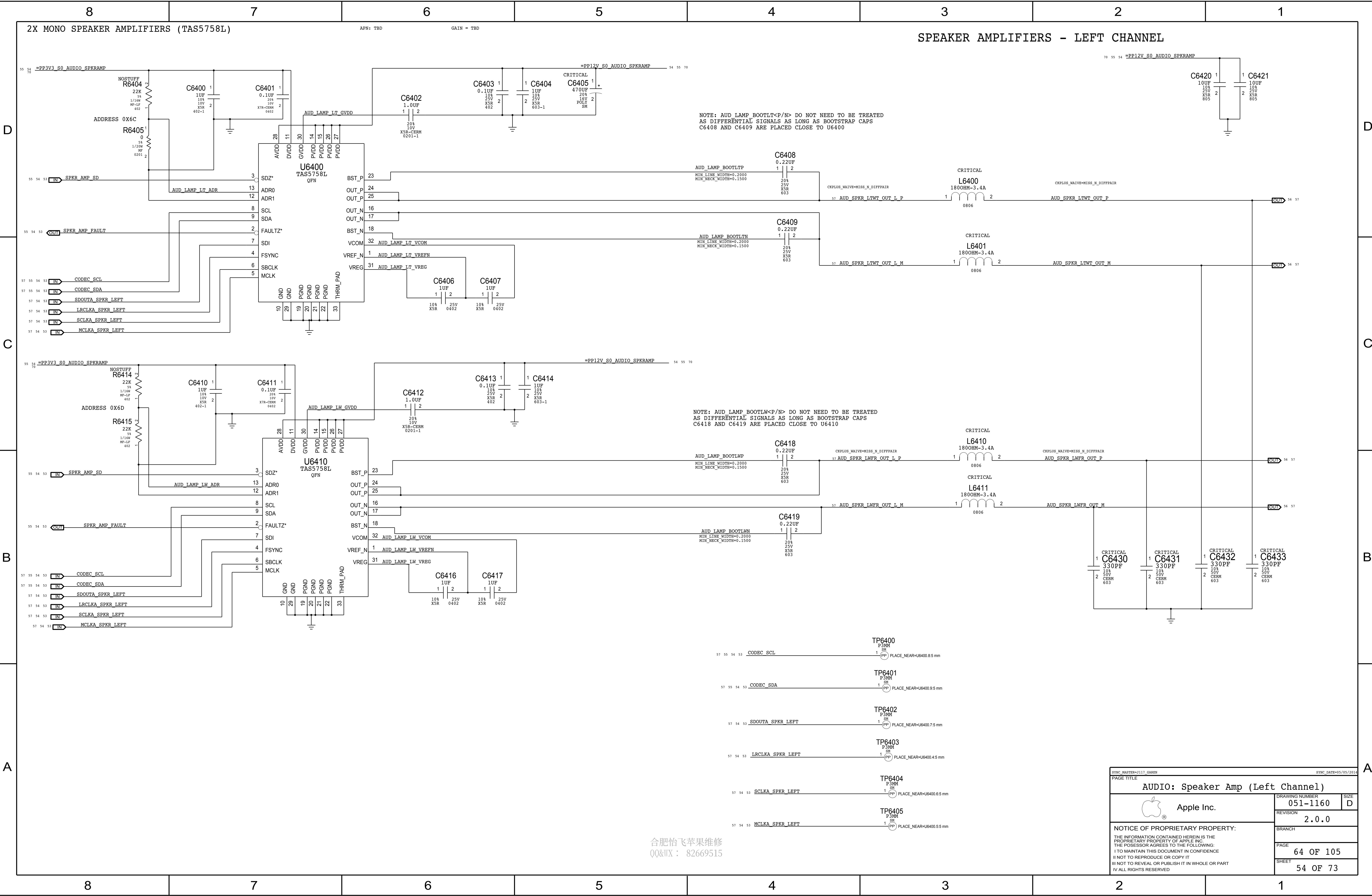
Add C6020 1000pF Cap, Change R6020 to 47K -- Radar 11661918 D8 Protol Fan Tach instability.

SMC Fan 1 (Unused)



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SYNC MASTER=J16 MLB IG		SYNC DATE=08/27/2013	
PAGE TITLE			
FAN: System Fan Connector			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-1160		D
	REVISION		
	2.0.0		
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		60 OF 105	
		SHEET	
		51 OF 73	

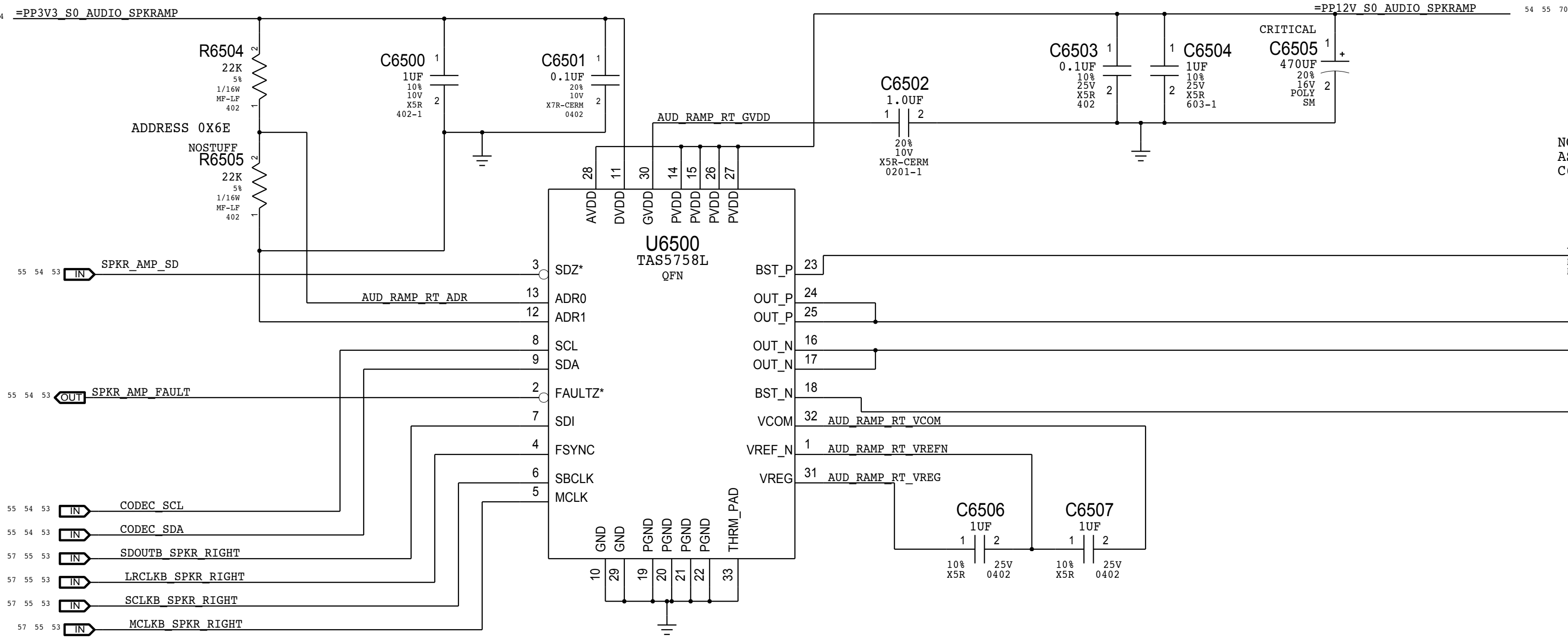


2X MONO SPEAKER AMPLIFIERS (TAS5758L)

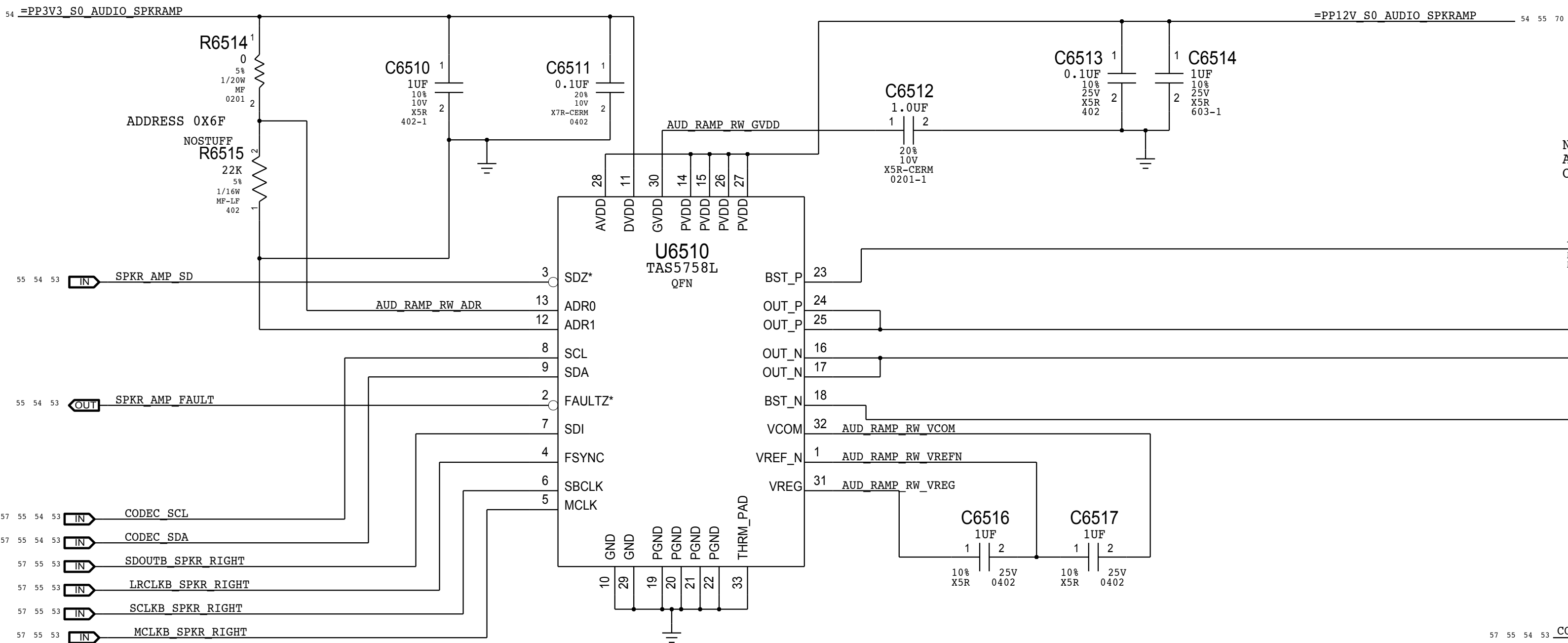
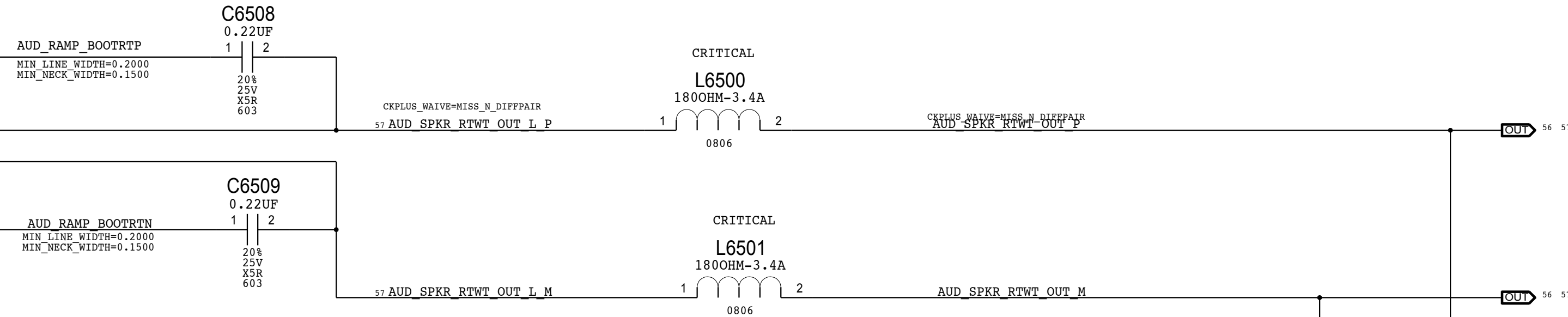
APN: TBD

GAIN = TBD

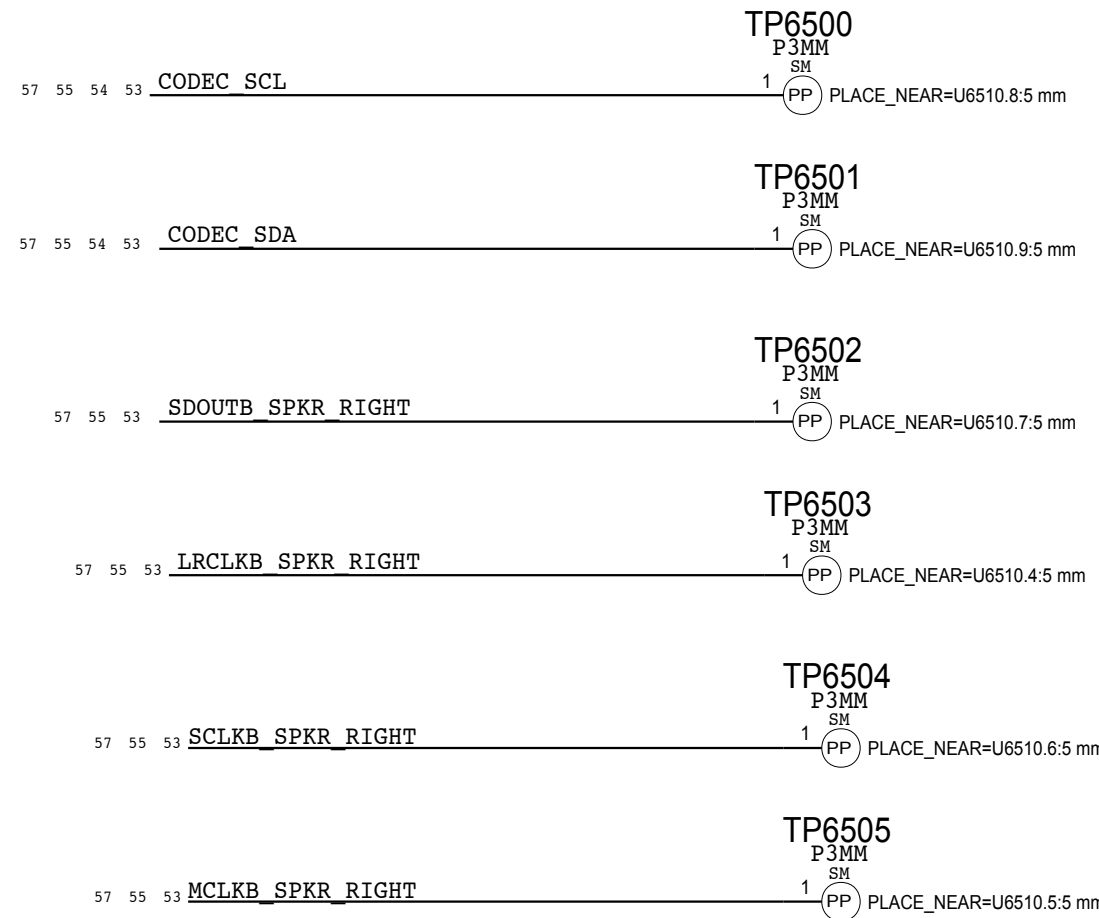
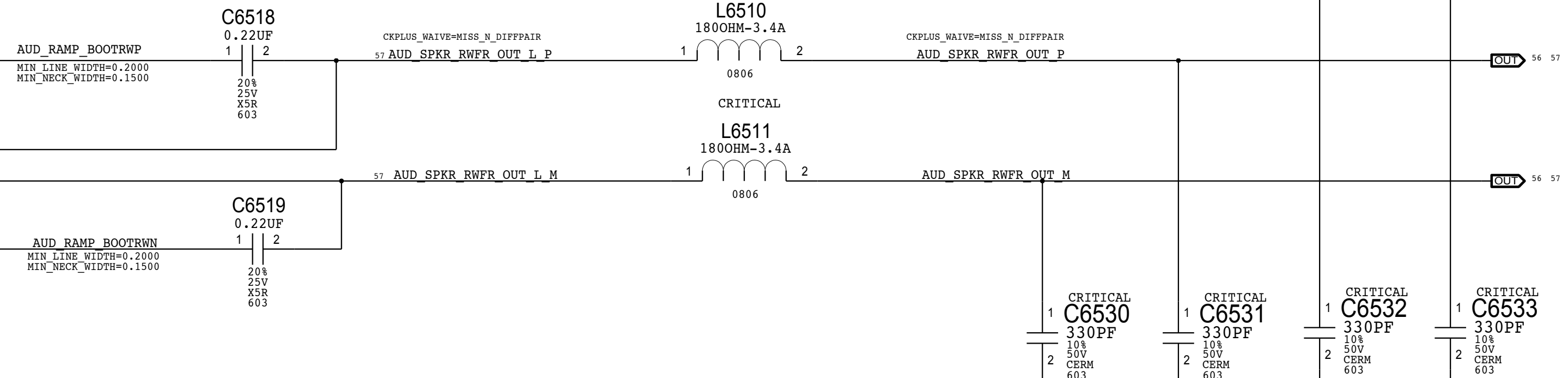
SPEAKER AMPLIFIERS - RIGHT CHANNEL



NOTE: AUD_LAMP_BOOTRT<P/N> DO NOT NEED TO BE TREATED AS DIFFERENTIAL SIGNALS AS LONG AS BOOTSTRAP CAPS C6508 AND C6509 ARE PLACED CLOSE TO U6500



NOTE: AUD_LAMP_BOOTRW<P/N> DO NOT NEED TO BE TREATED AS DIFFERENTIAL SIGNALS AS LONG AS BOOTSTRAP CAPS C6518 AND C6519 ARE PLACED CLOSE TO U6510



SYNC_MASTER=J117_GAREN		SYNC_DATE=05/05/2014	
PAGE TITLE			
AUDIO: Speaker Amp (Right Channel)			
 Apple Inc.	DRAWING NUMBER 051-1160		SIZE D
	REVISION 2.0.0		
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		BRANCH	
		PAGE	65 OF 105
		SHEET	55 OF 73

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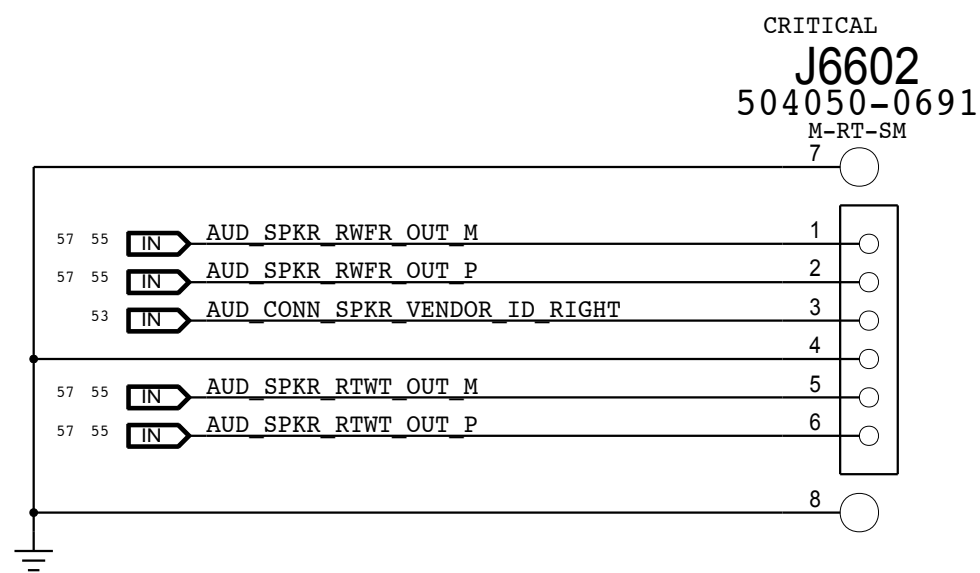
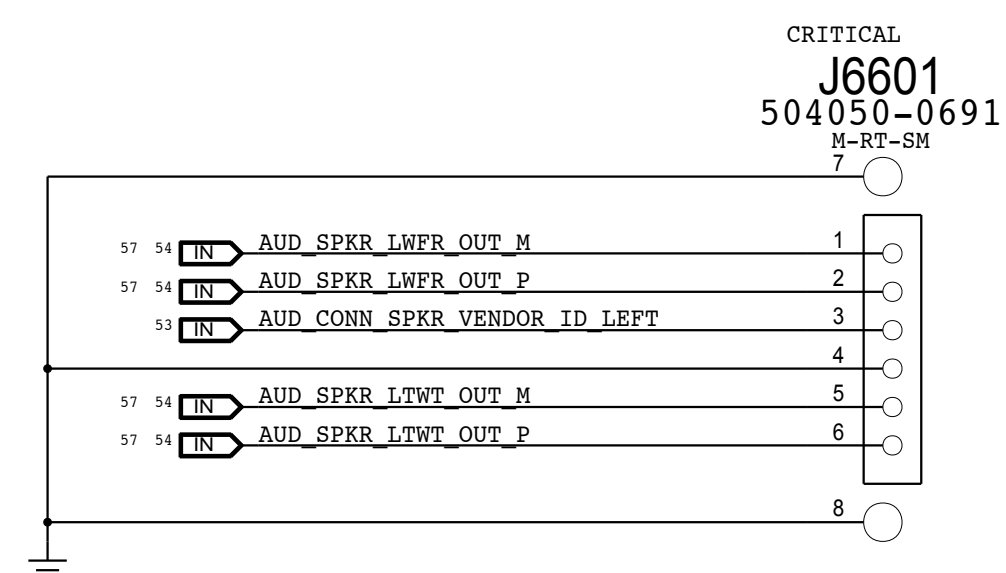
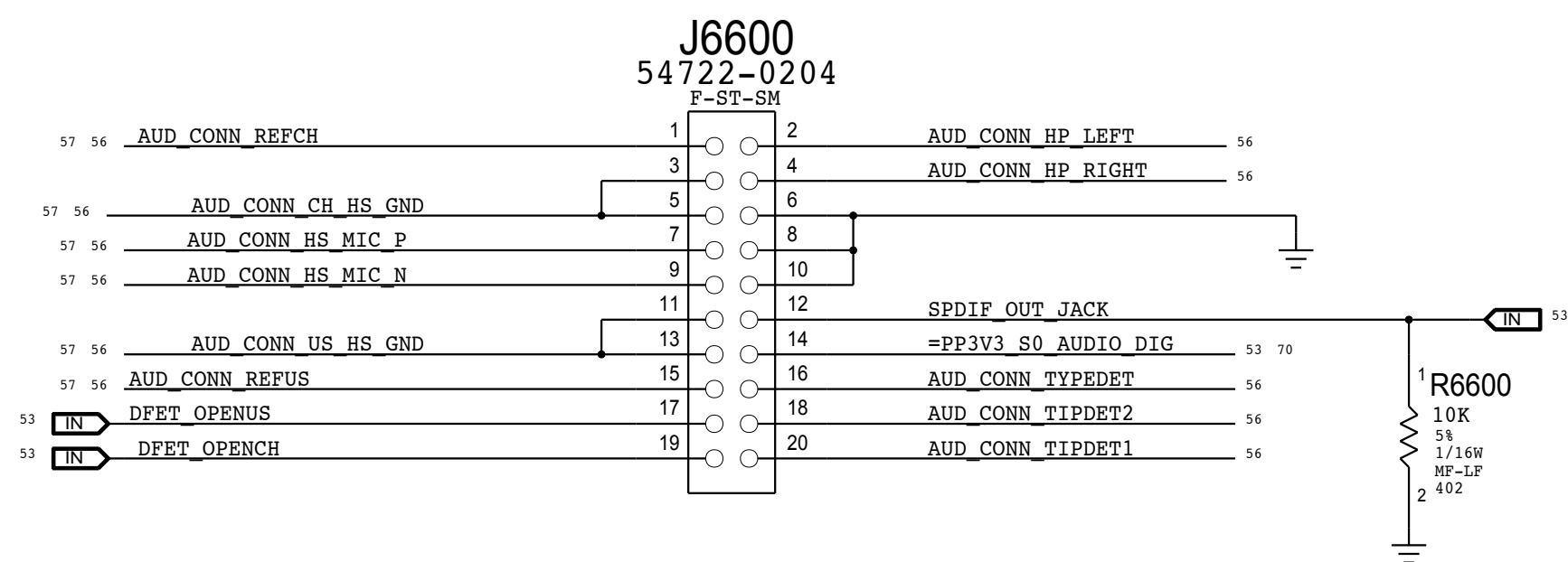
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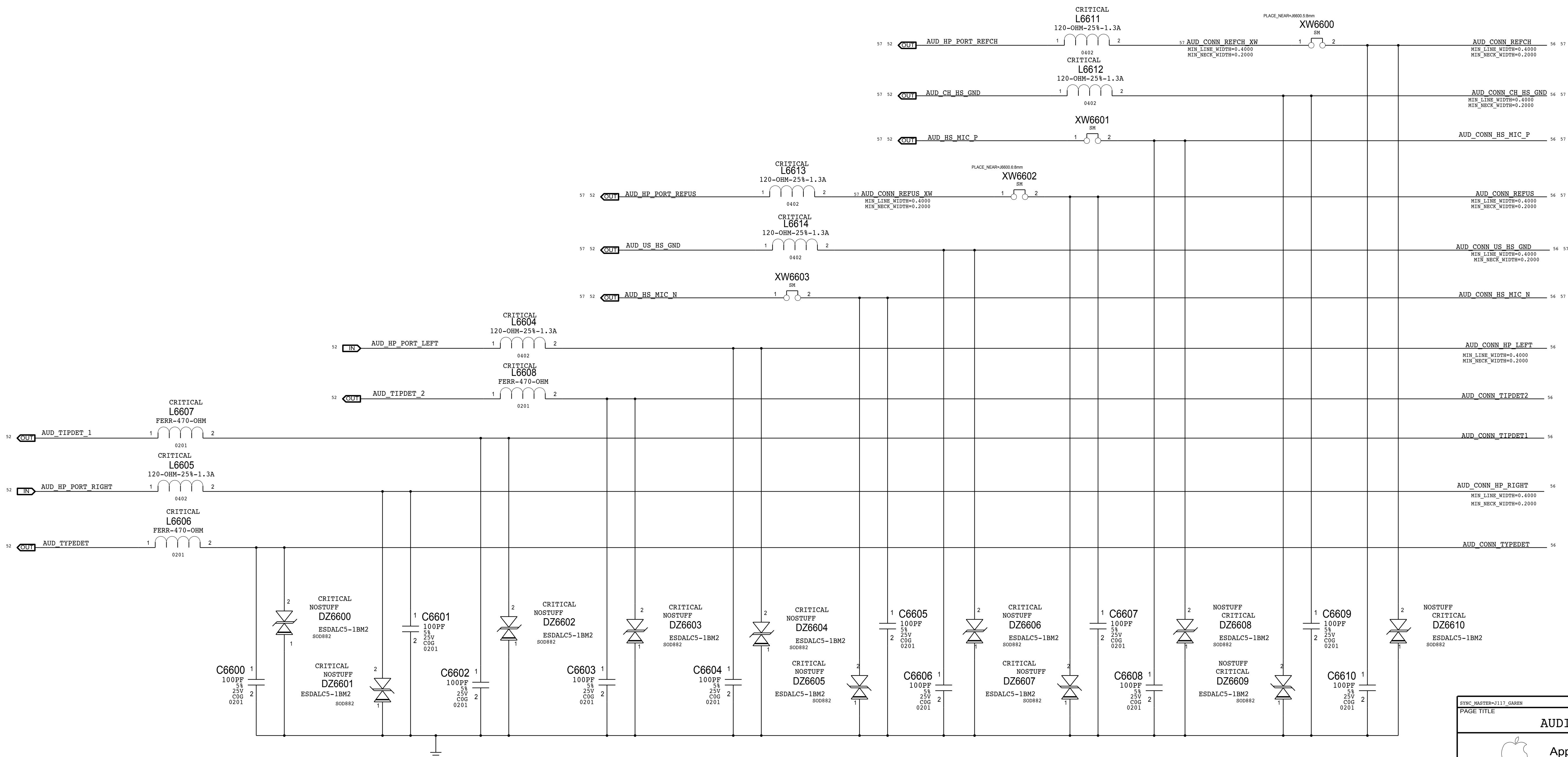
C

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A



Reversed Phases 1 & 2 and 5 & 6 on J6601 and J6602 Ref J16



ESD DIODES (DZ6600 -> DZ6610) NOW NOSTUFF - MAY BE ADDED FOLLOWING ESD TESTING

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SYNC_MASTER=J117_GAREN		SYNC_DATE=05/05/2014	
PAGE TITLE			
AUDIO: Jack Translators			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-1160		D
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	BRANCH		
	PAGE		
	66 OF 105		
SHEET			
56 OF 73			

D

D

CODEC OUTPUT SIGNAL PATHS						
FUNCTION	VOLUME/MUTE	CONVERTER	PIN COMPLEX	MAC SHDN	WIN SHDN	DET ASSIGNMENT
HP/LINE OUT	0X03 (3)	0X03 (3)	0X0A (10,D)	GPIO_2	GPIO_2	0X0A (DET D)
PRIMARY SPKRS (WFR)	0X04 (4)	0X04 (4)	0X0B (11)	MICBIAS	GPIO_3	N/A
SECONDARY SPKRS (TWT)	0X03 (3)	0X03 (3)	0X0A (10,V24)	N/A	N/A	N/A
SPDIF OUT	N/A	0X08 (8)	0x10 (16)	N/A	N/A	0X0D (DET B)
CODEC INPUT SIGNAL PATHS						
FUNCTION		CONVERTER	PIN COMPLEX	ENABLE/CONTROL		DET ASSIGNMENT
SPDIF IN		0X07 (7)	0x0F (15)	N/A		0X09 (DET A)
INTERNAL MIC ARRAY		0X06 (6)	0X0E (14,LEFT & RIGHT)	N/A		N/A
EXTERNAL MIC		0X05 (5)	0X12 (18,LEFT)			
		0X06 (6)	0X0D (13,V22,B,LEFT)	Lynx POINT GPIO 16		Lynx POINT GPIO 5 (RCVR INT)
						Lynx POINT GPIO 3 (PERIPH DET)
OTHER DETECT						
FUNCTION		CONVERTER	PIN COMPLEX	ENABLE/CONTROL		DET ASSIGNMENT
MULTIPLE SPKR VENDORS		N/A	N/A	N/A		0X0C (DET C)

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
AUDIO_ISO	*	0.1 MM	?
SPKROUT_ISO	*	0.2 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
AUDIODIFF	*	Y	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
AUDIODIFF_2MM	*	Y	0.2 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
SPKROUTDIFF	*	Y	0.6 MM	0.25 MM	10 MM	0.2 MM	0.2 MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
AUDIO	*	*	AUDIO_ISO
SPKROUT	*	*	SPKROUT_ISO

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
AUDIODIFF	*	AUDIODIFF
SPKROUTDIFF	*	SPKROUTDIFF

C

ELECTRICAL_CONSTRAINT_SET		NET_TYPE	
		PHYSICAL	SPACING
SPEAKER OUTPUTS			
1605	AUDIO_DIFFPAIR	SPKROUT_DIFF	SPKROUT AUD SPKR RWFR OUT P 53 56
1603	AUDIO_DIFFPAIR	SPKROUT_DIFF	SPKROUT AUD SPKR RWFR OUT M 53 56
1620	AUDIO_DIFFPAIR	SPKROUT_DIFF	SPKROUT AUD SPKR RWFR OUT L P 55
1619	AUDIO_DIFFPAIR	SPKROUT_DIFF	SPKROUT AUD SPKR RWFR OUT L M 55
1633	AUDIO_DIFFPAIR	SPKROUT_DIFF	SPKROUT AUD SPKR RTWT OUT P 53 56
1632	AUDIO_DIFFPAIR	SPKROUT_DIFF	SPKROUT AUD SPKR RTWT OUT M 53 56
1630	AUDIO_DIFFPAIR	SPKROUT_DIFF	SPKROUT AUD SPKR RTWT OUT L P 55
1631	AUDIO_DIFFPAIR	SPKROUT_DIFF	SPKROUT AUD SPKR RTWT OUT L M 55
1637	?	SPKROUT_DIFF	SPKROUT AUD SPKR LWFR OUT P 54 56
1636	?	SPKROUT_DIFF	SPKROUT AUD SPKR LWFR OUT M 54 56
1634		SPKROUT_DIFF	SPKROUT AUD SPKR LWFR OUT L P 54
1635		SPKROUT_DIFF	SPKROUT AUD SPKR LWFR OUT L M 54
1641	AUDIO_DIFFPAIR	SPKROUT_DIFF	SPKROUT AUD SPKR LTWT OUT P 54 56
1640	?	SPKROUT_DIFF	SPKROUT AUD SPKR LTWT OUT M 54 56
1638		SPKROUT_DIFF	SPKROUT AUD SPKR LTWT OUT L P 54
1639		SPKROUT_DIFF	SPKROUT AUD SPKR LTWT OUT L M 54
HEADSET MIC			
1634	AUDIO_DIFFPAIR	AUDIO_DIFF	AUDIO AUD_CONN_HS_MIC_P 56
1645	AUDIO_DIFFPAIR	AUDIO_DIFF	AUDIO AUD_CONN_HS_MIC_N 56
1646	AUDIO_DIFFPAIR	AUDIO_DIFF	AUDIO AUD_HS_MIC_P 52 56
1655	AUDIO_DIFFPAIR	AUDIO_DIFF	AUDIO AUD_HS_MIC_N 52 56
1624	AUDIO_DIFFPAIR	AUDIO_DIFF	AUDIO HS_MIC_P 52
1625	AUDIO_DIFFPAIR	AUDIO_DIFF	AUDIO HS_MIC_N 52
1632	AUDIO_DIFFPAIR	AUDIO_DIFF	AUDIO CODEC_HS_MIC_P 52
1637	AUDIO_DIFFPAIR	AUDIO_DIFF	AUDIO CODEC_HS_MIC_N 52
1634		AUDIO	AUDIO AUD_CONN_CH_HS_GND 56
1635		AUDIO	AUDIO AUD_CONN_US_HS_GND 56
1636		AUDIO	AUDIO AUD_CH_HS_GND 52 56
1635		AUDIO	AUDIO AUD_US_HS_GND 52 56
1634		AUDIO	AUDIO AUD_CONN_REFCH 56
1637		AUDIO	AUDIO AUD_CONN_REFUS 56
1631		AUDIO	AUDIO AUD_CONN_REFCH_KW 56
1630		AUDIO	AUDIO AUD_CONN_REFUS_KW 56
1632		AUDIO	AUDIO AUD_HP_PORT_REFCH 52 56
1632		AUDIO	AUDIO AUD_HP_PORT_REFUS 52 56

B

ELECTRICAL_CONSTRAINT_SET		NET_TYPE	
		PHYSICAL	SPACING
CODEC SCL & SDA			
1635			CODEC_SCL 53 54 55
1655			CODEC_SDA 53 54 55
TDM			
1626		?	MCLKA_SPKR_LEFT 53 54
1631		?	SCLKA_SPKR_LEFT 53 54
1625		?	LRCCLKB_SPKR_LEFT 53 54
1653		?	SDOUTA_SPKR_LEFT 53 54
1632			MCLKA_SPKR_LEFT_R 53
1631			SCLKA_SPKR_LEFT_R 53
1632			LRCCLKB_SPKR_LEFT_R 53
1638			SDOUTA_SPKR_LEFT_R 53
1634		?	MCLKB_SPKR_RIGHT 53 55
1655		?	SCLKB_SPKR_RIGHT 53 55
1655		?	LRCCLKB_SPKR_RIGHT 53 55
1634		?	SDOUTB_SPKR_RIGHT 53 55
1632			MCLKB_SPKR_RIGHT_R 53
1632			SCLKB_SPKR_RIGHT_R 53
1635			LRCCLKB_SPKR_RIGHT_R 53
1632			SDOUTB_SPKR_RIGHT_R 53

A

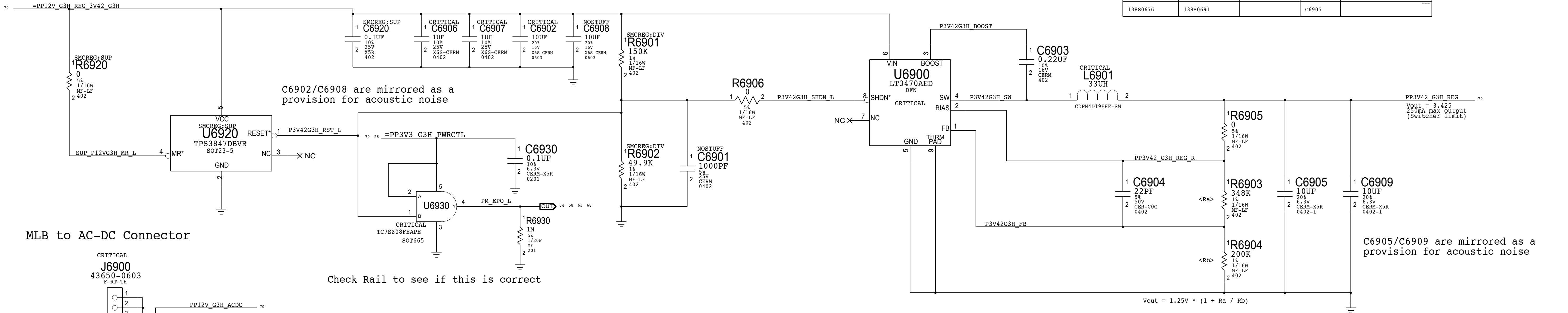
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SYNC_MASTER=J117 ANDRES		SYNC_DATE=03/24/2014	
PAGE TITLE			
AUDIO: Speaker ID			
	Apple Inc.	DRAWING NUMBER	051-1160
		REVISION	2.0.0
		BRANCH	
		PAGE	
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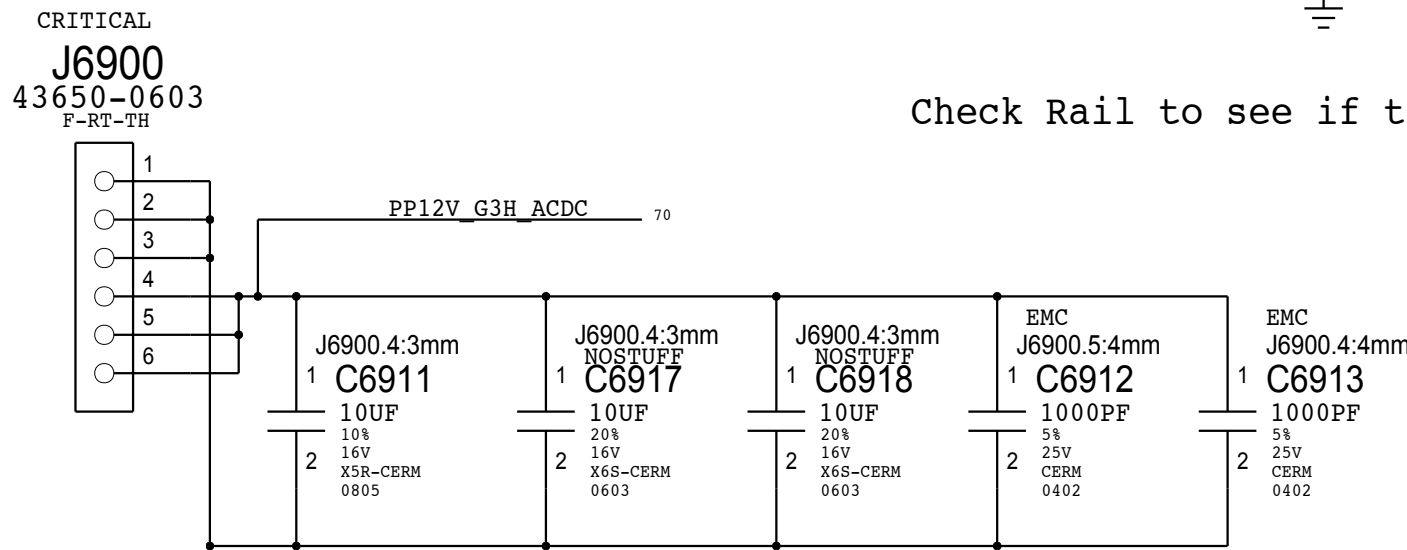
3.425V "G3Hot" Regulator

Switching freq: 409 kHz = $\frac{13.5}{L6901}$

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
13880676	13880691		C6905	

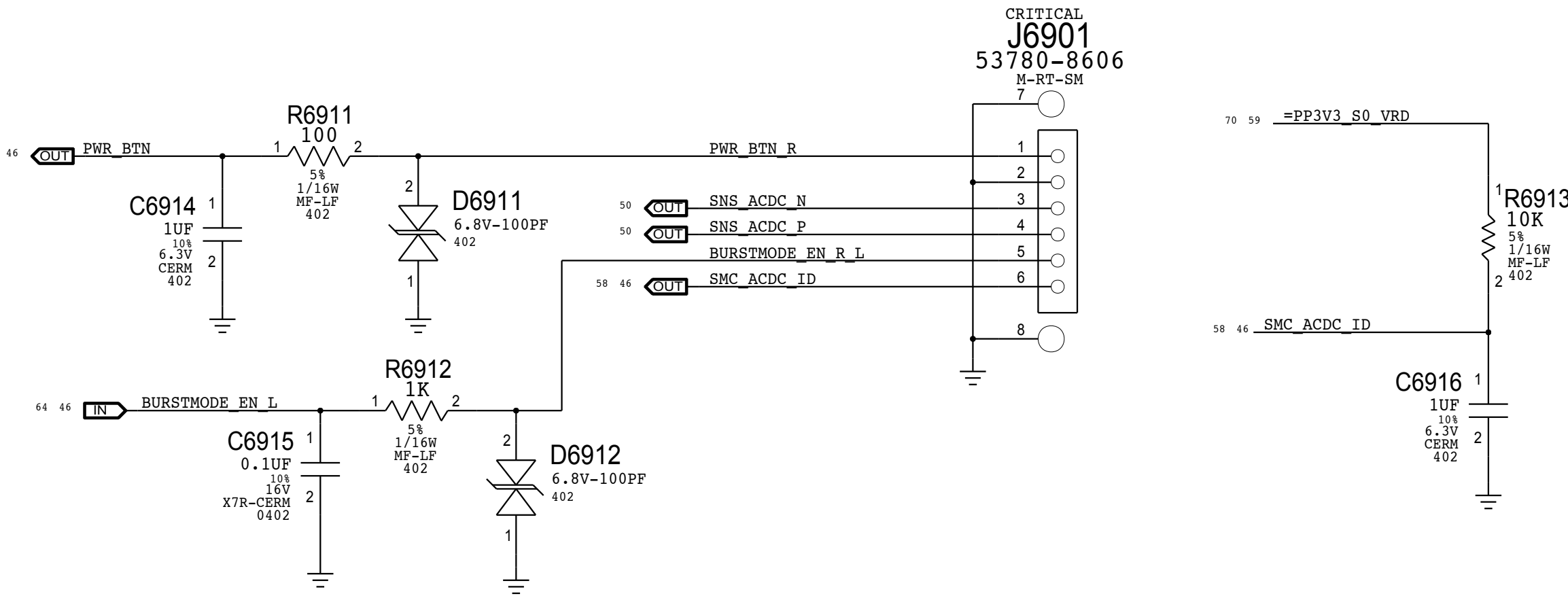


MLB to AC-DC Connector

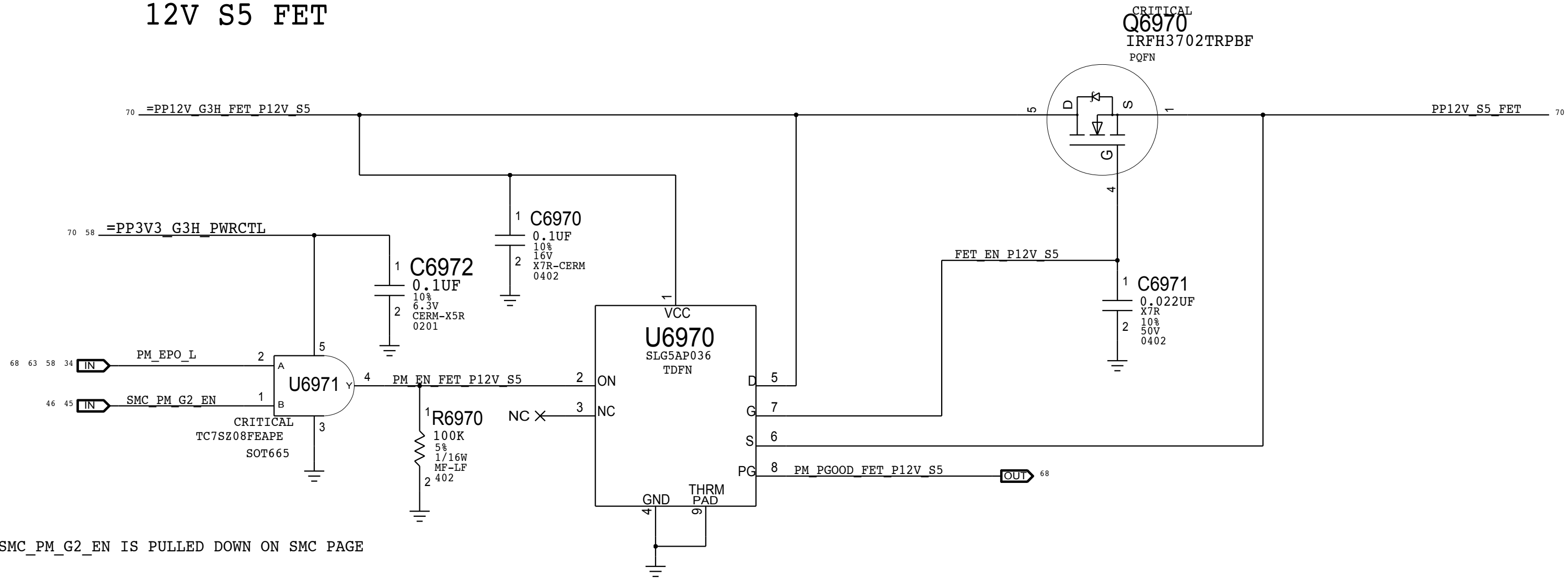


Check Rail to see if this is correct

MLB to AC-DC Supplemental Signal Connector



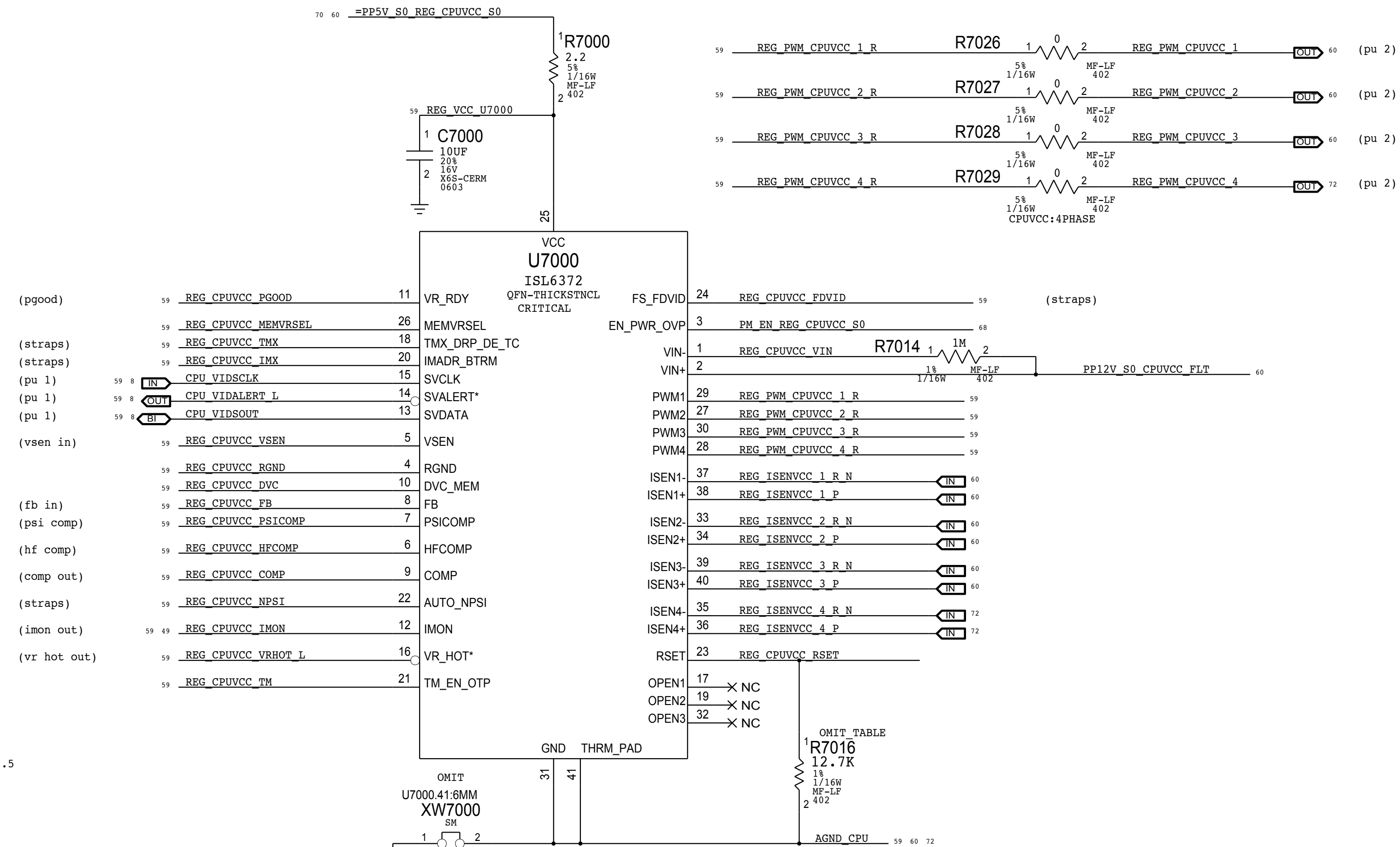
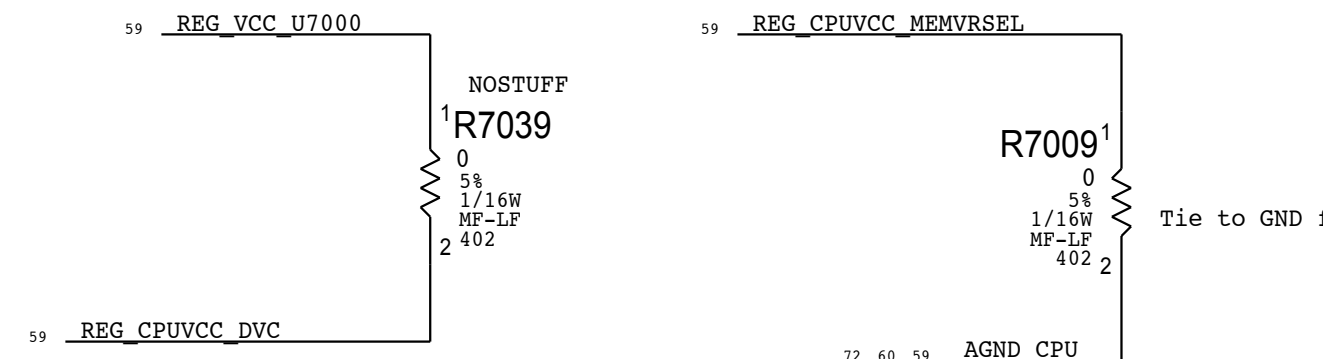
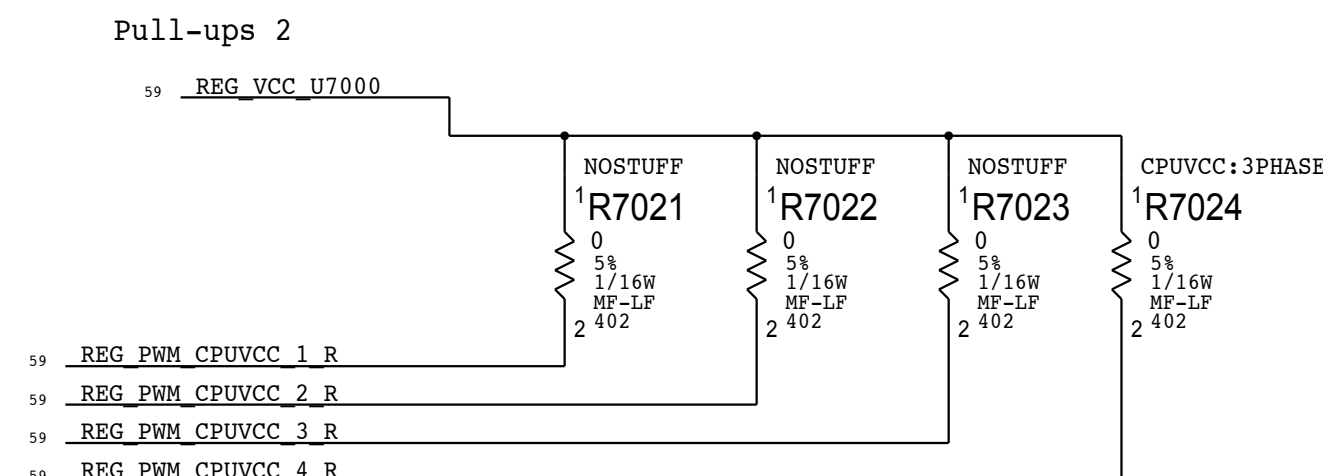
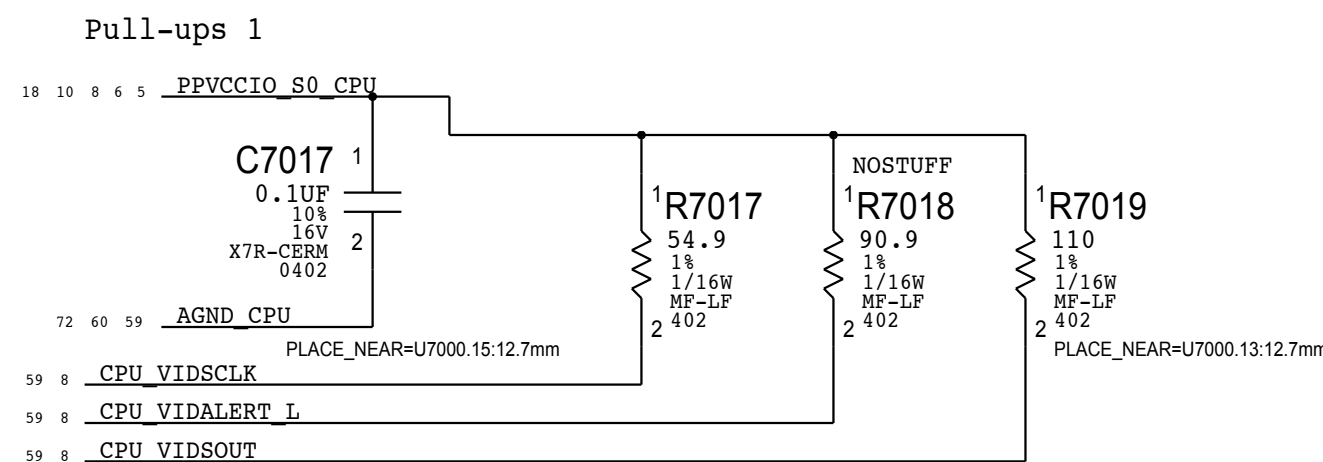
12V S5 FET



SYNC_MASTER=J117 ANDRES		SYNC_DATE=03/24/2014	
PAGE TITLE			
PLATFORM POWER: Connectors / VReg G3Hot			
 Apple Inc.	DRAWING NUMBER	051-1160	SIZE D
	REVISION	2.0.0	
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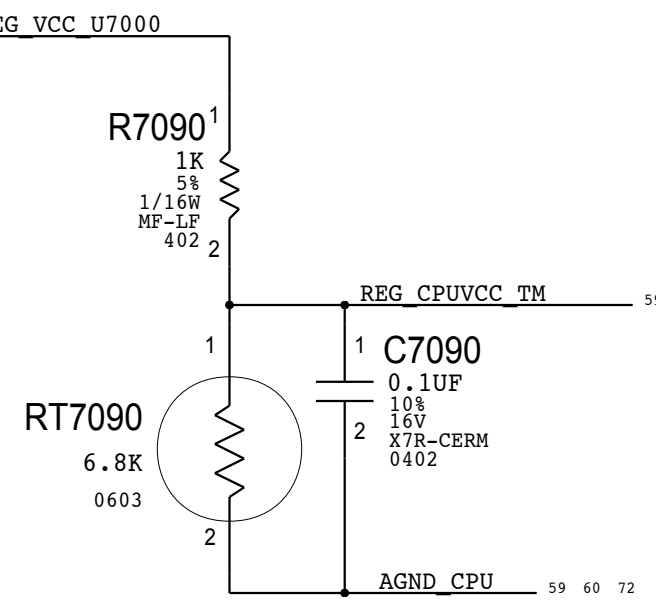
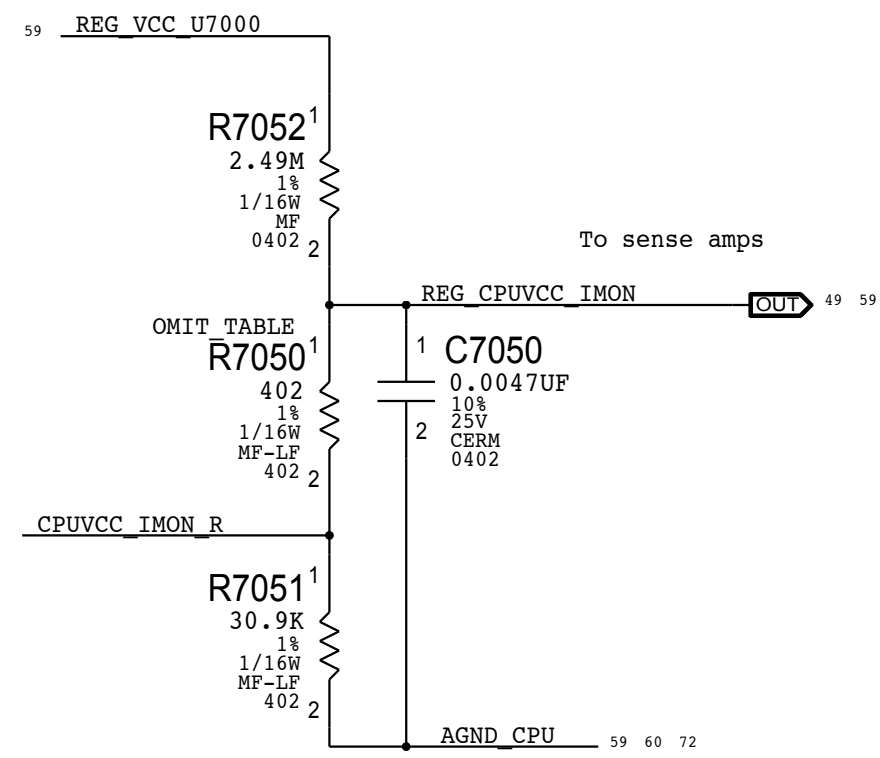
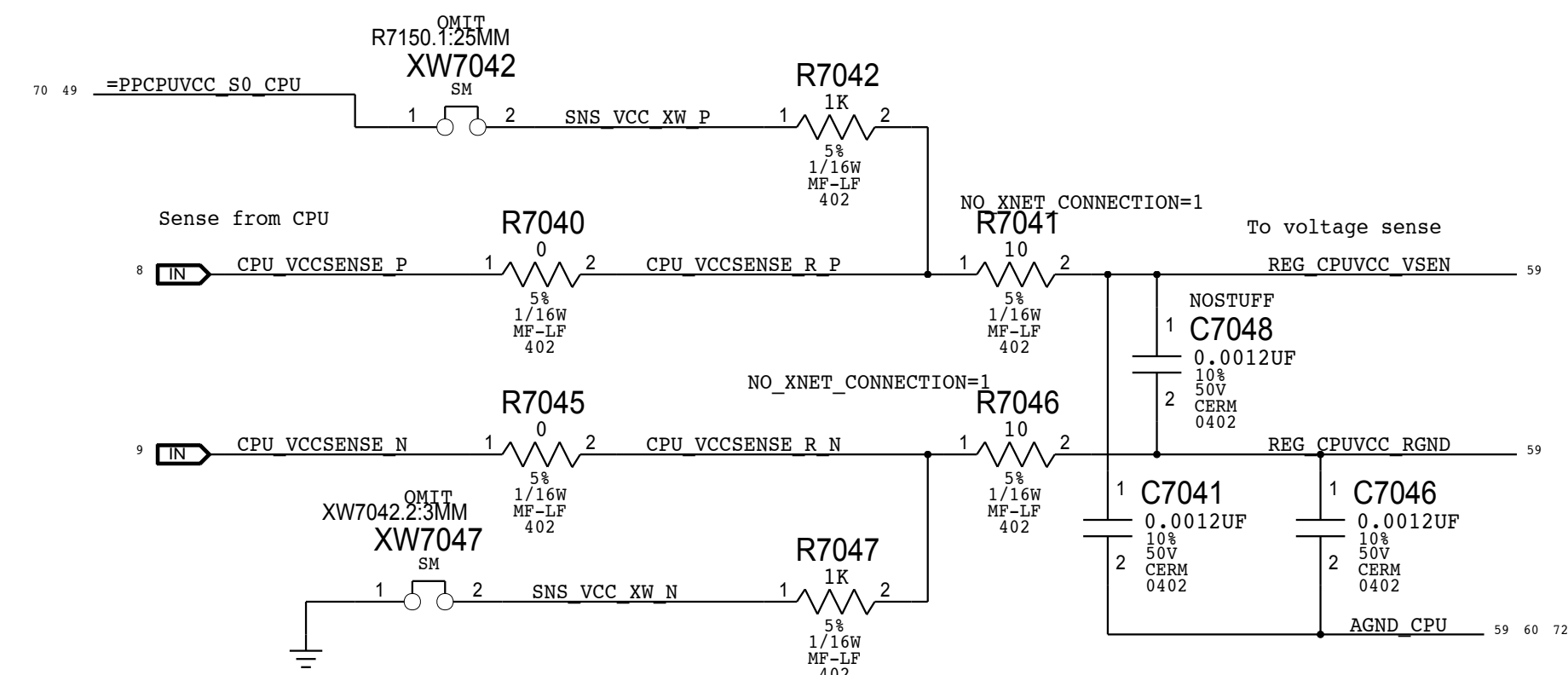
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QQ&WX: 82669515

Switching freq: 403 kHz = $\frac{5 \text{ E10}}{\text{R7003}}$

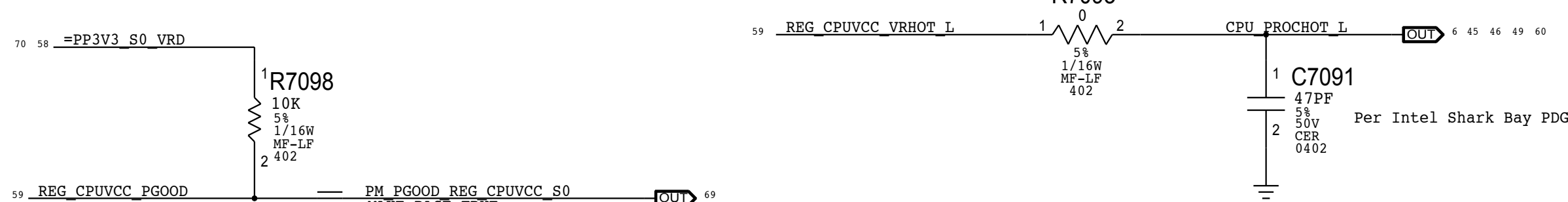
[illegible]

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
114S0324	1	RES, 12.7K, 402	R7016	CPUVCC:3PHASE
114S0316	1	RES, 10.2K, 402	R7016	CPUVCC:4PHASE

Temp measurement

[illegible]

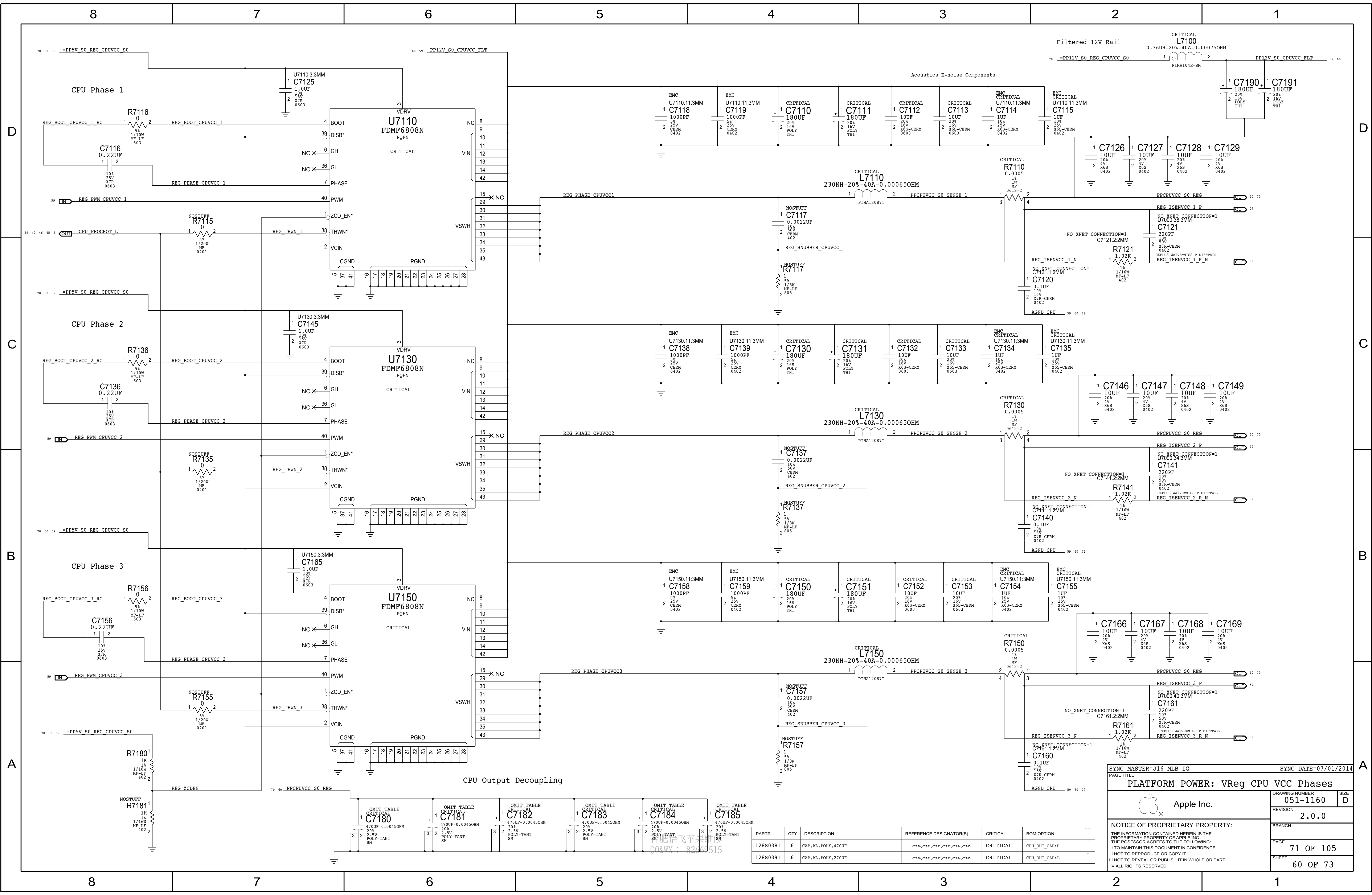
VRHot to ProcHot



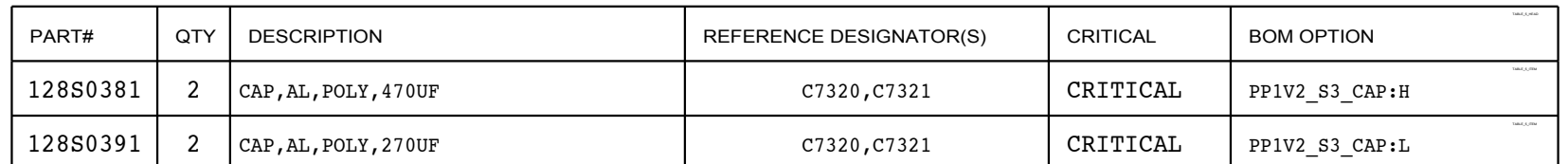
J16: 3PHASE		J17: 4PHASE		
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
11880311	1	RES, 340K, 201	R7001	CPUVCC: 3PHASE
11880116	1	RES, 158K, 201	R7001	CPUVCC: 4PHASE
11860575	1	RES, 95.3K, 201	R7002	CPUVCC: 3PHASE
11880380	1	RES, 44.2K, 201	R7002	CPUVCC: 4PHASE
11480206	1	RES, 750 OHM, 402	R7032	CPUVCC: 3PHASE
11480210	1	RES, 825 OHM, 402	R7032	CPUVCC: 4PHASE
11480208	1	RES, 787 OHM, 402	R7034	CPUVCC: 3PHASE
11480189	1	RES, 499 OHM, 402	R7034	CPUVCC: 4PHASE
11480179	1	RES, 402 OHM, 402	R7050	CPUVCC: 3PHASE
11480184	1	RES, 453 OHM, 402	R7050	CPUVCC: 4PHASE

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SYNC MASTER=J16 MLB IG		SYNC DATE=07/01/2014	
PAGE TITLE			
PLATFORM POWER: VReg CPU VCC Cntl			
 Apple Inc.		DRAWING NUMBER 051-1160	
		SIZE D	
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		BRANCH	
		PAGE 70 OF 105	
		SHEET 59 OF 73	



Switching freq: 500 kHz



Vout = 1.794V
Max Current = 1.8A
Freq = 1 MHz

C7355 and C7357 must be mirrored.

SYNC MASTER=J70 ROSSANA		SYNC DATE=09/12/2013	
PAGE TITLE			
PLATFORM POWER: VReg VDDQ S3 / 1.8V S3			
	Apple Inc.		DRAWING NUMBER 051-1160
			REVISION 2.0.0
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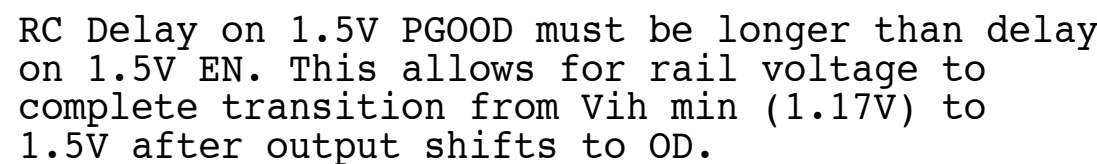
D

C

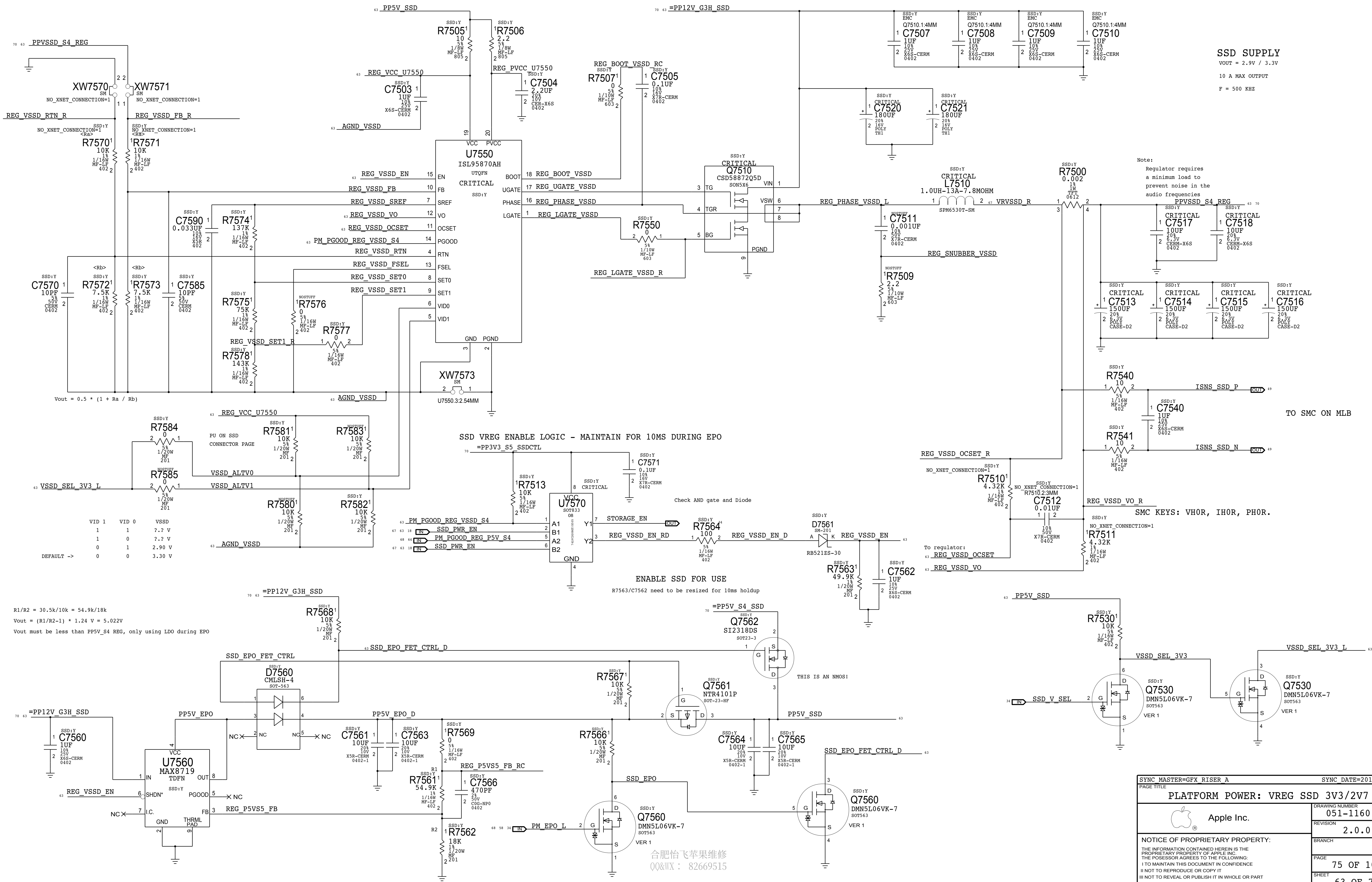
B



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$$V_{ih} = 0.65 \times V_{CC} = 1.17V.$$

SYNC MASTER=J117 ANDRES		SYNC DATE=03/24/2014	
PAGE TITLE			
PLATFORM POWER: VREG 1V05 S0 / 1V5 S0			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-1160	D
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		BRANCH	
		PAGE	74 OF 105
		SHEET	62 OF 73



OC trip point:	12.5 A =	$\frac{R7618 * 10 \text{ E-6}}{\text{DCR(L7610)}}$
Switching freq:	356 kHz =	$\frac{1}{170 \text{ E-12} * R7633}$

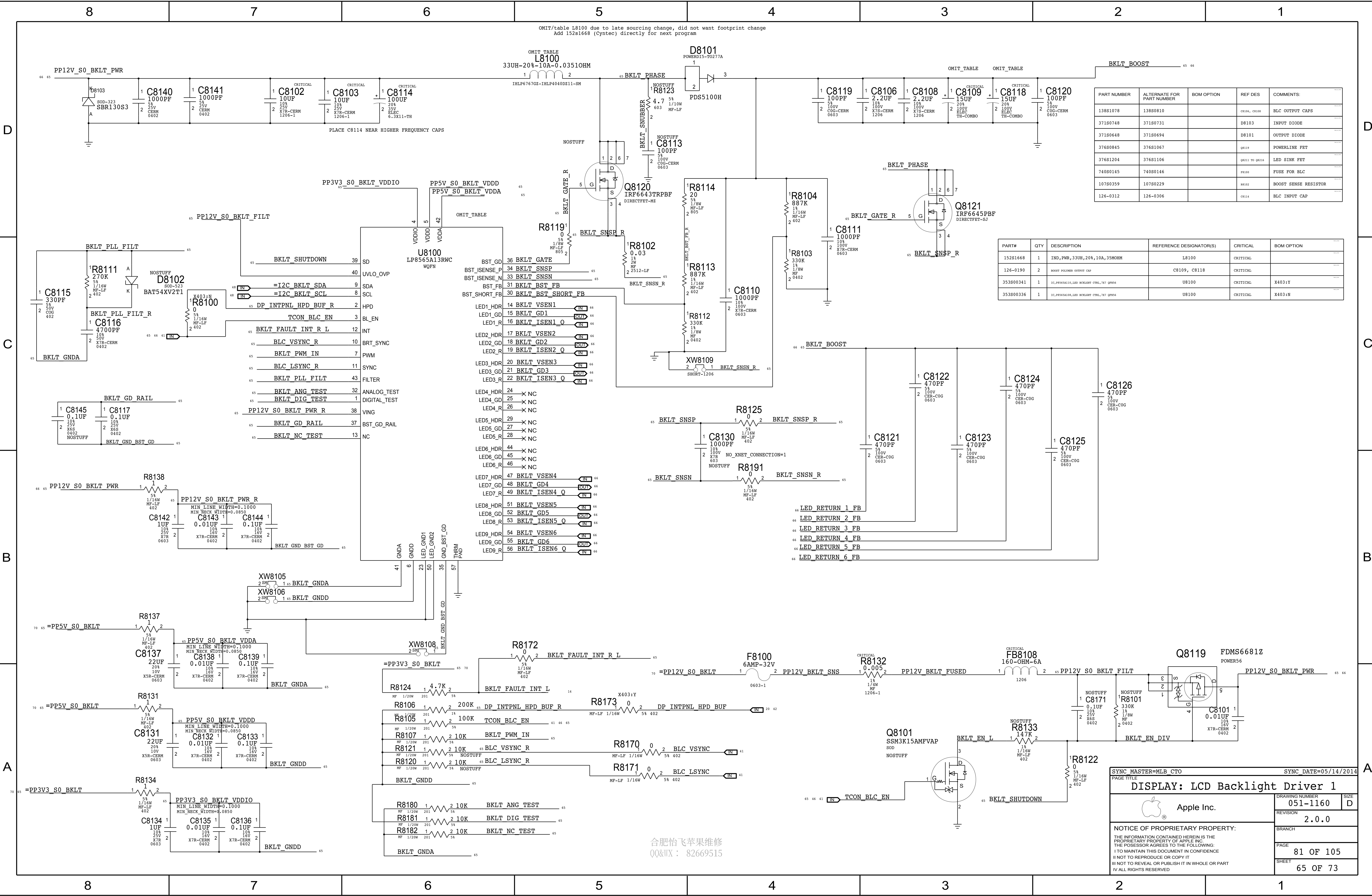
OC trip point:	14.1 A =	$\frac{R7658 * 10 \text{ E-6}}{\text{DCR(L7650)}}$
Switching freq:	356 kHz =	$\frac{1}{170 \text{ E-12} * R7673}$



BURSTMODE_EN_L	Vreg Mode
0	PWM
1	DCM

This circuit toggles the Vreg between PWM and ultrasonic DCM modes based on load requirements.

SYNC MASTER=J117 ANDRES PAGE TITLE		SYNC DATE=03/24/2014	
PLATFORM POWER: VReg 3.3V S5/5V S4			
 Apple Inc.		DRAWING NUMBER 051-1160	SIZE D
		REVISION 2.0.0	
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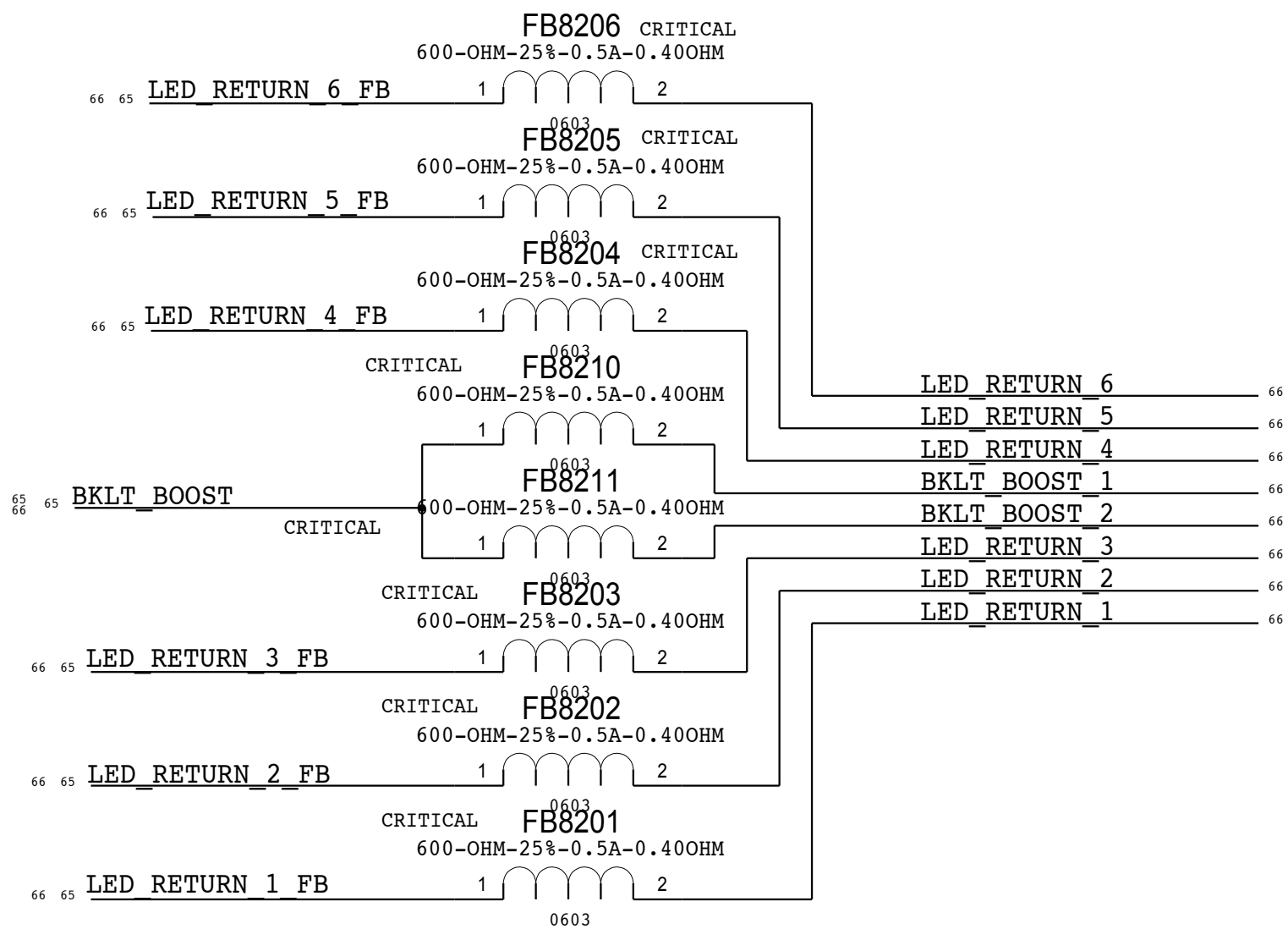
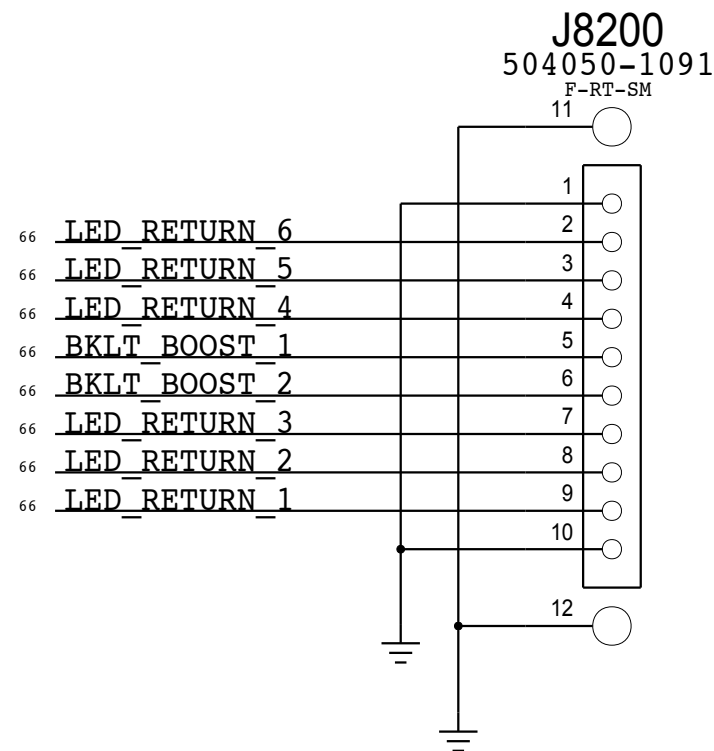
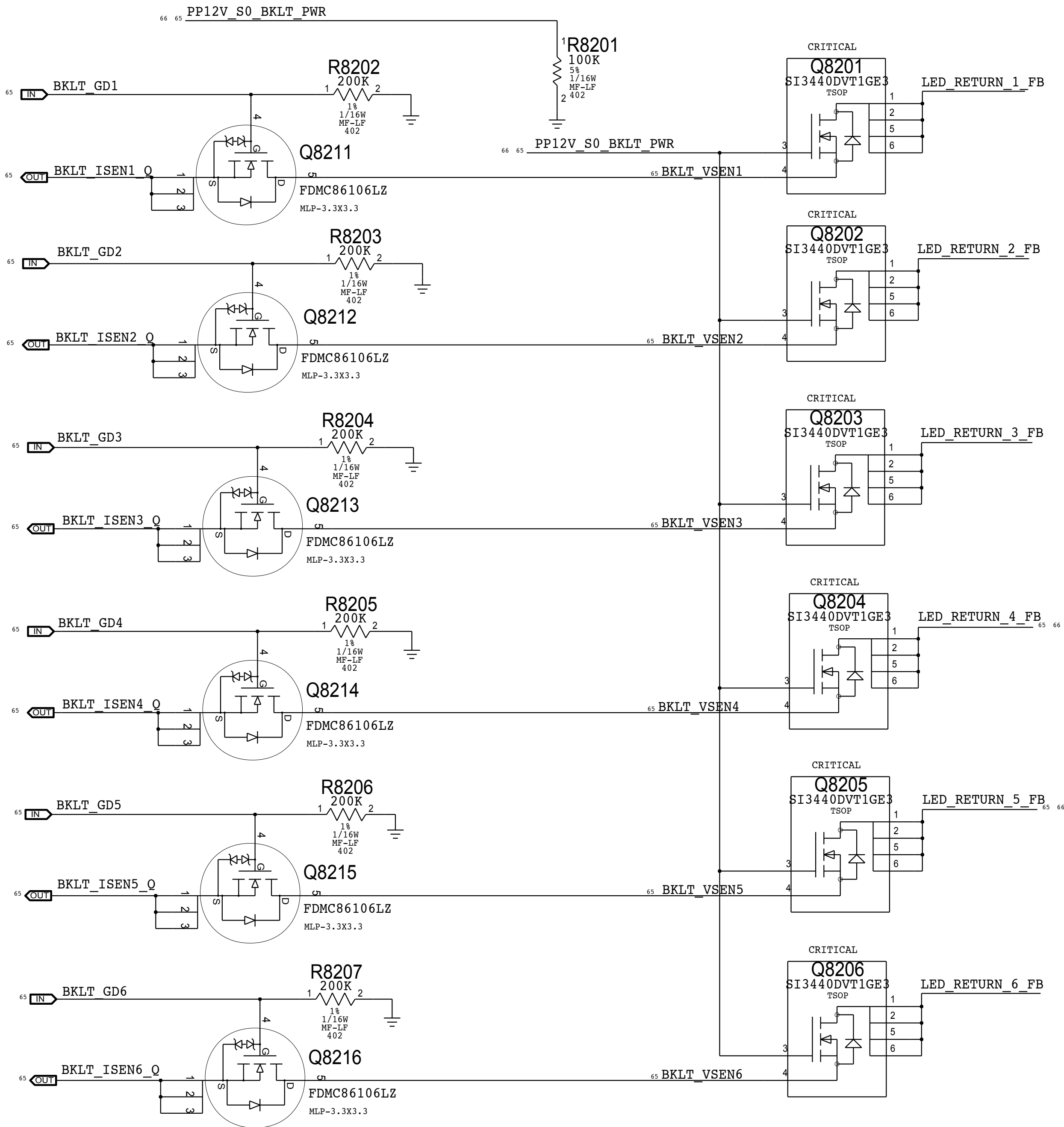
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
136S1078	136S0810		CB106, CB108	BLC OUTPUT CAPS
371S0748	371S0731		D8103	INPUT DIODE
371S0648	371S0694		D8101	OUTPUT DIODE
376S0845	376S1067		QB119	POWERLINE FET
376S1204	376S1106		QB211 TO QB214	LED SINK FET
740S0145	740S0146		FB100	FUSE FOR BLC
107S0359	107S0229		RB102	BOOST SENSE RESISTOR
126-0312	126-0306		CB114	BLC INPUT CAP

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S1668	1	IND, PWR, 33UH, 20%, 10A, 35MORM	L8100	CRITICAL	
126-0190	2	BOOST POLYMER OUTPUT CAP	CB109, CB118	CRITICAL	
353S00341	1	IC, PPSAS110, LED BACKLIGHT CTRL, 757 QP056	U8100	CRITICAL	X403:Y
353S00336	1	IC, PPSAS108, LED BACKLIGHT CTRL, 757 QP056	U8100	CRITICAL	X403:N

SYNC_MASTER=MLB CTO		SYNC_DATE=05/14/2014	
PAGE TITLE			
DISPLAY: LCD Backlight Driver 1			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-1160	D
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		PAGE	81 OF 105
		SHEET	65 OF 73

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PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
37681296	37681073		ALL	Short Protection FET
15580831	15580797		ALL	FB8201 TO FB8211



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SYNC MASTER=MLB CTO		SYNC DATE=05/14/2014	
PAGE TITLE			
DISPLAY: LCD Backlight Driver 2			
 Apple Inc.	DRAWING NUMBER	051-1160	SIZE D
	REVISION	2.0.0	
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		SHEET	66 OF 73

D

C

B

A

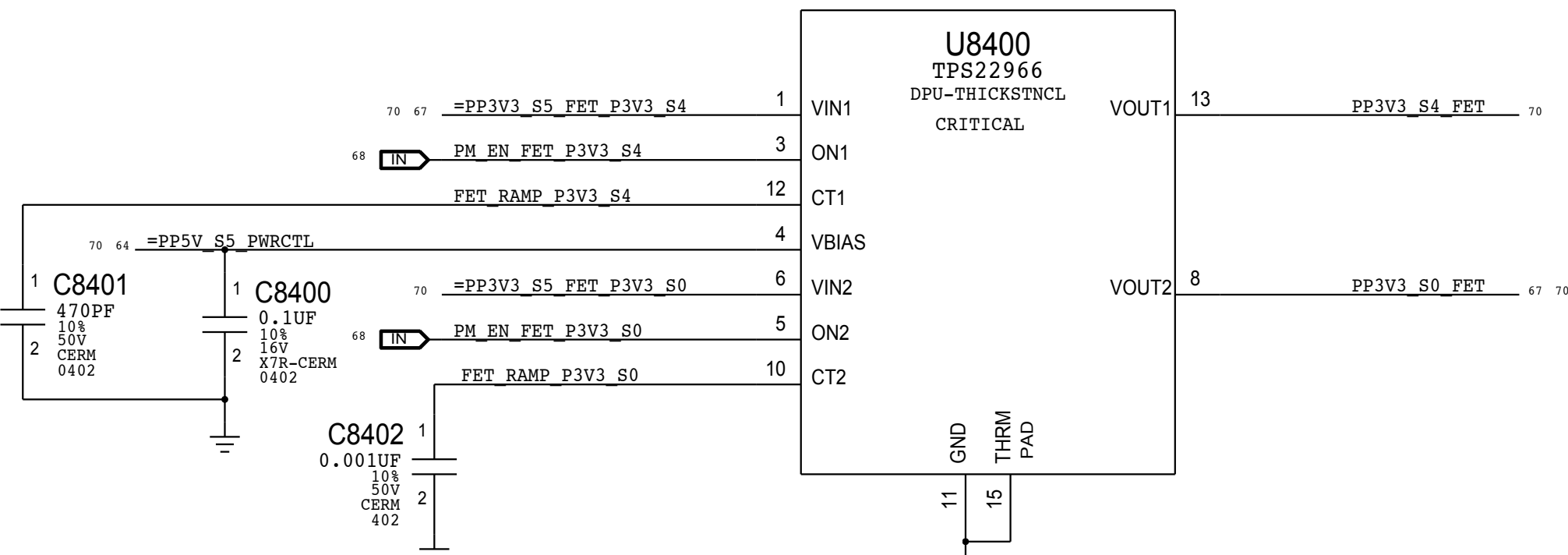
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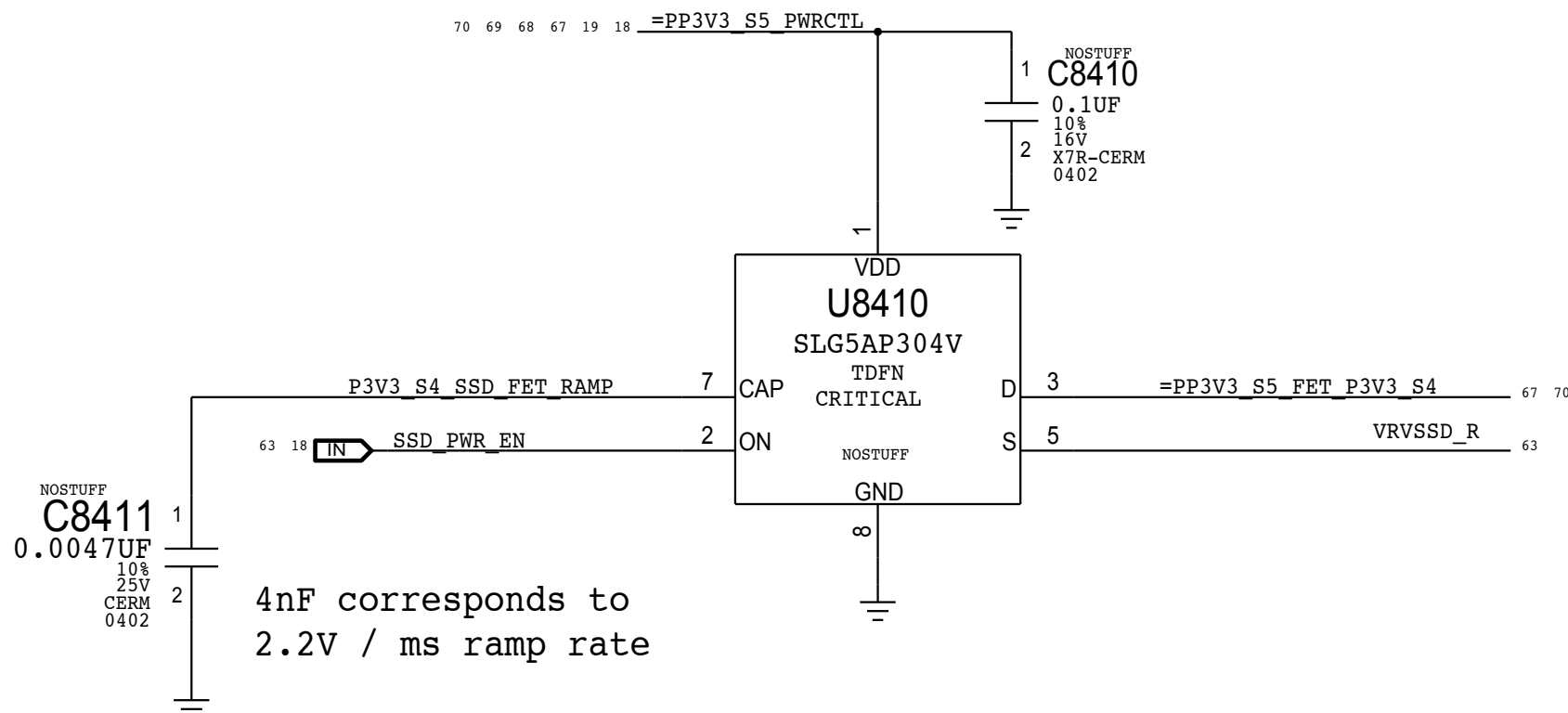
B

A

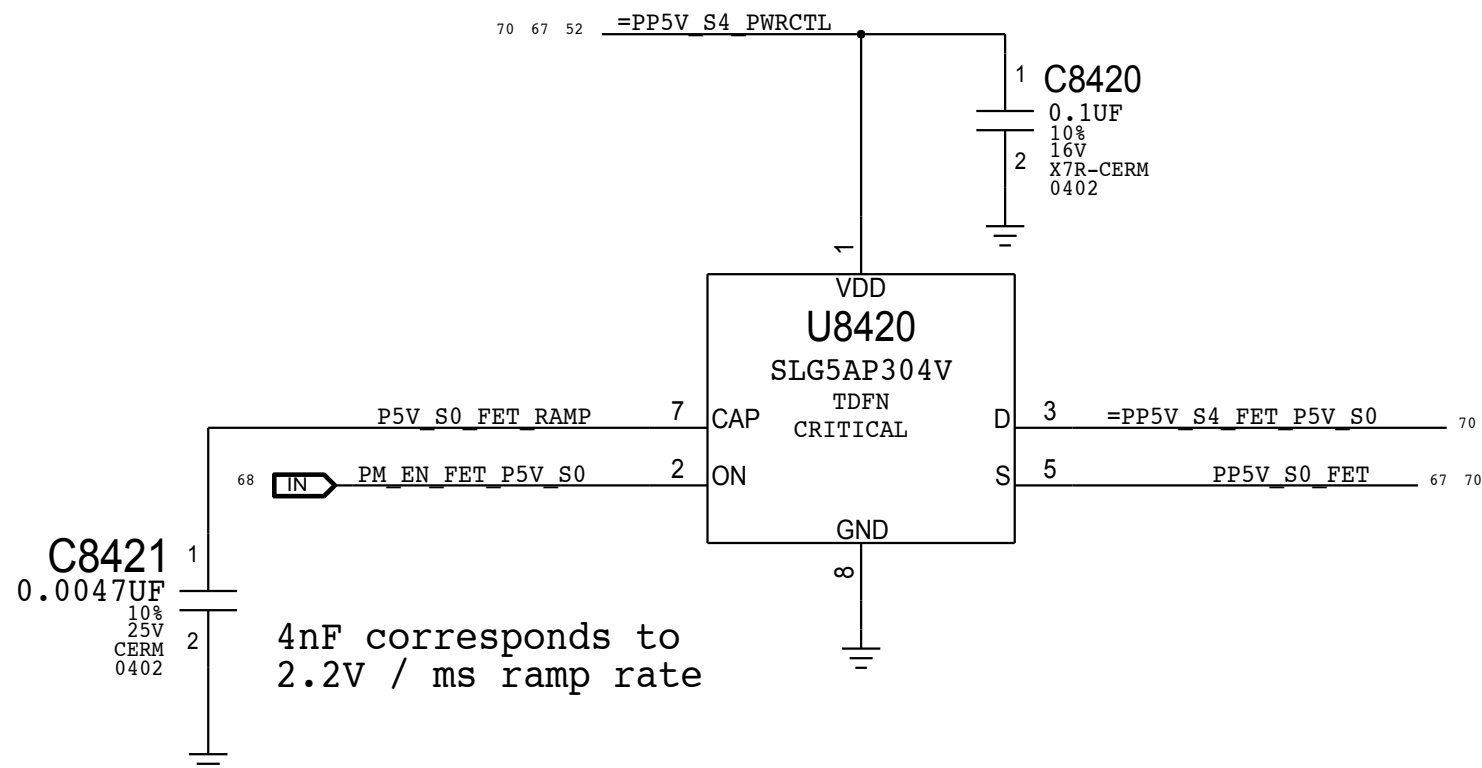
3.3V S4/S0 FET



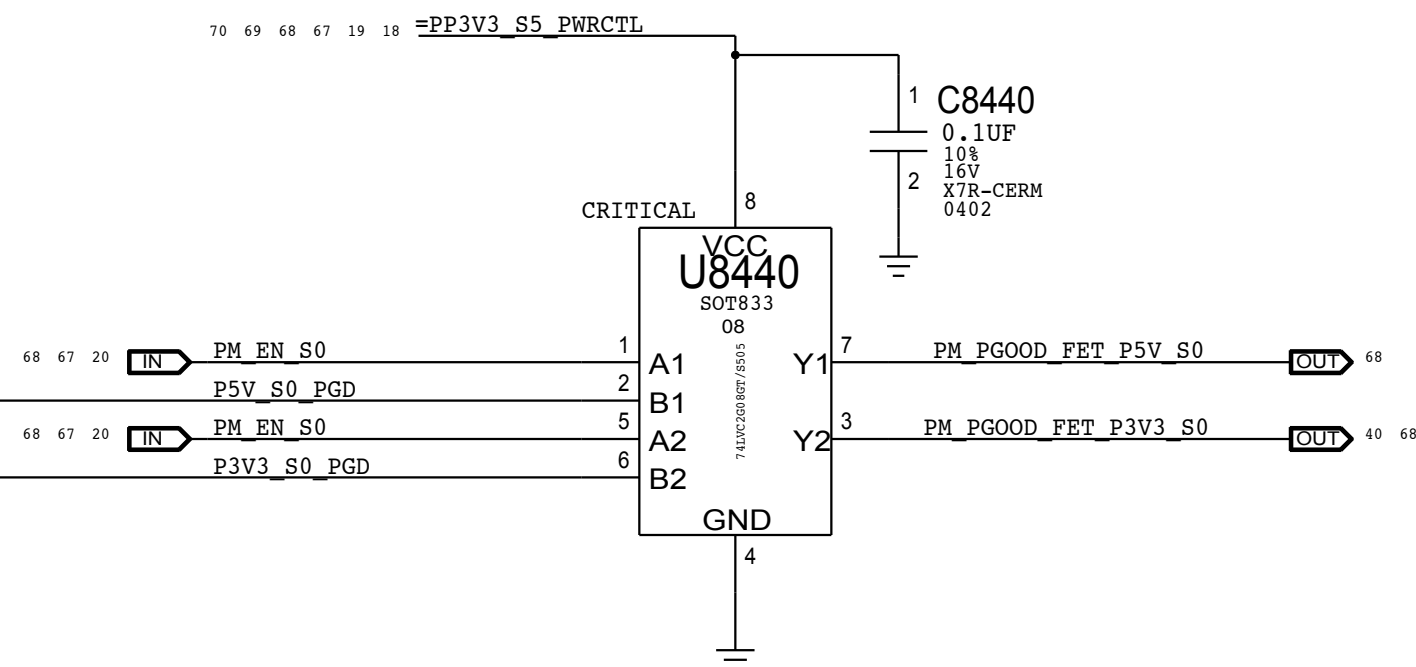
3V3 S4 SSD



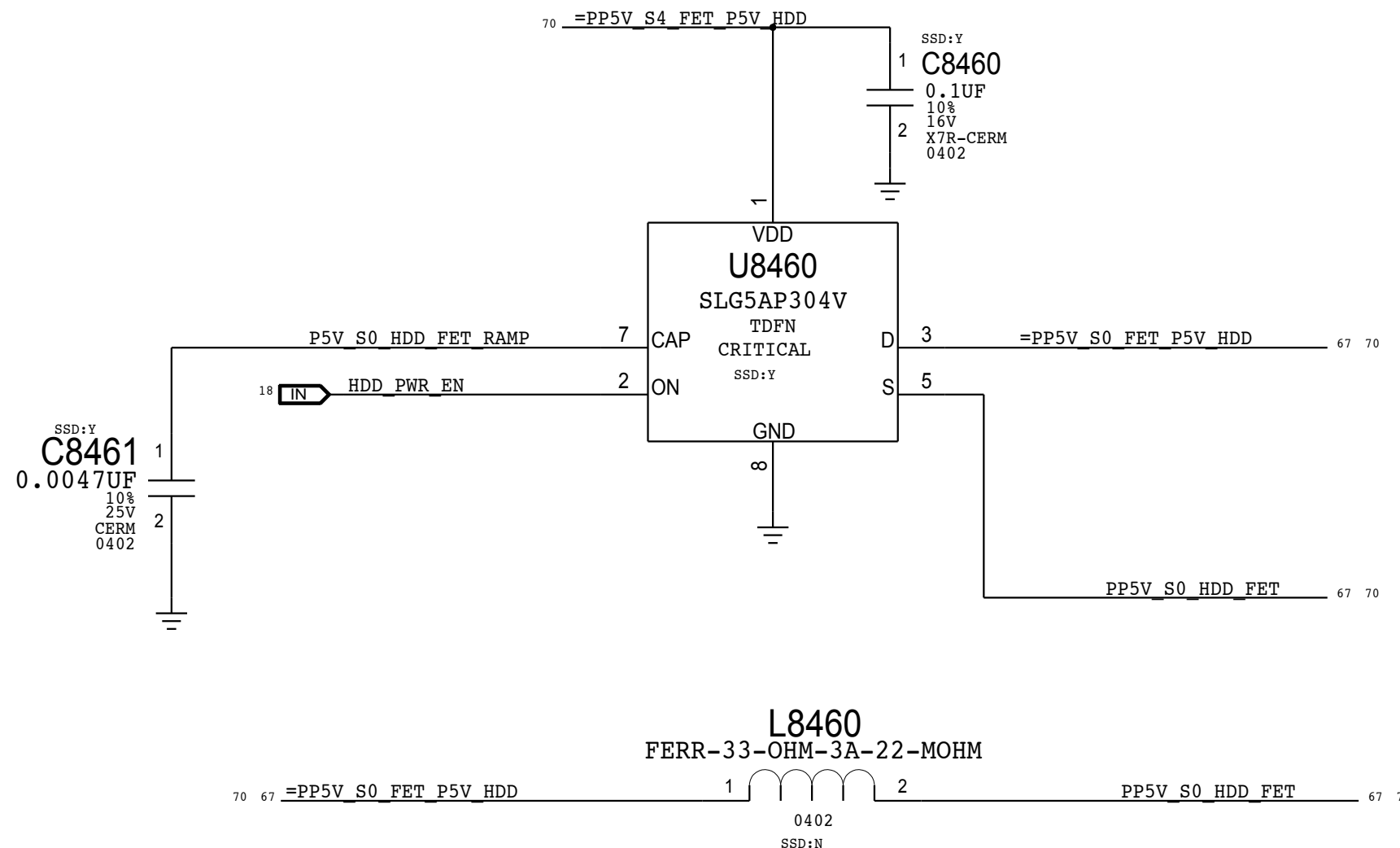
5V S0 FET



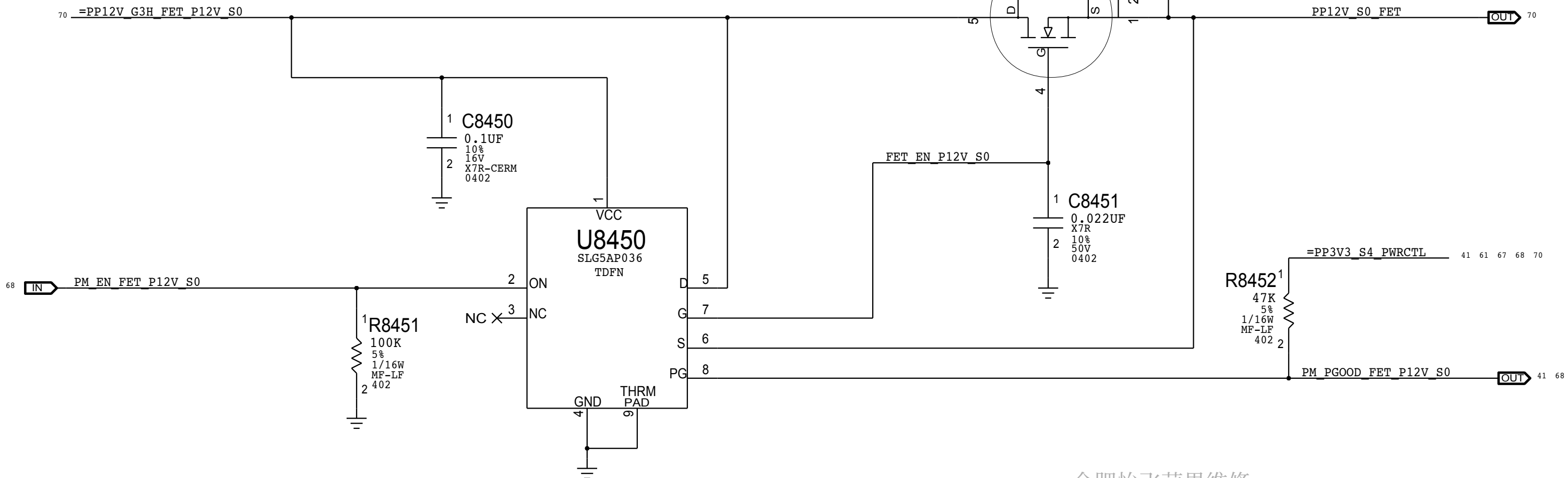
5V / 3V3 S0 PGOODs



5V HDD FET



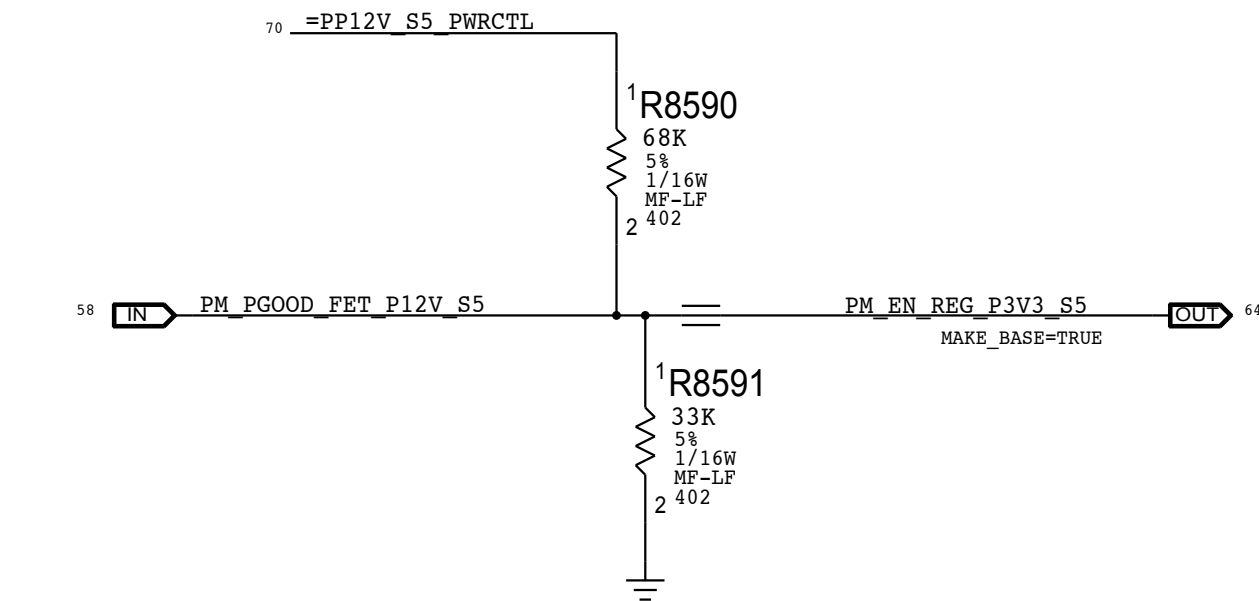
12V S0 FET



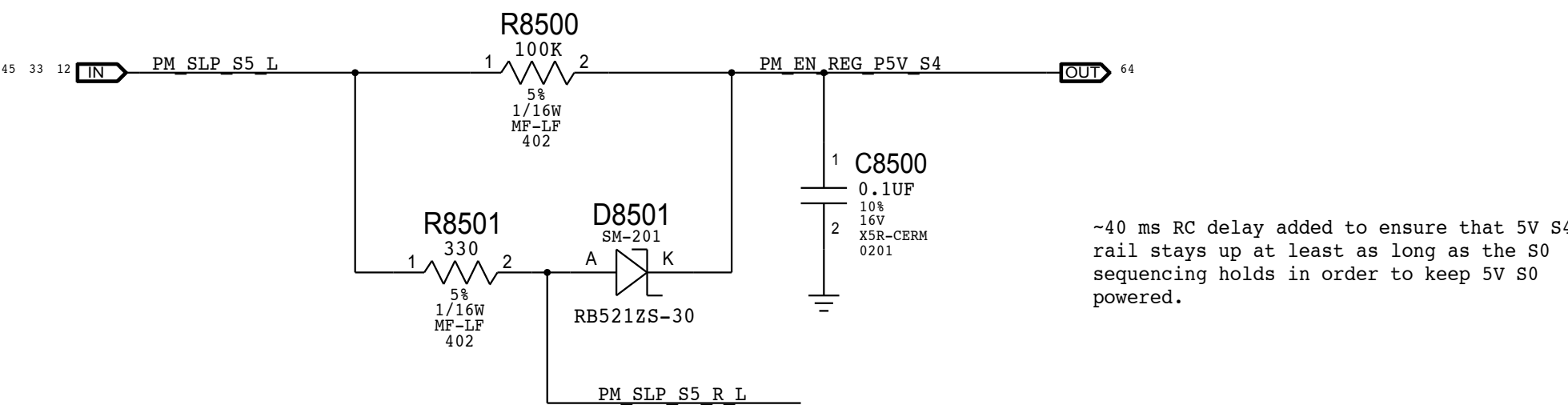
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SYNC_MASTER=J16 MLB_IG		SYNC_DATE=08/27/2013	
PAGE TITLE			
PLATFORM POWER: PM FETs/LDOs			
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	REVISION	2.0.0	
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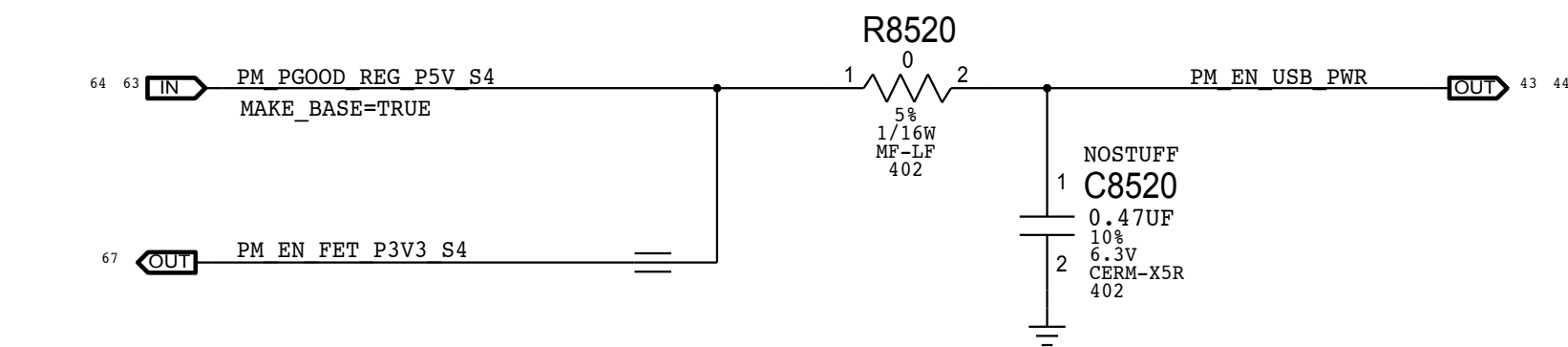
S5 Enable



S4 Enables



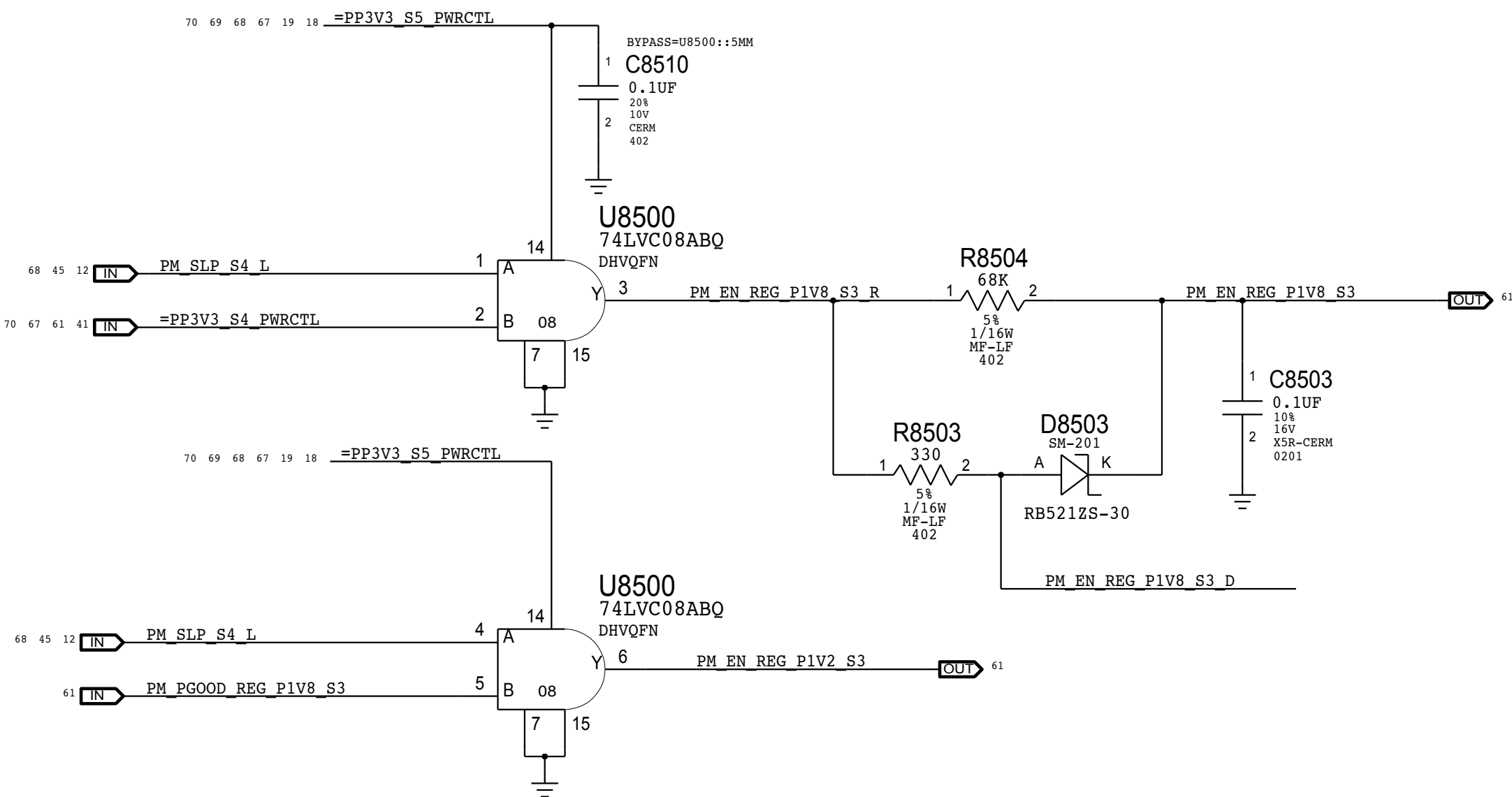
S4 USB Enable



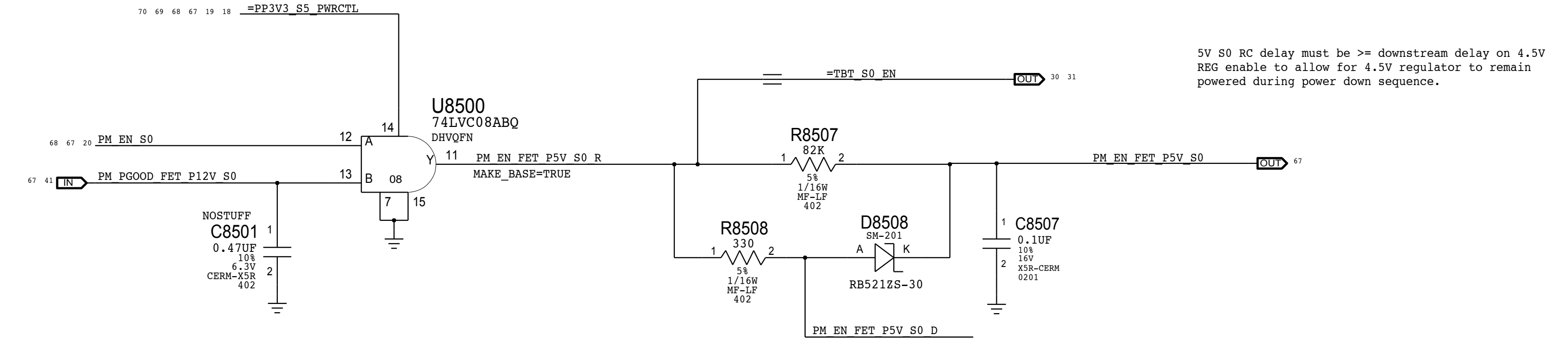
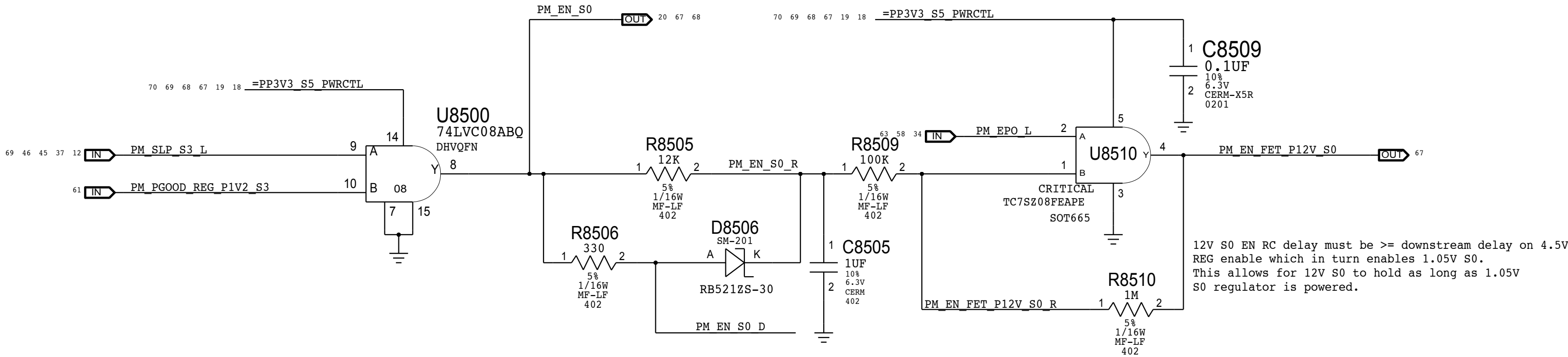
S4 TBT Port Enable



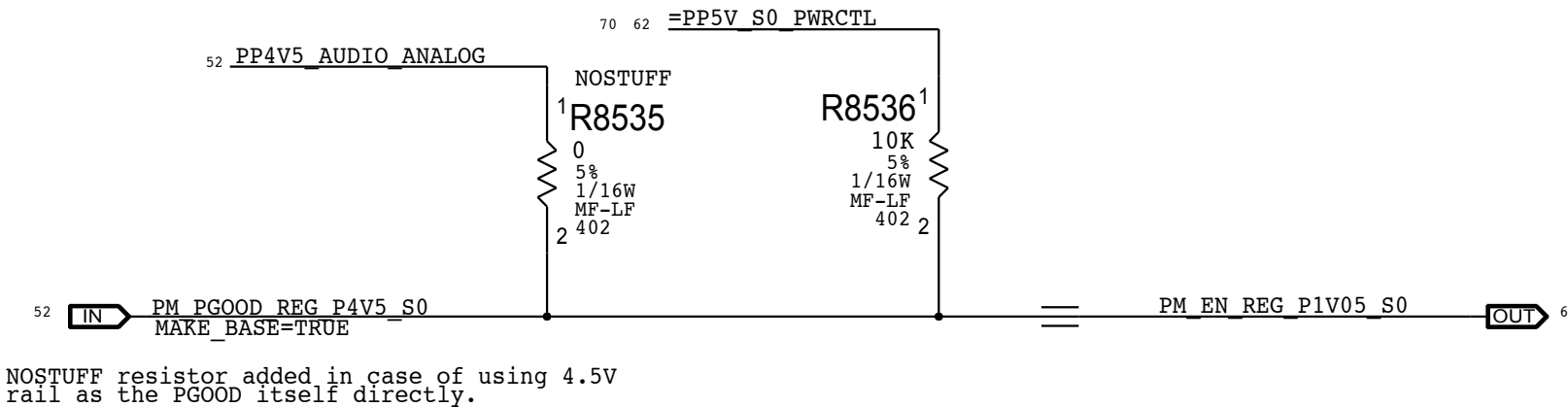
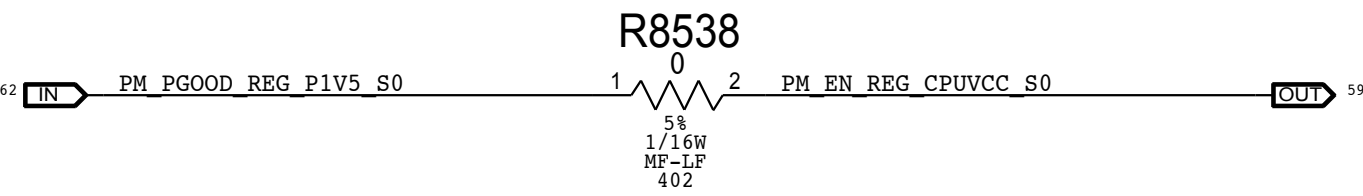
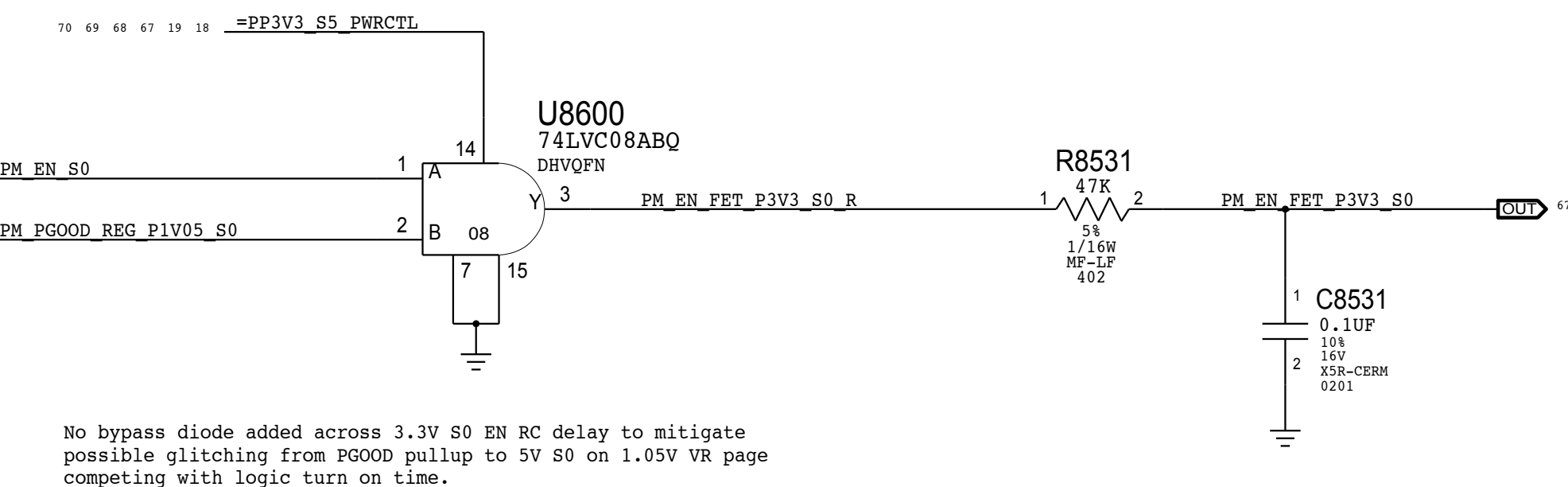
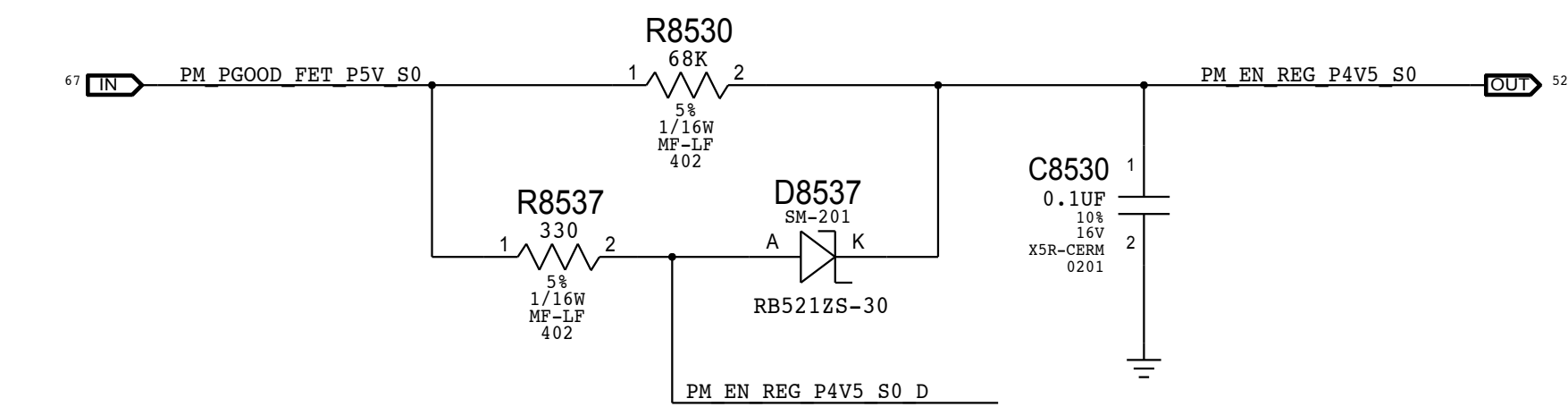
S3 Enables



S0 Enables



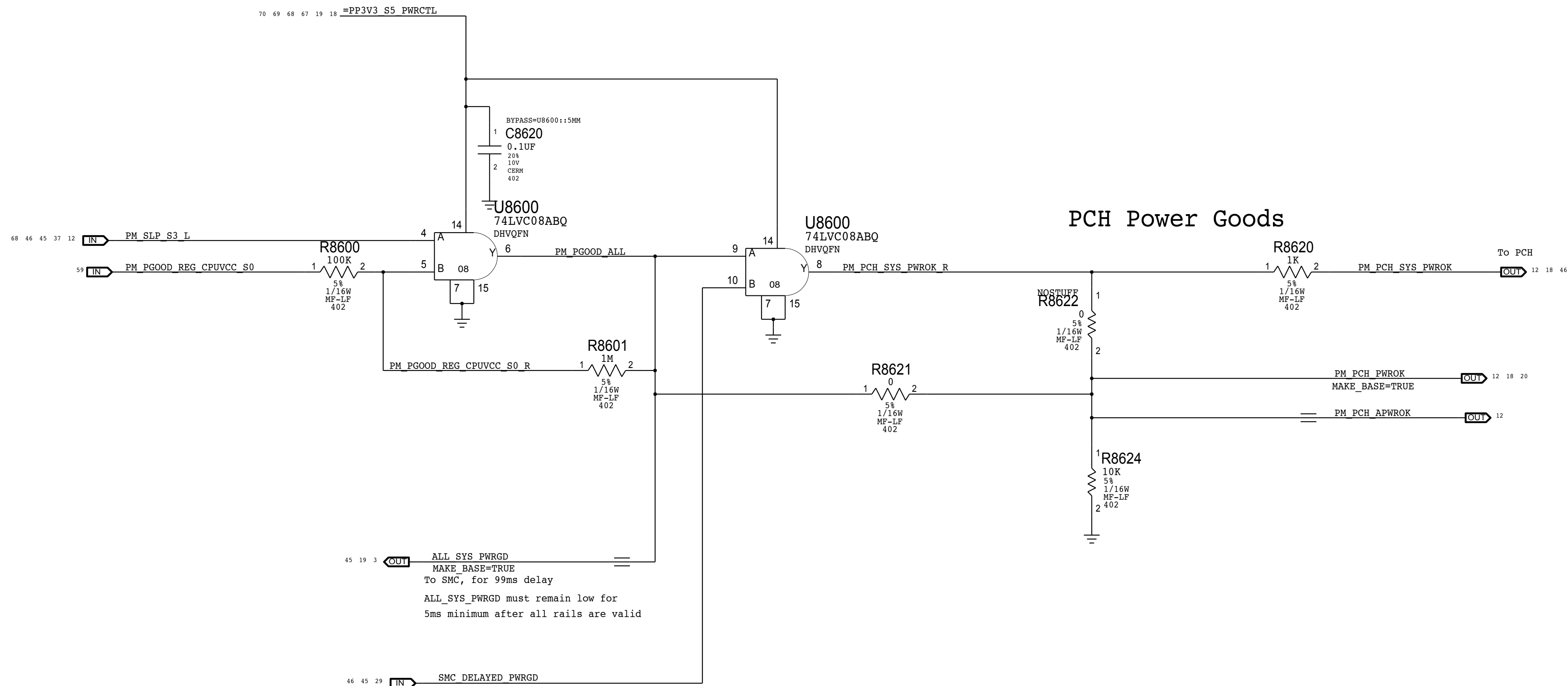
Audio + PCH Sequencing Requirements:
4.5V -> 1.05V -> 3.3V -> 1.5V -> ALL SYS GOOD



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SYNC_MASTER=J117_ANDRES		SYNC_DATE=03/24/2014	
PAGE TITLE			
PLATFORM POWER: PM Regulator Enables			
	DRAWING NUMBER		SIZE
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ALL_SYS_PWRGD, PCH_PWROK & SYS_PWROK Generation



Rail definitions

Platform: All processor non-Core and non-Graphics (5V, 3.3V, 1.5V, 1.05V for PCH/TBT/GPU)
Uncore: 1.8V and 1.2V for DDR3

Notes on sequencing requirements

Intel:

1. No hard specification on platform rails
2. SMC guarantees timing on PCH DPWROK and PWROK
3. VCC3_3 may power up before VCC3, VCC must ramp to 0.6V within 25ms of VCC3V3 ramping to 2.6V
4. VCC1_5 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC1V5 ramping to 1.35V
5. VCC may power down before VCC3_3, VCC3_3 must ramp down to 2.6V within 35ms
6. VCC may power down before VCC1_5, VCC1_5 must ramp down to 1.35V within 35ms

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Resume Reset

Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

Note:

The iMac J70 design does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together

Requirements:

Power on:
Asserted at least 10 ms after all suspend well power is valid

Power off or loss of AC:
Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V
to allow PCH to switch suspend well to battery without excessive loading

Method:

The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.

RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.

